PETROBRAS	MATERIAL REQUISITION	No.	I-RM-3010.2G-5330-667-KES-301
	CLIENT:	E & P	SHEET 1 of 38
	JOB:	BUZIOS FIELD AREA DEVELOPMENT	
	AREA:	PETROBRAS 78 (P-78)	
SRGE	TITLE:	MATERIAL REQUISITION FOR OILY WATER SEPARATOR	(I)
			SCP78

Keppel Shipyard / Hyundai Heavy Industries [File: MICROSOFT WORD/V.2003/ I-RM-3010.2G-5330-667-KES-301_A]	CONTRACT: 5900.0117818.21.2		
	CONTRACTOR Nº		
	TECH.RESP.:: Shravan Kamat	CREA Nº:	INITIALS:

INDEX OF REVISIONS

REV.	DESCRIPTION AND/OR REVISED SHEETS
0	FOR PURCHASE
A	APPROVED FOR PURCHASE

	REV. 0	REV. A	REV. B	REV. C	REV. D	REV. E	REV. F	REV. G	REV. H
DATE	21/05/2021	01/09/2021							
DESIGN	KES	KES							
EXECUTION	DongkwanLee.HHI	JongwonKim.HHI							
CHECK	YounchulChoi.HHI	YounchulChoi.HHI							
APPROVAL	Choong Kim Whye	Choong Kim Whye							

INFORMATION IN THIS DOCUMENT IS PROPERTY OF PETROBRAS, BEING PROHIBITED OUTSIDE OF THEIR PURPOSE

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APPENDIX 13: INTENTIOANLLY LEFT BLANK

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1.0 INTRODUCTION

1.1 PURPOSE OF THIS DOCUMENT

The purpose of this document is to describe the minimum requirements for supplying, Oily Water Separator Package as listed on the attached specifications, of proven design, for installation on the Floating Production, Storage and Offloading Unit (FPSO) in the Buzios field in the Santos Basin, Brazil (Hereinafter referred as “P-78 FPSO”).

1.2 PROJECT DESCRIPTION

Petrobras (Client) has launched a tender to supply a floating production, storage and offloading vessel (FPSO) with a topside crude oil process plant and gas process plant for field development in the Buzios field in the Santos Basin. The FPSO, referred to as P-78, will have the capacity to process 180,000 barrels of oil and 7.2 million m3 of gas daily.

COMPANY	:	PETROBRAS
MODULE SUPPLIER or TOPSIDES CONTRACTOR	:	KEPPEL SHIPYARD LTD. (KSL)
BUYER or HULL CONTRACTOR	:	HYUNDAI HEAVY INDUSTRIES CO., LTD (HHI)
PROJECT or CONTRACT	:	PETROBRAS P-78 Buzios FPSO (P-78 FPSO)
PACKAGER, SUPPLIER OR VENDER	:	The party that manufactures or supplies equipment and services to perform duties specified by the HULL CONTRACTOR / COMPANY and bears full responsibility for the equipment and service supplied.
SUB-SUPPLIER	:	The party that manufactures or supplies equipment and serviced to perform duties specified by the VENDOR / SUPPLIER.
CLASSICIATION SOCIETY	:	ABS (American Bureau of Shipping) described in Sec.3 Class Notation of “Exhibit XV (Directives For FPSO Classification)”
THIRD PARTY	:	An Independent 3rd Party Certifying Authority for certifying specific item/Equipment fabricated at VENDOR’s shop.
INSPECTOR	:	COMPANY / HULL CONTRACTOR appointed person, group or organization acting in behalf of COMPANY / CONTRACTOR responsible for inspection and witness of them and/or Equipment at VENDOR’s shop

2.0 GENERAL

All materials and equipment supplied shall be brand new (not overhauled), field proven, free from defects and accepted by COMPANY and the Classification Society. Materials and equipment shall be manufactured according to internationally recognized standards for the offshore oil drilling and Production industries, and shall be in conformance with the Basic Design, Agreement Specification and requirements.

A definition of Field Proven is as follows; Systems and equipment shall demonstrate satisfactory operation at least in 3 offshore floating production units, operating under similar conditions of the FPSO to be supplied, for a minimum of 24,000 (twenty-four thousand) hours in each unit. Unproven designs or prototypes (including components) without offshore service will not be accepted. Project required minimum service life is 30 years while operating in the environment specified on the Data Sheets, specifications and the other appendices in this material requisition.

The VENDOR shall identify any aspects of its standard components or packages that do not meet this requirement and submit descriptions to the BUYER for consideration with the Submission. Consumables and components that cannot achieve the full service life of the equipment and require periodic replacement (e.g. filter elements, pump seals, gaskets etc) shall be listed in design data submitted for COMPANY approval. Consumables and replacement components shall be provided to allow completion of all package commissioning and use during:
- Hull Commissioning, Topside integration; tow voyage; FPSO start-up.

2.1 CONFLICTS

Should VENDOR find conflict between documents or has any doubts regarding the specified requirements, these shall be communicated immediately to BUYER in writing. Related WORK shall not proceed until VENDOR receives clarification from BUYER in writing.

In the event of any conflict arising among the Specifications, Data Sheets and other Documents listed here-in, refer comments to the BUYER for clarification before design or fabrication commence. The general order of precedence is as follows.

a. Brazilian Regulatory Standards (NRs) from Ministry of Economy (Labour Department)
b. Classification Society(ABS) Rules
c. This Material Requisition and Equipment Technical Specification
d. Data Sheets
e. Project Drawings
f. Project Specifications
g. International Codes and Standards

Should a conflict exist between any documents referenced in this attachment, it shall be the responsibility of the VENDOR to call attention to the discrepancy and request a written ruling or interpretation from the BUYER's Engineer responsible for the equipment. VENDOR is not at liberty to assume which instruction shall govern.

2.2 ABBREVIATIONS AND ACRONYMS

Abbreviations	Descriptions
AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
AO	Analog Output

ASTM	American Society for Testing and Materials
AVR	Automatic Voltage Regulator
CSS	Control & Safety System
CS	Classification Society
CT	Current Transformer
DO	Digital Output
EGC	Emergency Generator Controller
ESA	Electrical System Automation
EGCP	Emergency Generator Power and Control Panel
ESD	Emergency Shutdown
HMI	Human-Machine Interface (current designation for MMI)
IEC	International Electro-technical Commission
ISO	International Organization for Standardization
PMG	Permanent Magnetic Generator
VT	Voltage Transformer
AGC	Auxiliary Generator Controller
AGCP	Auxiliary Generator Power and Control Panel
AI	Analog Input
DI	Digital Input
AC	Alternating Current
CPY	Company
AMS	Asset Management System
CCR	Central Control Room (located in the Hull Accommodation)
EMC	Electromagnetic Compatibility
FAT	Factory Acceptance Test
FGS	Fire and Gas System
FRP	Fiberglass Reinforced Plastics
HCS	Hull Control System
HSD	Hull Shutdown
I/O	Input / Output
IP	Ingress Protection Ratings
IS	Intrinsically Safe
MMS	Machinery Monitoring System
OPC	Open Platform Communication
PCS	Process Control System
PLC	Programmable Logic Control
PSD	Process Shutdown
SAT	Site Acceptance Test
SIT	Site Integration Test
SNMP	Simple Network Management Protocol
SOS	Supervision and Operation System
TCP/IP	Transmission Control Protocol/Internet Protocol
UCP	Unit Control Panel (Package Control Panel)
USB	Unit Switch Blowdown
USM	Unit Switch Malfunction
USS	Unit Switch Shutdown
DDA	Document Design Analysis
VDA	Vendor Design Analysis
TQF	Technical Query Form
DOL	Direct On Line starting
ETA	Estimated Time of Arrival
IRC	Inspection Release Certificate
ITP	Inspection and Test Plan

MC	Mechanical Completion
MFR	Manufacturer
MRB	Manufacturing Record Book
NDT	Non-Destructive Testing
PMI	Positive Material Identification
PIM	Pre-Inspection Meeting
QMS	Quality Management System
SPIR	Spare Parts Interchangeability Report
VDRL	VENDOR Document Requirement List
VMDL	VENDOR Master Document List
QA / QC	Quality Assurance / Quality Control
HUC	Hook-Up and Commissioning
OPS	Operation and Start-up
PIMS	Electronic Document Management System(EDMS) by OMEGA
FPSO	Floating Production Storage and Offloading Unit
INMETRO	Instituto Nacional de Metrologia, Normalização e Qualidade Industrial

2.3 REVIEW OF DOCUMENTS

VENDOR is responsible for complying with all requirements from Brazilian rule & regulations, Project specifications, Classification Society Rules and international standards. VENDOR shall obtain all registrations and pay all fees as applicable by national rules covering these items.

Review / approval of VENDOR documents shall not relieve VENDOR of his guarantees regarding performance of the unit or the responsibility to ensure the equipment is safe for operation and fit for purpose.

~~2.4 INFORMATION REQUIRED WITH BID~~

~~The following information shall be submitted in accordance with Appendix 2 and Appendix 12, Essential documents list for Bid as part of the bid package. Any bid with missing information or not using the Project format will be turned down and not be evaluated by BUYER. If VENDOR cannot issue within Bid Closing, VENDOR shall propose the possible date. But it will be negative condition for short listing activity~~

~~1) A complete description of the scope of supply~~

~~2) A list of deviations from technical specifications and standards, including reasons for those deviations; this list shall be with the following columns: Technical Specification / Standard, Item Description, Deviation and Performance Impact~~

~~3) A complete data sheet, including painting scheme, that shall be in accordance to COMPANY Painting Technical Specification (I-ET-3010.00-1200-956-P4X-002)~~

~~4) An outline drawing showing the key dimensions and maintenance area~~

~~5) Completed Bid Tabulation sheets~~

~~6) The weight and the centre of gravity~~

~~7) Data on noise and vibration~~

~~8) The delivery schedule~~

~~9) A list of all spare parts (containing its complete technical specification and original manufacturer part number reference) for commissioning and start-up, as well as another list for two years of operation~~

~~10) A list of all special tools~~

~~11) The name, address, telephone, e-mail address and service network of a contact person from main vendor and its sub-vendors~~

~~12) All documents and drawings required by this material requisition and project specification~~
~~13) A reference list of similar installations~~
~~14) Required inspections and/or tests (ITP template)~~
~~15) The manufacturer schedule~~
~~16) Quality/inspection levels~~
~~17) A description of warranties~~
~~18) Technical assistance and training to be provided~~
~~19) Drawings of the layout~~
~~20) Performance curves~~
~~21) A list of sub-suppliers~~
~~22) Quality plan~~
~~23) List of Lube/Hydraulic Oils~~
~~24) List of Consumable Materials needed during commissioning, pre-operation and start-up~~
~~25) Store and preservation recommendations~~
~~26) List of service centres (Technical Assistance)~~
~~27) The name, address, telephone, fax number, e-mail address & service network of contact person~~

3.0 APPLICABLE DOCUMENTS

The following documents are referenced herein and shall form part of the technical requirements. Current editions of the industry standards including all mandatory addenda in effect at the time of the order shall apply unless otherwise indicated.

All deviations from this specification or the documents referred to herein shall be stated in writing. In the absence of such a statement, it is assumed that the requirements of the specification are met without exception. Failure to meet the requirements of this specification and referenced documents shall be corrected at BIDDER's cost.

Other than those approved deviations / exceptions / comments listed in Appendix 8, exceptions to or deviations from this specification and its' attachments if any along with applicable project specification, codes, standards and Brazilian regulations as specified in Appendix 2, are not permitted unless accepted in writing by BUYER and COMPANY.

3.1 DATASHEETS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
1	I-FD-3010.2G-5330-660-KES-303	A	DATASHEET – OILY WATER SEPARATOR PACKAGE

3.2 PACKAGE SPECIFICATIONS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
1	I-ET-3010.1M-5330-667-P4X-001	C	SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501)

3.3 PROJECT SPECIFICATIONS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
~~1~~	~~I-ET-3010.00-1200-940-P4X-002~~	~~C~~	~~GENERAL TECHNICAL TERMS~~
~~2~~	~~I-ET-3010.00-5140-797-P4X-001~~	~~D~~	~~ELECTRICAL SYSTEM AUTOMATION ARCHITECTURE~~
~~3~~	~~I-ET-3010.00-5140-700-P4X-001~~	~~F~~	~~SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS~~
4	I-ET-3010.00-5140-700-P4X-002	E	SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS
5	I-ET-3010.00-5140-700-P4X-003	D	ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS
~~6~~	~~I-ET-3010.00-5140-700-P4X-005~~	~~A~~	~~REQUIREMENTS FOR HUMAN ENGINEERING DESIGN FOR ELECTRICAL SYSTEMS OF OFFSHORE UNITS~~
7	I-ET-3010.00-1200-956-P4X-002	F	GENERAL PAINTING
~~8~~	~~I-ET-3010.00-5143-700-P4X-001~~	~~D~~	~~ELECTRICAL SYSTEM PROTECTION CRITERIA~~
~~9~~	~~I-LI-3010.00-5140-700-P4X-001~~	~~B~~	~~ELECTRICAL EQUIPMENT DATA SHEET MODELS~~
~~10~~	~~I-LI-3010.00-5140-797-P4X-001~~	~~D~~	~~ELECTRICAL SYSTEM AUTOMATION INTERFACE SIGNALS LIST~~
11	I-ET-3010.1M-1200-300-P4X-002	A	NOISE CONTROL REQUIREMENTS FOR ACCOMMODATION / SHIPSIDE
~~12~~	~~DR-ENGP-M-I-1.3-R.5~~	~~0~~	~~SAFETY ENGINEERING~~
13	I-ET-3A36.00-1000-941-PPC-001	D	METOCEAN DATA
14	I-ET-3000.00-0000-940-P4X-002	A	SYMBOLS FOR PRODUCTION UNITS DESIGN
15	I-ET-3000.00-1200-940-P4X-001	C	TAGGING PROCEDURE FOR PRODUCTION UNITS DESIGN
16	DR-ENGP-I-1.15	0	COLOR CODING
~~17~~	~~I-ET-3010.00-5140-712-P4X-001~~	~~C~~	~~LOW VOLTAGE INDUCTION MOTORS FOR OFFSHORE UNITS~~
18	I-ET-3010.00-1200-800-P4X-002	F	AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS
19	I-ET-3010.1M-1200-800-P4X-014	F	AUTOMATION INTERFACE OF PACKAGE UNITS PACKAGE INTERFACE
~~20~~	~~I-RL-3010.1M-1200-940-P4X-001~~	~~B~~	~~GENERAL SPECIFICATION FOR AVAILABLE UTILITIES~~
~~21~~	~~I-ET-3010.00-1200-251- P4X-001~~	~~B~~	~~REQUIREMENTS FOR BOLT MATERIALS~~
~~22~~	~~I-ET-3010.00-1200-200-P4X-116~~	~~A~~	~~REQUIREMENTS FOR BOLTED JOINTS ASSEMBLY AND MANAGEMENT~~
23	I-ET-3010.00-5400-947-P4X-002	0	SAFETY SIGNALING
24	I-ET-3010.1M-1200-200-P4X-002	A	PIPING SPECIFICATION FOR HULL
25	I-RL-3010.1M-1350-960-P4X-009	C	MOTION ANALYSIS
~~26~~	~~I-ET-3000.00-1350-940-P4X-013~~	~~F~~	~~TECHNICAL REQUIREMENTS FOR DIGITAL ENGINEERING~~
~~27~~	~~I-ET-3010.1M-1200-300-P57-001~~	~~0~~	~~COMPLIANCE WITH NR-12 REQUIREMENTS FOR MACHINES AND EQUIPMENT~~
~~28~~	~~I-ET-3010.1M-1200-500-PAH-001~~	~~A~~	~~COMPLIANCE WITH NR-13~~

			~~REQUIREMENTS FOR PRESSURE VESSELS~~
~~29~~	~~EXHIBIT III w/ appendices~~	~~17~~	~~DIRECTIVES FOR ENGINEERING~~
~~30~~	~~I-ET-3010.00-1200-431-P4X-001~~	~~D~~	~~THERMAL INSULATION FOR MARITIME INSTALLATIONS~~
~~31~~	~~I-ET-3010.00-1000-970-P4X-002~~	~~0~~	~~REQUIREMENTS FOR NDT~~
~~32~~	~~I-ET-3010.00-1200-200-P4X-001~~	~~C~~	~~MINIMUM REQUIREMENTS FOR PIPING MECHANICAL DESIGN AND LAYOUT~~
~~33~~	~~I-ET-3010.00-5520-888-P4X-001~~	~~C~~	~~AUTOMATION PANELS~~
~~34~~	~~I-ET-3010.00-1200-800-P4X-013~~	~~G~~	~~GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS~~
~~35~~	~~I-ET-3010.00-1200-955-P4X-001~~	~~B~~	~~WELDING~~
~~36~~	~~I-ET-3010.00-1200-955-P4X-002~~	~~B~~	~~REQUIREMENTS FOR WELDING INSPECTION~~
~~37~~	~~I-ET-3010.00-1200-956-P4X-001~~	~~A~~	~~QUALIFICATION TESTS FOR PAINT SYSTEMS~~
~~38~~	~~I-ET-3010.00-0000-970-P4X-001~~	~~A~~	~~REQUIREMENTS FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION~~
~~39~~	~~EXHIBIT V (Applicable Sections)~~	~~8~~	~~DIRECTIVES FOR PROCUREMENT~~
~~40~~	~~EXHIBIT VII (Applicable Secions)~~	~~1~~	~~DIRECTIVES FOR QUALITY ASSURANCE SYSTEM~~
~~41~~	~~EXHIBIT VIII (Applicable Sections and Appendix II and III)~~	~~M~~	~~DIRECTIVES FOR COMMISSIONING~~
~~42~~	~~EXHIBIT XV (Applicable Sections)~~	~~2~~	~~DIRECTIVES FOR FPSO CLASSIFICATION~~
~~43~~	~~I-ET-3010.2G-9000-911-KSL-001~~	~~A~~	~~DOCUMENT CONTROL PROCEDURE~~
~~44~~	~~I-ET-3010.1M-1350-140-P4X-001~~	~~C~~	~~HULL STRUCTURAL REQUIREMENTS~~
~~45~~	~~I-ET-3010.00-1352-130-P4X-001~~	~~A~~	~~FLOOR GRATINGS, TRAY SYSTEMS AND GUARDRAILS MADE OF COMPOSITE MATERIALS~~
~~46~~	~~I-ET-3010.1M-1200-800-P4X-001~~	~~A~~	~~INSTRUMENTATION ADDITIONAL TECHNICAL REQUIREMENTS~~
~~47~~	~~I-ET-3010.00-1200-588-P4X-001~~	~~C~~	~~SAMPLE CONNECTIONS~~
~~48~~	~~I-ET-3010.1M-1200-800-P4X-005~~	~~F~~	~~FIELD INSTRUMENTATION~~
~~49~~	~~I-ET-3010.00-1200-800-P4X-01~~	~~A~~	~~CRITERIA FOR DETAILING DESIGN CAUSE & EFFECT MATRIX~~
~~50~~	~~I-ET-3010.00-1200-800-P4X-010~~	~~B~~	~~CRITERIA FOR ESTABLISHING CABLE CODES AND CABLE GLAND CODES~~
~~51~~	~~I-MD-3010.1M-1200-940-P4X-02~~	~~K~~	~~DESCRIPTIVE MEMORANDUM - AUTOMATION & CONTROL~~

3.4 PROJECT DRAWINGS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
1	I-DE-3010.2G-1200-944-KES-301	0	P&ID GENERAL NOTES
2	I-DE-3010.2G-5330-944-KES-301	A	PID for BILGE, SLUDGE, BALLAST (AFT) AND GENERAL SERVICE SEAWATER SYSTEM
3	I-DE-3010.2G-6124-944-KES-301	A	P&ID FOR HULL SERVICE AND INSTRUMENT AIR DISTRIBUTION
4	I-DE-3010.2G-5115-944-KES-302	A	P&ID FOR FRESH HOT AND POTABLE WATER SYSTEM
5	I-DE-3010.00-5140-700-P4X-003	E	GROUNDING INSTALLATION TYPICAL

			DETAILS
6	I-DE-3010.2G-5140-700-KES-452	0	POWER INSTALLATION TYPICAL DETAILS
~~7~~	~~I-DE-3010.1M-1200-942-P4X-002~~	~~E~~	~~GENERAL ARRANGEMENT~~
8	I-DE-3010.2G-1351-942-KES-309	A	ENGINE ROOM LAYOUT
9	I-DE-3010.1M-5400-94A-P4X-001	A	AREA CLASSIFICATION - GENERAL
10	I-LI-3010.2G-5400-940-KES-301	0	AREA CLASSIFICATION DATA LIST
11	SHK-ST-DA-801-0151	B	STANDARD EQUIPMENT SEAT TYPE 3 & 4

3.5 INTERNATIONAL CODES AND STANDARDS

S/N	CODE NUMBER	DESCRIPTION
1	DIN	German National Standard Code
2	ISO	International Standard Organization
3	EN	European Standards
4	VDE / IEC	German National Electric Standard Codes / International Electric Codes
5	BGV	German Safety Regulations
6	ASME	American Society Of Mechanical Engineers
7	ANSI	American National Standards Institute
8	Classification Society	Classification Society defined for the Hull scope
9	Marpol	Marpol 73/78 - Annex I or most updated
10	IMO	IMO Resolution MEPC 107 (49) or most updated
11	IOGP	International Association of Oil & Gas Producers
12	Classification Society defined for the Hull scope	

Note: Equipment governed by this requisition shall be designed, manufactured and tested in accordance with the latest edition of Codes and Standards as specified in Package Specification of APPENDIX 2 and above 3.5 international code and standards in this material requisition.

3.6 BRAZILIAN REGULATIONS

S/N	REGULATION NUMBER	DESCRIPTION
1	NR-10	Brazilian Federal Government Regulatory Standards (Norma Regulamentadora Nº 10, Segurança em Instalações e Serviços em Eletricidade) - Safety in Electrical Facilities and Services;
2	NR-12	Brazilian Federal Government Standards– (Norma Regulamentadora Nº 12, Segurança em Instalações e Serviços em Máquinas e Equipamentos) - Safety on Installation and Services with Machines and Equipment;
3	NR-13	Brazilian Federal Government Standards– (Norma Regulamentadora Nº 13, Segurança em Inspeção de Segurança e Operação de Vasos de Pressão, Caldeiras e Tubulações) - Safety on Boilers and Pressure Vessels;
4	NR-26	Brazilian Federal Government Standards– (Norma Regulamentadora Nº 26, Sinalização de Segurança) - Safety Signaling;

5	NR-37	Brazilian Federal Government Standards– (Norma Regulamentadora Nº 37, Segurança e Saúde em Plataformas de Petróleo) - Safety and Health onboard Offshore Platforms;
6	NORMAM-01	Normas da Autoridade Marítima para Embarcações Empregadas na Navegação em Mar Aberto;
7	IBAMA	NOTA TÉCNICA CGPEG/DILIC/IBAMA Nº 01/11 or most updated

3.7 CLASSIFICATION SOCIETY

S/N	Class Name	DESCRIPTION
1	ABS (American Bureau of Shipping)	✠ A1, Floating Production, Storage and Offloading System (Ship-Type), (S) Buzios Field Santos Basin Brazil, FL(30), 205X, ✠ AMCC, EXP, CRC(OC), HAB++(OS), UWILD, HELIDK, OHCM - ABS VEC(VENDOR Equipment Coordination): Ms Sirin Safie (Tel: +65 6371 2653, ssafie@eagle.org) - RIG/SITE/FPSO info. : *Application: Floating Production, Storage and Offloading Syste m *Rig/FPSO Type: ship-type *Owner/Operator: Petrobras *Rig/Site/FPSO Name: (Site) Buzios Field Santos Basin Brazil, (name) P-78 FPSO - Offshore Info. : *Vessel Flag: To be advised (Marshall Islands or Liberia – to be decided) *IMO Number: To be advised *MMSI(Maritime Mobile Service Identity) Number: To be advised *Hull ID Number: Q554 *Class Society: YY283791 *Commissioning Date(est.): 1st Nov 2024

4.0 SCOPE OF WORK

The Work consists of design, manufacture, testing, inspection, preservation, packing, commissioning assistance, services and documentation for the Goods fully in accordance with the Requisition, project specifications (and drawings), Classification Society Rule and with relevant Brazilian legislation & standards. Unless specifically stated in the Requisition, the Work shall include all items, accessories and features that are necessary for the satisfactory performance and safe operation of the equipment, as well as all items, accessories and features required to comply with applicable rules and regulations.

The equipment shall, to the greatest practicable extent, be delivered fully erected and assembled and mechanically complete, including punched by QC/HUC/OPS, where applicable. The Goods shall be delivered as independent and complete packages (unless specifically noted otherwise). The PACKAGER shall be responsible to the lay-out of the equipment, accessories and piping around the skid, taking into account the space required for operation, maintenance and considering that the spaces and weights shall be the least possible.

Vendor’s minimum scope of supply shall include, but not be limited to: process and

mechanical design, detailed engineering, equipment procurement / expediting / inspection, fabrication, assembly, painting, insulation, instrument and control system integration, shop testing, inspection, documentation, identification, preservation, packing, service and documentation for shipment. The details of scope of supply shall be based on based on section 3.1 package datasheet and 3.2 package specification.

S/N	Q'ty	Item No.	Description
1	**Equipment, Materials and relative activities**		
1.1	**1 Unit**	**SAO-5330501**	**Oily Water Separator Package**
1.1.1	-	-	One(1) Completed Package is intended to perform the bilge water suction from the Engine Room Bilge Water Settling Tank, with a flow rate of 5 m³/h with the maximum oil content of 500 ppm, and discharge this bilge water flow rate to sea overboard with a maximum oil content of 15 ppm, attending IMO MEPC 107 (49) requirements and Marpol Annex I – Reg. 21.
			Bilge water oil monitoring system - One(1) sampling collecting point in the Oil Water Separator overboard discharge; - One(1) oil cell with range between 0 and 30 ppm of oil;
			Control Panel (PN-SAO-5336501)
			Automation as per Package Classification "P1" as defined in I-ET-3010.00-1200-800-P4X-002(AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS) and I-ET-3010.1M-1200-800-P4X-014(AUTOMATION INTERFACE OF PACKAGE UNITS PACKAGE INTERFACE)
			All required sub-equipment and/or componets in the Package based on PACKAGER's proposed technology.
1.1.2	-	-	All required valves, PSV and piping;
1.1.3	-	-	Junction boxes and control panels, as required;
1.1.4	-	-	Drain collector connected to the drain system of the unit;
1.1.5	-	-	Complete one(1) common skid assembly with resilient mounting for diesel oil purifier package - HHI standard type 4, specified in 5.2.3 Equipment support type selection - Self-supporting structural skid with drip pan, grating for access ways etc. along with earthing (grounding) / bonding.
1.1.6	-	-	On-skid access platforms and ladders, including ladder and stairway to upper levels for any skids with multiple levels as applicable – Requirement in Brazil Regulation NR 12 shall be complied with.
1.1.7	-	-	Access steps/platforms or ladders to be mounted outside of the skid edge where applicable shall be provided with fixing brackets and bolts/nuts as shipped loose. Mounting pads for the fixing bolts shall be welded at the skid edge.
1.1.8	-	-	All on-skid piping and tubing system including valve and supports terminating in flanged connections at skid edge. (All shipped loose piping spools and valves (if any) shall be completely welded and tested.)
1.1.9	-	-	All on-skid electrical equipment required for the safe, automatic, continuous operation and monitoring of the package including cable trays/supports for BIDDER and BUYER cables and

			earthing (grounding) within the package.
1.1.10	-	-	All on-skid instrumentation and controls required for the safe, automatic, continuous operation and monitoring of the package, including cabling, cable trays and junction boxes supplied via the platform control system BIDDER in accordance with the approved vendor list. Refer to APPENDIX 14 E&I TIE-IN DOCUMENT GUIDANCE TO BIDDERS and provide required interface documents as per the guidance.
1.1.11	-	-	Pre-Inspection Meeting(PIM), Material Identification & Traceability, PMI reports, Welding, Inspection, NDT, Heat Treatment as required and Hydrostatic Testing, Bolt tightening Record, Leak Test, Mechanical Completion(MC), Factory Acceptance Test, Punch Close, Final Inspection, Packing & Preservation and Inspection Release Certificate(IRC)
1.1.12	-	-	Calculation Reports – for system, capacity, selection and etc.
1.1.13	-	-	Classification Society Certification (Classification & Certification) VENDOR / SUB-VENDOR to allow CS(ABS) to share VENDOR Comment List with HHI (Hull Contractor) and his Consortium lead KSL. VENDOR to enable BUYER oversight access in Eagle Construct (ABS). VENDOR to note that ABS VEC in Singapore can and will direct individual VENDOR to the nearest design review and Survey office.
1.1.14	-	-	VENDOR / Sub-vendor to proceed with specific contracts (purchase order) allowing CS to send to CPY the correspondence.
~~1.1.15~~	~~-~~	~~-~~	~~INMETRO compliance Certification~~
1.1.16	-		Surface preparation and painting based on I-ET-3010.00-1200-956-P4X-002 – (PETROBRAS – General Painting Technical Specification).
1.1.17	-		Tagging identification with stencil and Nameplates (dual language: English and Portuguese, respectively) for Package / Equipment / Component
1.1.18	-	-	Safety sign per specification I-ET-3010.00-5400-947-P4X-002
1.1.19	-	-	Transport restraints, lifting spreader bars or frames and slings for each package for transportation and installation, complete with slings and shackles, materials, welding and NDE certification and proof load test certification.
1.1.22	-	-	VENDOR's standard mechanical completion check sheets are to be detailed for the overall package, as well each individual component (e.g. Pump, Motor, Filters, Instruments, Control panel, etc.) as pre section 8.7 in the MR. The mechanical completion check sheets shall be included in FAT procedure which will be submitted as part of vendor document during execution stage for SELLER review and approval.
1.1.23	-	-	Piping flow marking(stripes for identification) shall be provided as per Section 11 in DR-ENGP-I-1.15-R.4_0_COLOR CODING. The details for piping flow marking shall be included in Painting procedure which will be submitted as part of vendor document during execution stage for HHI review and approval.
1.1.24	-	-	When cables are subjected to mechanical impacts, proper additional protection by covers shall be foreseen. Cable trays covers shall be of the same material and the same thickness of cable trays they are

			protecting.
1.1.25	-	-	Flange isolation kits shall be provided where applicable as per ANNEX E in I-ET-3010.1M-1200-200-P4X-002_0_PIPING SPECIFICATION FOR HULL
1.1.26	-	-	Structural isolation kits shall be provided where applicable. (e.g. Between Stainless Steel Equipment's Support and Carbon Steel Skid's beam
1.1.27	-	-	I-ET-3010.00-1200-200-P4X-004_D_REQUIREMENTS FOR PIPING SUPPORT. Refer to Attachment 73-1.
2	**Spare Parts**		
2.1	1 lot	-	All equipment's/material's consumable and spare parts recommended by PACKAGER/MANUFACTURER for the construction, testing, commissioning, pre- operation and start-up phases. - Gaskets - Bolt/Nuts, Fasteners - Others as recommended
2.2	1 lot	-	Spare parts for start-up, commissioning and others as recommended by Classification Society
2.3	1 lot	-	Spare Part list for two (2) years of operation, including a detailed part description, price and delivery time of each part, lay-out and sectional drawings indicating the location of part and TAG / reference identification (original manufacturer part number) and subsystem
3	**Special Tools**		
3.1	1 lot	-	Special / Unusual tools for assembly, disassembly, installation, commissioning, operation and maintenance
4	**Documentation**		
4.1	1 lot	-	Documentation as per VDRL, this material requisition and project specifications
4.2	1 lot	-	Vendor Data Book as per this material requisition chapter 11.0 Document Requirement based on project Exhibit III-Directives for Engineerin, Exhibit V – Directives for Procurement ~~and I-ET-3010.2G-9000-911-KSL-001~~ (DOCUMENT CONTROL PROCEDURE in APPENDIX 3)
5	**Inspection, Testing and Certification**		
5.1	1 lot	-	Inspection and testing during manufacture in accordance with Approved "Inspection and Test Plan" (ITP) including sub-vendors.
5.2	1 lot		Mechanical Completion with documentations in accordance with project specification, P&ID and code/standards
5.3	1 lot		Factory Acceptance Test
5.4	1 lot		Full functional test shall be implemented for each Sub-vendor's package in conjunction with other related mechanical and electrical equipment. All interface connections shall be fully terminated and function checked with associated documentation
5.5	1 lot		The thickness measurement on all equipment and piping would be required using Ultrasound Equipment.
6	**Others to be in the base scope of supply**		
6.1	1 lot	-	Engineering services
6.2	1 lot	-	Process, mechanical, electrical and instrumental (automation) design and performance guarantees
6.3	1 lot	-	The following meeting shall be included in VENDOR's base scope of work

			1) Kick Off Meeting – Conference Call (1 time, 2 working days) 2) Pre-Inspection Meeting at VENDOR's Premises (1 working day) ~~3) Clarification Meeting at BUYER's Office (2 times, 2 working days each)~~ 4) HAZOP Meeting - Conference Call (1 time, 2 working days) 5) Technical Clarification Meeting at VENDOR's Premises or Conference Call (anytime when required)
6.4	-	-	PDMS 3D Modelling Submission in Native Files at least four times - For 30% review, - For 60% review, - For 90% review. - Final as built model
6.5	1 lot	-	Anchor bolts / setting bolts / hold-down bolts/nuts/washer for equipment/items supplied by BIDDER (based on I-ET-3010.00-1200-251-P4X-001 – (PETROBRAS – Requirements for Bolting Materials)
6.6	1 lot	-	All required project dossiers shall be made available as per specified schedule mentioned in the project Exhibit, the project specifications and this Material Requisition
6.8	1 lot	(Based on Exhibit VIII – Directives for Comm'g, Appendix 3 for Vendor Technical Assistance)	Supervision service for installation, pre/commissioning & start-up ; Supervision services for listed activities with daily rates shall be proposed by VENDOR as optional scope. - During Installation at Fabrication Site - During Pre-commissioning at Fabrication Site - During Commissioning at Offshore Field
6.9	1 lot	(Based on Exhibit VIII – Directives for Comm'g, Appendix 2)	Training of for BUYER Operating and Maintenance Teams **as optional scope** ; VENDOR shall provide 3 optional proposals for each 3 locations of Korea(On-shore), Singapore(On-shore), Brazil(Off-shore). ; Training Program for each piece of equipment shall cover maintenance and operation aspects, since the public to attend the training shall be presented by two (2) separated groups of fifteen (15) technicians, being one group for maintenance staff and other for operation staff. Attendees shall be awarded certificates of training, upon completion of the course. ; The training shall be performed in Brazilian Portuguese language or in English with translation to Brazilian Portuguese language. ; Training
6.10	1 lot	-	Travel and subsistence to and from the site location

5.0 ENGINEERING NOTES

5.1 GENERAL

5.1.1 SAFETY FEATURES

VENDOR shall provide all safety features as required for operation of this facility as an integral part of the total facility. The equipment (package) shall be designed with necessary pressure relief and automatic controls to allow safe, continuous and unattended operation.

5.1.2 MEASUREMENTS

SI units of measurement shall be used for all measurements and calibration, except for piping.

5.1.3 UTILITIES AVAILABLE

VENDOR to refer to I-RL-3010.1M-1200-940-P4X-001 General Specifications for Available Utilities. VENDOR shall state the specific utility requirements for the Package in its Submission. The VENDOR shall also state and detail any additional utility requirements.

5.2 BASIC DESIGN

5.2.1 SEISMIC FORCES, BLAST LOADS, LIFTING AND TRANSPORTAION ACCELERATIONS

BIDDER shall design equipment and structural members suitable for seismic forces, blast loads, lifting and transportation load, and provide recommendations for equipment support, tie down, and fluid levels in order to reduce design acceleration loads of the equipment during transportation. Refer to the I-ET-3A36.00-1000-941-PPC-001 METOCEAN DATA and I-RL-3010.1M-1350-960-P4X-009 MOTION ANALYSIS.

5.2.2 DIMENSION REQUIREMENTS

VENDOR shall put in place a procedure on dimensional control and strictly adhere to it. Method for verification of dimension shall be described in the procedure clearly and the procedure shall be submitted for BUYER, CS and CPY's review and approval. Fabrication tolerance approved by BUYER, CS and CPY, which has to be shown VENDOR's drawing, shall be checked at every stage and at the final survey. Especially, Dimension (X, Y and Z coordination) with agreed tolerance for interface points connected to BUYER such as Piping and Cable tray tie-in point dimension, Baseplate Bolt Hole dimension, Baseplate footprint shall be described on approved drawings, submitted with technical proposal and carefully measured.

For accurate measurement, VENDOR shall utilize Digital Dimension Measuring Equipment, e.g. Total Station.

The tolerance for location of tie-in nozzles shall be +/- 3mm of the dimensions provided in on the approved equipment general arrangement drawing.

The tolerance for location of foundation bolt holes shall be +/- 1.5mm of the dimensions including diagonal dimensions provided in on the approved equipment general arrangement drawing.

VENDOR shall submit confirmed dimension records and weight records right after the completion of corresponding/relevant inspection and test to BUYER by using project standard format in APPENDIX 7.

In case any non-conformity issues out of tolerance is found against the approved drawings, VENDOR shall take corrective actions before shipment else VENDOR shall do the corrections at BUYER yard at his cost.

5.2.3 EQUIPMENT SUPPORT TYPE SELECTION

Equipment support will be designed in accordance with Type 4 in SHK-ST-DA-801-0151 (Bolting Type Equipment Seat and Support)

TYPE	Support Height (mm)	Deck Opening	Drawing Description	Installation Method
4	200	Open	Package Item	Bolting

In case any non-conformity on standard equipment seat and support, the VENDOR's proposed equipment's support TYPE will be discussed during Technical Bid Evaluation

5.3 PIPING

VENDOR shall follow the project requirement of piping system in the package based on I-ET-3010.00-1200-200-P4X-001 (MINIMUM REQUIREMENTS FOR PIPING MECHANICAL DESIGN AND LAYOUT) and I-ET-3010.1M-1200-200-P4X-002 (PIPING SPECIFICATION FOR HULL).

VENDOR shall provide all piping, tubing and fittings within the boundary limits of Skid. BUYER's connections shall be terminated with flange connection at the edge of Equipment. Interconnecting piping between skids, if required, shall be sized by the VENDOR and shall be provided by the BUYER. Where applicable, common services shall be manifold configuration within the skid limits and a single, common customer connection shall be provided. The type of flanges shall be ASME rating.

VENDOR has to advise if the vents can be left locally. In case, BIDDER recommends to take away from the local area, then all such vents shall be combined to one and the piping / tanks etc. are to be designed for static head based on the BUYER's piping data. This termination shall be at the skid edge for easier BUYER's interface.

All auxiliary tubing or piping shall be adequately supported to prevent damage from vibration.

5.4 ELECTRICAL

VENDOR shall follow the project requirement of electrical system in the package based on I-ET-3010.00-5140-700-P4X-003 (ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS), I-ET-3010.00-5140-700-P4X-001 (SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS) and the all referenced project specifications in the aforementioned specifications.

VENDOR is responsible for detailed electrical design and engineering within the Package and shall perform all functions required to interface with the design of electrical system, as well as guarantee the control and monitoring by A&C (Automation and Control System), when required.

All electrical installations within the Package shall be VENDOR responsibility. VENDOR is also responsible for supervising the installation of equipment that will be installed out of the Package skid, if any. For electrical equipment, it shall be considered the reference temperatures stated in specific Technical Specifications included in its Material Requisition (RM), standard IEC 61892-1 and applicable rules and regulations of Classification Society

The application of electrical equipment, instruments and accessories on hazardous areas shall follow the requirements of IEC series 60079, 61892-6 and 61892-7, as well as have their conformity certificates according to INMETRO Portaria 179, May 18th 2010 and INMETRO Portaria 89, Feb 23rd 2012 and approved by Classification Society.

Electrical equipment and accessories installed in external safe or hazardous areas, which shall be kept operating during emergency shutdown ESD-3P or ESD-3T shall be certified for installation in hazardous areas Zone 2 Group IIA temperature T3 (IEC 61892-7).

All equipment and the outgoing feeders of the Package shall be supplied with connectors and terminals for power, control, heating and grounding.

Each Package shall have a field emergency stop push-button, located in safe area (area with easy access in case of emergency condition in Package) in Package skid, or closer to Package skid, to stop the Package, as required in NR-12.

5.5 INSTRUMENTATION

VENDOR shall follow the project requirement of Automation, Control and Instrument system in the package based on I-ET-3010.00-1200-800-P4X-002 (AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS), and the all referenced project specifications in the aforementioned specifications. Packages shall have their operating controls and shutdown systems defined according to the Package Technical Specification.

Electronic instruments shall meet the requirements of IEC 61000-6-1/2 regarding electromagnetic compatibility. All instruments, junction box, panels, materials and equipment proper to be used in hazardous areas, shall have conformity certificates complying with PORTARIA INMETRO Nº 179, DE 18/maio/2010, and its annexes, changed by PORTARIA INMETRO Nº 89, de 23/fereveiro/2012, and shall be approved by Classification Society. Electrical equipment installed in external areas, that shall be kept operating during emergency shutdown ESD-3P and ESD-3T shall be certified for installation in hazardous areas Zone 2 Group IIA temperature T3.

All instruments and their accessories shall be at least IP-56 (IEC 60529) and according to Classification Society rules. By the time the Package is delivered, all its components shall not be obsolete. Warranty and spare parts policies are described in the respective Material Requisition, Purchase Order or Contract. Continuity of spare parts supply and maintenance shall be given for a 5-year period. Instrument shall be painted to withstand the environment conditions specified in item 3. Colors shall be in accordance with DR-ENGP-I-1.15-R.3 – COLOR CODING.

All electrical and electronic devices, beyond mechanical parts of the equipment, shall be designed and constructed in a tropicalized version. Tropicalization process comprises application of reinforced protective resin Class 2 according to IEC 61086 and fungus proof according to ASTM G21 on all printed circuit boards, use of anti-rust materials and accessories and other implementations according to manufacturers' experiences and related rules, aiming to provide a robust and reliable construction. All junction boxes shall be according to I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS.

All UNIT equipment and systems that requires PLC (Programmable Logical Controller), HMI (Human Machine Interface), servers, workstations, industrial switches, special tools according to their respective Technical Specification and related documents, shall be supplied with the complete software licenses, as well as their access passwords to all functionalities, for all kinds of possible programming users.

APPENDIX 14 E&I TIE-IN DOCUMENT GUIDANCE TO VENDORS and the E&I tie in document will be priority to determine the E&I interface between VENDOR and BUYER. Therefore, any conflicts between vendor documents and E&I tie-in document must be resolved by VENDOR prior to the submission of any E&I interface documents to BUYER. Any of discrepancy between tie in document and other vendor data/actual equipment shall be rectified before the 1st submission.

5.6 STRUCTURAL REQUIREMENTS

Vendor shall design equipment and structural members suitable for lifting and transportation load, seismic loads, blast overpressure, wind and provide recommendations for equipment support, tie down, and fluid levels in order to reduce design acceleration loads of the equipment during transportation. Refer to the I-ET-3A36.00-1000-941-PPC-001 METOCEAN DATA and I-RL-3010.1M-1350-960-P4X-009 MOTION ANALYSIS.

5.7 PROTECTIVE COATINGS & INSULATION

VENDOR shall follow the project requirement of painting for the package based on I-ET-3010.00-1200-956-P4X-002 (GENERAL PAINTING), I-ET-3010.00-1200-956-P4X-001 (QUALIFICATION TESTS FOR PAINT SYSTEMS) and the all referenced project specifications in the aforementioned specifications.

Before commencement of any painting work, Coating Procedure Test (CPT) shall be firstly completed including adhesion test (Min 5MPa) for each painting system applied for the equipment and materials.

BIDDER shall follow the project requirement of qualification of personnel carrying out painting work and inspection based on I-ET-3010.00-000-970-P4X-001 (REQUIREMENTS FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION) and the all referenced project specifications in the aforementioned specifications.

Painting Shop Audit by BUYER's coating inspector shall be carried out before starting Painting work including sub-vendor's painting shop.

Where insulation is required, equipment, vessels and piping that require insulation shall be insulated in accordance with I-ET-3010.00-1200-431-P4X-001 (THERMAL INSULATION FOR MARITIME INSTALLATIONS). Insulation shall be suitably protected to prevent transport related damage.

6.0 VENDOR'S RESPONSIBILITIES

VENDOR is responsible for coordinating the technical aspects and required inspection and test of the equipment and all auxiliary systems included in the scope of the order. This includes expediting, and assurance that all equipment, including that furnished by Sub-vendor, is supplied to the applicable specifications referenced on the data sheets or requisition. It also includes a guarantee that the performance of the complete package shall be satisfactory to the BUYER. It is the responsibility of the VENDOR to notify all Sub-vendors of applicable specifications, Codes, and Industry Standards.

The VENDOR shall ensure that reference specifications are part of any requisition that are placed on a third party for supply of equipment or service under this requisition.

VENDOR shall be responsible to transfer all applicable specifications to Sub-vendor to comply with project requirement. Drawings and information, i.e. WPS, ITP and WPQT, NDE

Procedure, Weld Map, NDE Map, Coating Procedure and Test Procedure, etc., from Sub-vendor shall be submitted for BUYER's review & approval prior to commence of any work by Sub-vendor.

VENDOR shall bear all facilitating cost to have regular design review meetings with BUYER, COMPANY and other relevant parties excluding attendee's cost of transportation and accommodation.

BUYER's review of and comments to drawings, specification and other documents prepared by VENDOR shall in no way be construed as an "approval" (even if the word "approval" is inadvertently employed), and shall in no way relieve from the responsibility to comply with requirements set forth in the Contract.

VENDOR shall use Project legend, symbols when drafting PFD's, P&ID's and electrical drawings. BIDDER standard legend and symbols will not be accepted.

VENDOR shall participate in Hazard and Operability sessions (HAZOP) for his supplied process and utility system equipment in accordance with Project HAZOP Strategy. VENDOR shall bear ALL costs associated with participants and implement the outcome into equipment design.

VENDOR shall upon request provide assistance to the BUYER's commissioning team. The commissioning of relevant items and systems shall be performed according to project procedures and VENDOR's specific procedures.

VENDOR shall provide/devise some sort of method/software/tool/technology which should be used for inspection/meeting in case of the access to facility is restricted to external personnel due to COVID-19 emergency lockdown and/or similar cases.

6.1 APPROVED VENDOR LIST

Equipment and components within the package shall be from VENDORs listed on the Approved Vendor List. VENDOR shall submit a list of Sub-vendors indicating the items to be furnished.

VENDOR shall secure BUYER approval before making an Order if VENDOR wants to change any Sub-vendor.

VENDOR shall provide list for the proposed Sub-vendor for fabricating and/or manufacturing of the component and/or parts, i.e. Skid, Structure, Vessels, Piping, etc., for Package (Equipment) for BUYER's review & approval.

And, VENDOR has responsibility to forward all relevant specification to their Sub-vendor to meet Project requirement, and submit all drawings and information, i.e. WPS, ITP, WPQT, etc., from Sub-vendor for BUYER's review & approval prior to start any work by Sub-vendor.

6.2 SHIPPED LOOSE ITEM CONTROL

VENDOR shall perform trial-fit of all their supplied items before FAT. If any items need to be disassembled from the equipment after the test and shipped as shipped loose due to practical reasons, VENDOR shall submit shipped loose item list and assembly drawings indication item number matching with the packing list and get a written approval from BUYER.

All shipped loose item shall have self-standing supports mounted on the equipment, structure, duct, piping, except VENDOR Furnished (VF - Inline/Online) Instrument or Valve

installed outside battery limit. Without this protocol, shipped loose supply of any kind of component will not be accepted. All responsibility of guaranteed performance related to this subject is under VENDOR's scope.

6.3 WEIGHT CONTROL

The installation of equipment on an offshore platform requires the accurate consideration of the weight involved. The VENDOR shall exercise an active weight control system throughout the design and fabrication of the units.

A weight control data sheet is included in mechanical data sheet of the package for this purpose, and this shall be completed by the VENDOR in a timely manner.

A preliminary copy of the weight control data sheet shall be submitted with VENDOR's submission. Options for saving weight shall also be included in the Submission.

VENDOR shall ensure that the equipment weight is properly documented by a weight report and weight certificate. The certificate shall where applicable indicate outline dimensions and CoG related to a defined origin. The weight record shall be used to finalize and maintain the overall weight report of the equipment.

VENDOR shall provide lifting arrangement drawings for the installation. The arrangement shall be a single point lifting of fully assembled equipment on ground.

VENDOR shall comply with ISO 19901-5 (2016 edition) Class A requirements. More specifically, during prior equipment delivery, VENDOR shall fully comply with Section 9 and issue officially Annex A (weight control datasheet) and Annex B (Weighing certificate) fully consistent deliverables based on frequency defined in sub-section 9.2.

Notes:

1) The weight control datasheet shall be reissued officially each time the item weight variation is exceeding +/- 5% from the first revision.
2) Weight control datasheet local CoG coordinates shall indicate either the North platform direction or provide a clear picture of the equipment. (Objective is to allow Engineering contractor to orientate the equipment on the platform and perform relevant CoG coordinates conversions to match with Project 3D model coordinates system)
3) All equipment including loose delivered items, mobile equipment and auxiliaries shall be reported in a dedicated weight control datasheet (one sheet per item except for identical ones (number of similar items and/or tag numbers to be defined)). A weight or CoG input field with a "HOLD" or "no value" is not acceptable. If value is not available at weight control datasheet issuance date, Supplier/Vendor shall provide weights and CoGs preliminary estimates and clearly indicate the status "Preliminary Estimate" on the weight control datasheet.

Sub-contractors by VENDOR shall fully comply with ISO 19901-5 (2016 edition) Class A requirements. Depending of their scope of work, weight control activities can be limited to VENDOR ones after formal agreement with BUYER.

Sub-contractor Weight control procedure shall be issued by VENDOR at least 2 times in order to implement Contractor & Company comments. Sub-contractor Weight control reports shall be issued by VENDOR at least 4 times along the Project (2 months after POI, 1 month after 30% model review, 1 month after 60% Model review and 1 month after 90% model review). Database in excel native format shall be issued officially with the report.

6.4 MATERIAL HANDLING REQUIREMENTS

VENDOR shall perform a handling study covering all components need its maintenance. The weight and recommended lifting procedure and dimensional maintenance area requirement for components shall be documented and submitted to BUYER. The description shall include all suggested maintenance activities, including preventive based, planned activities, and breakdown repairs.

VENDOR shall provide material handling equipment such as monorail with hoists or pad-eyes inside the package for equipment removal from the skid. VENDOR shall carry out the assessment for their lifting appliances and submit the test certificate prior to FAT.

All lifting attachment designs and calculations are the responsibility of the VENDOR and must be reviewed by the BUYER prior to fabrication. All engineered lifting components shall be designed for a lift equal to two (2) times the maximum load to be lifted, unless larger factors are required by applicable codes or standards. NDT for lifting attachment shall be carried out 100%.

7.0 NOISE AND VIBRATION LIMITATION

Noise and vibrations limits shall conform to I-ET-3010.1M-1200-300-P4X-002(NOISE CONTROL REQUIREMENTS FOR ACCOMMODATION / SHIPSIDE).

Noise and vibration data is required for the final proposal and after the Factory Acceptance Tests (FAT).

The maximum sound level at any location 1m from the equipment shall not exceed 85 dB(A). This is applicable to all of the operating conditions for which the equipment is used.

8.0 INSPECTION AND TESTING

VENDOR is responsible for compliance with Minimum Quality Requirements in Appendix 15 as well as follow sections.

8.1 INSPECTION AND TEST PLAN

VENDOR shall provide and implement an Inspection and Test Plan (ITP) for each inspection and testing required in Project. The ITP shall be submitted to BUYER, CS (If required) and CPY for review and approval. The ITP shall contain inspection involvement to be witnessed by BUYER, schedule of inspection and test, test procedures.

VENDOR shall ensure that inspection will be carried out for witness and hold activities of BUYER, IVB and CPY as per approved ITP. At its discretion, BUYER may conduct detailed inspections in addition to those mentioned in the approved Test and Inspection Plan.

ITP which VENDOR submits shall be satisfied with requirement stated in this section as minimum. Legend of inspection involvement to be used on ITP shall be as follows;

1) “Review–R”: A point at which BUYER and/or CS, COMPANY shall review the relevant document as per the approved Inspection & Test Plan (ITP).
2) "Spot Witness-SW": A point at which BUYER and/or CS, COMPANY shall attend or witness the initial activity only. In case any discrepancy is found during initial inspection, the number of BUYER, CS and CPY’s visit will be increased on a random basis. If there is no abnormal issue at the initial activity, further inspection may be waived with declaration of BUYER and/or IVB, Company then Inspection notification shall be

provided continuously.

3) "Witness–W": A point at which BUYER and/or CS, Company shall attend or witness the activity as per approved Inspection & Test Plan (ITP). If there is no notice for participation from BUYER and/or CS, Company until appointed inspection date, inspection can be performed without BUYER and/or CS, COMPANY.
4) "Hold–H": A point at which BUYER and/or CS, Company shall inspect and complete the activity with participation as per approved Inspection & Test Plan (ITP). Otherwise, no further activity shall proceed without prior permission by BUYER and/or CS, Company. BUYER and/or CS, COMPANY shall provide notice of attendance intention when inspection notification has been provided.
5) "Monitoring–M": A point at which BUYER and/or CS, Company may visit to observe fabrication progress going on anytime as per approved Inspection & Test Plan (ITP). BUYER and/or IVB, COMPANY should provide advance notice to VENDOR for the monitoring on desired schedule.

* Note (1): When a conflict exists between Material Requisition, Company specification and ITP, Material Requisition and Company specification have priority. VENDOR is not at liberty to assume which instruction shall govern.

VENDOR shall develop and implement processes and procedures which efficiently and effectively monitor, verify and document Quality of Work scope deliverables. VENDOR shall ensure that VENDOR and sub-vendors ITP are prepared at a level of detail sufficient to address all Quality Control related activities in chronological order, from contract review through manufacturing, fabrication, required inspection, assembly, required testing, documentation, and certification. In addition, VENDOR shall ensure compliance with the following requirements to be incorporated on ITP as minimum:

1) To ensure the compatibility of the project, procedures and applicable internal documents to the equipment or material which constitutes the object of supply;
2) To indicate the fixtures and inspection equipment items (including measurement uncertainties) required to achieve quality by checking critical dimensions, functional tests, etc;
3) To indicate the personnel qualification performing inspection and checking activities
4) To identify the inspection points foreseen throughout the equipment or material production cycle (for example: raw material delivery, intermediate fabrication steps, painting, intermediate testing, final testing, transport preparation, delivery at site), where checking and inspection shall be performed (mandatory hold points, documents review and witness points), including checking and inspection of sub-suppliers. To indicate the type of examination, trial, test or inspection to be performed.
5) To indicate the execution procedures and acceptability standards for all quality features and requirements, including those, which contain a subjective element. To consider, when so determined in the project, the reference quality level (e.g. API/PSL)
6) To identify and develop quality records, mentioning the type of record: report, certificate, graph, etc. Such records shall allow the identification and traceability of critical components along the entire manufacturing process.
7) Each Inspection and Test Plan shall comprise, in detail, all the manufacturing stages of the respective equipment or material, as well as their critical components, including those adjudicate by sub-suppliers.
8) The aforementioned documents, the documents concerning the treatment of nonconformities and the qualification records regarding welding and welder's procedures shall be available for inspection during the manufacturing process.
9) Special attention shall be taken to all equipment / skids / material that are subject to

coating application, metallizing, lining and / or material specification, as to ensure that these points are considered in the Inspection Test Plan

10) Supplier shall indicate in the Inspection and Test Plans the extension of its own participation, or by an inspection company which it may have contracted, in the performance of manufacturing inspection, pointing out the document review, the mandatory hold points and the witness points.
11) All VENDOR and sub-vendors ITP activity shall be performed using a BUYER, CS and CPY approved VENDOR and sub-vendors ITP.
12) Inspection and test activity shall be initiated after PIM & ITP approved.
13) BUYER and/or CS, COMPANY reserves the right to select witness and hold point within all developed VENDOR and sub-vendors QC/ITPs for BUYER and/or CS, COMPANY oversight of selected functions and to perform surveillance or audits of the Work.
14) When applicable, the following sampling criterion may be adopted by the inspection company: STANDARD NBR 5426 sampling procedures and tables for inspections by attributes.
    i. Inspection Level II
    ii. Simple sampling plan
    iii. Normal Inspection
    iv. Quality limit: 10%
    v. Consumer's risk: 5%

Especially, the 1st official revision of The vendors' Inspection and Test Plan (ITP) shall be issued and submitted for BUYER and CPY`s analysis within ten (10) Days from Purchase Order signature. The ITP shall be presented at Kick Off Meeting to be held among CPY, BUYER and Vendor.

VENDOR shall perform the thickness measurement on all equipment and piping under NR-13 and SPIE(Serviço Próprio de Inspeção de ServiçoEquipamentos). The Ultrasonic method shall be applied. The specific points on the main parts of equipment and piping where ultrasonic measurements will be identified and communicated during execution stage.
In general, the scope of SPIE includes following items as minimum but specific points will be fixed during execution stage:

1)Pressure vessels and piping covered by item 13.2.1 of NR-13.
2)Pressure vessels and piping NOT covered by item 13.2.1 of NR-13.
3)Other equipment, like non-metallic vessels, tanks, pig launcher/receiver, PSV, plate heat exchangers and so on.
4)For example, piping containing compressed air (class C as per NR-13), water (class D as per NR-13) is included in the scope of thickness measurement.

8.2 INSPECTION AUTHORITY

VENDOR shall accept the presence of representatives from COMPANY, BUYER, other nominated representatives and CS and BUYER at VENDOR's facilities and at any Sub vendors premises, for the purpose of verifying quality, progress check, document review and that Work is in accordance with authority requirements and the requirements in this Contract. BUYER reserves the right to participate of any meeting or activities to be held among Suppliers, Classification Society and any other Sub-Suppliers.

BUYER and/or CS, COMPANY or its duly authorized representatives shall have access, at all reasonable times, to VENDOR and sub-vendors books, records, correspondence, instruction, invoices, plans, drawings, receipts, vouchers and memoranda of every description pertaining to work for the purpose of auditing and verifying the cost of the work or for any other reason associated with work.

At the BUYER, CS and CPY's discretion, an in-plant inspection may be performed on the equipment purchased. BUYER, CS and CPY shall have free entry, at all times while work on this contract is being performed, to all parts of the VENDOR's and Sub-vendor's premises which involve the manufacture, inspection, or testing of the equipment. VENDOR shall provide all required inspection tools, instruments, facilities, utility and all up to date engineering / manufacturing documents at free of charge to the Inspector(s). VENDOR shall also instruct his Sub-vendors of this requirement. VENDOR shall ensure responsibility for costs associated with BUYER, CS and CPY participation in aborted "Hold," "Witness" point and equivalent work progress visits which are same applied to sub-vendors.

The aborted visit is what caused, from VENDOR and sub-vendor, by unprepared inspection, rejected inspection and late notice on inspection postponement and cancellation.

Supplier shall get approval the fabrication procedures and PURCHASE shall inspect all materials and equipment supplied by suppliers and sub-suppliers, checking the compliance with Purchase Orders and/or Contract specifications and requirements. Supplier shall make available field inspection teams upon issuance of the Purchase Order and make sure that all contractual specifications applicable to the equipment / system be submitted to the team. Supplier shall issue a report of each inspection performed.

At any time, BUYER can reject any materials or equipment supplied by vendors, sub-vendors or subcontractors that do not meet the contract requirements. In case of rejected materials or equipment, SELLER shall replace them, at SELLER's account, within a time period compatible with the time schedule of the contract.

At any time, BUYER may request the substitution of any material and/or equipment not complying with Agreement specifications and requirements, without any additional charge to BUYER.

BUYER, CS and CPY's participation or not in any of these inspections and witnessing shall not relieve VENDOR of its obligations. VENDOR shall make provisions and facilitate BUYER, CS and CPY's visits.

8.3 INSPECTION NOTICE

VENDOR shall notify the authorized inspector and the BUYER and Company at least twelve (12) calendar days with respect to vendors located in Brazil, and thirty two (32) Days prior to each inspection for vendors located outside Brazil, in advance of the date when the material, equipment, package will be ready for inspection.

Should any postponement/cancellation become necessary, VENDOR shall provide written notice at least 5 working days prior to the originally scheduled date for inspection.

A certified copy of the manufacturer's reports on inspection and testing carried out prior to Company / CS / BUYER's visit shall be available before or at the latest, at the time of the inspection or verification. Copies of these reports must be retained by the manufacturer for at least five years.

8.4 PRE-INSPECTION MEETING (PIM)

The Pre-Inspection Meeting (PIM) shall be held with BUYER, CS and CPY at VENDOR and/or main sub-vendor facility where main fabrication shop is. PIM should be held to clarify the manufacturer's Inspection & Testing Plan / Quality Plan and to define hold and witness points, held before start of production.

The PIM shall be held after approval of the following documents as minimum Code 2 (Comments Added) or Code 4 (Not Commented).

1) WPS/PQR requested
2) Approved Welder List requested
3) GAD requested
4) Data Sheet requested
5) ITP

8.5 WELDING AND WELDING INSPECTION

VENDOR shall follow the project requirement of welding in the package based on I-ET-3010.00-1200-955-P4X-001 (WELDING) and I-ET-3010.00-1200-955-P4X-002 (REQUIREMENTS FOR WELDING INSPECTION) and the all referenced project specifications in the aforementioned specifications.

Welding inspector qualification shall meet requirement in I-ET-3010.00-000-970-P4X-001 (REQUIREMENTS FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION) and the all referenced project specifications in the aforementioned specifications.

All Welding shall be visually inspected by BUYER.

Each welding joint shall be traced on actual product in comparison to welding traceability sheet including the following information;

- WPS No. / Welding Date / Joint No. / Welder No. / NDE date / NDE Result / NDE Type

8.6 BOLT TIGHTENING AND LEAK TEST

Bolted jints asembly and management shall meet the requirment in I-ET-3010.00-1200-200-P4X-116, REQUIREMENTS FOR BOLTED JOINTS ASSEMBLY AND MANAGEMENT.

Bolt tightening and Leak Test shall be verified with BUYER through torque tightening inspection using torque tool on flanged joints. A report on Bolt tightening for each joint and Leak Test shall be made available and signed off by BUYER and submitted.

8.7 MECHANICAL COMPLETION (MC)

Mechanical Completion (MC) shall be carried out prior to Factory Acceptance Test (FAT). Piping system installation, including valves, and E&I installation shall be walked down using latest version of approved drawing such as P&ID, GA drawing, Detail drawing, Wiring diagram etc. VENDOR's standard mechanical completion check sheets are to be detailed for the overall package, as well each individual component (e.g. Vessel, Filter, Pump, Motor, Instruments, Control panel(s), etc.).

Before visit of BUYER & COMPANY for MC, VENDOR has obligation to provide readiness for inspection. VENDOR shall provide internal MC records in advance.

Inspection on inside of vessels/tanks which are installed on skid shall be carried out with BUYER at this stage then a report shall make available with photos.

Punch items which are found during MC shall be compiled/recorded with relevant photos

using BUYER's format.

8.8 FACTORY ACCEPTANCE TEST (FAT)

FAT procedure (including performance test) shall be developed by vendors or sub-vendors and submitted to BUYER for approval, at least three (3) months prior to a FAT planned. FAT shall be a part of the vendors or sub-vendors scope of supply and shall be witnessed by BUYER and Classification Society surveyor, as applicable.

Before performing FAT, punch lists which are found during MC shall be closed with BUYER witness.

Pre-FAT by VENDOR for readiness provision to BUYER shall be carried out internally.

All test required as a part of FAT shall be tested as a part of Pre-FAT tests to ensure that the entire system complies with project requirements and is ready for FAT. All findings shall be recorded in a punch list and shall be rectified prior to FAT. The entire document including punch list clearance shall be submitted BUYER when submitting FAT inspection notification. Pre-FAT shall be observed by BUYER if required.

All inspection/test reports for the FAT shall be endorsed/signed by all relevant parties.

Punch items which are found during FAT shall be compiled/recorded with relevant photos using BUYER's format. All punch lists generated from the FAT shall be inspected under witness of BUYER and signed off for close out.

8.9 NON CONFORMANCE REPORT (NCR)

Non-conformance Report (NCR) for repair, acceptance or re-grade resolution strategies, together with relevant procedures, must be formally submitted to and accepted by BUYER, CS and CPY.

NCR shall include root cause, corrective action and preventive action for the non-conformance issue. All repair work for NCR shall be started after getting approval of Supplier's NCR with repair procedure. NCR shall be formally closed after inspection by BUYER, CS and CPY for repaired non-conformance issue.

NCR registration shall be maintained as live document and submit to BUYER when requested.

8.10 INSPECTION REAEASE CERTIFICATE (IRC)

BUYER shall inspect equipment / material at the vendors or sub-vendors site/factory and issue the Inspection Release Certificate (IRC), before shipment, complying with all Purchase Order requirements. In case of any punch lists, vendors and sub-vendors shall present to BUYER's approval an action plan with a deadline to clear the issues.

This Punch List and its action plan shall be attached to the IRC and incorporated at Integrated Punch List.

VENDOR shall issue a photographic report, demonstrating all items of the packing list, before packing and transportation.

VENDOR shall inform BUYER when the equipment/skid/package is ready for transportation. BUYER at its discretion will attend the release inspection before transportation. The advanced

notice for the inspection herein shall be forwarded to BUYER thirty (30) days before the event. Additionally, The IRC shall also be part of project records.

VENDOR shall ensure, before packing for shipment IRC submission, that equipment / skids / material is painted according to PETROBRAS Painting Technical Specification (I-ET-3010.00-1200-956-P4X-002) or other painting scheme accepted by Petrobras, so they can withstand with weather and marine environment (Unit operation conditions).

All Quality records shall be reviewed before issuance of IRC. Unless there are quality records not reviewed by BUYER, IRC will not be issued. If there is additional visit required due to not reviewed quality records after equipment release, the visit cost will be charged to VENDOR.

Even though IRC issued amicably, VENDOR's responsibility is not relieved for findings or issues at BUYER yard and offshore which is not complied to this Contract and Specification.

8.11 SURVEY INSPECTION TYPES

The definition of survey inspection events to be followed by VENDOR, as determined in the respective Quality Plans shall consider the type of inspection determined by COMPANY for each supply of the items – Packaged item, its main equipment and sub-components. VENDOR to follow the table 1, table 2 and relevant remarks in Exhibit V chapter 6 (page 46 to 49 of the same) which determines the minimum activities for each type of survey inspection. Type of inspection for Oily Water Separator Package is globally 'B', ~~however this inspection type to be further clarified during TBE stage.~~

8.12 QUALITY ASSURANCE

The VENDOR shall demonstrate that they operate a quality management system in accordance with the ISO 9001: 2015 series of requirements. The effectiveness of the quality system and the VENDOR 's compliance with it shall be subject to monitoring by BUYER and in addition, may be audited following an agreed period of notice.

The quality system (quality control program) shall provide for the planned and systematic control of all quality related activities performed during design / development, production, installation or servicing (as appropriate to the given system). Implementation of the system shall be in accordance with Quality Manual and Project Specific Quality Plan, which shall together with all related / referenced procedures, be submitted to BUYER for review, comment and approval as required by the Purchase Order. The VENDOR shall provide facilities for and cooperate with BUYER and statutory authority inspectors during manufacturing, assembly and testing.

BUYER, CS and CPY reserve the right to conduct a VENDOR Quality Audit to verify. VENDOR QMS complies with relevant quality standards. Should VENDOR QMS be found deficient during audits or verification visits, VENDOR may be given the opportunity to carry out corrective action within a defined period of time. BUYER, CS and CPY reserve the right to conduct additional audit visits to verify VENDOR compliance with corrective actions.

Additional process and deliverables quality verification visits may be employed by BUYER, CS and COMPANY to monitor VENDOR Quality Control.

VENDOR shall provide 12 (Vendors in Brazil) or 32 working days (Vendors outside Brazil) advance notification to BUYER regarding actual date and location for all the audits to be undertaken to permit BUYER, CS and CPY. If BUYER, CS and CPY decides to re-audit any portion of the VENDOR's quality system, VENDOR shall cooperate all necessary assistance and access.

All quality records such as Material Certificate, PMI report, welding record, NDE record, pressure test report, coating report, MC check sheet, bolt tightening record, leak test report, FAT report, packing & preservation report and packing list shall be documented by VENDOR and reviewed by BUYER.

Fabrication shall not commence until all quality procedures such as PMI, welding, NDE, pressure test, coating, are reviewed by BUYER and have been given a status of “A” or “B”.

8.13 TRACEABILITY OF MATERIAL / MANUFACTURING WORK AND MARKING

All materials and equipment supplied by VENDOR shall be brand new (not overhauled), unused, fully traceable, field proven, free from defects and procured from acceptable Origin of Manufacturer by Project. The materials and equipment shall be accepted by BUYER and the Classification Society. Materials and equipment shall be manufactured according to internationally recognized standards for the offshore oil drilling and production industries, and shall be in conformance with the Basic Design and Agreement specifications and requirements.

All permanent materials and equipment received at the site must come with the proper material certificates. Materials of unknown origin or materials without any certification shall not be used unless agreed otherwise in writing by Company authorized representative.

Full material traceability with traceability sheet shall be provided by VENDOR. For various systems consisting of numerous parts with possibly many different types and grades of materials, it is important that only the correct materials are connected and installed together in a system and equipment. For this to be achieved there must be auditable traceability of all components. There shall be a system in place that allows full traceability to an original material certificate with heat no. and Lot no. of every component part. The cast / heat number as marked, provides traceability back to the relevant material certification. Traceability sheet of material shall be kept to trace where the materials have been used along with installation location, basic description, material grade, heat number, certificate number, NDE record and welding records.

All permanent materials and equipment shall be traceable to the source via identification and marking inscribed or low pressure die stamped on the work piece. VENDOR shall put in place a traceability procedure to ensure the proper traceability of the materials and equipment.

All materials purchased by and received by VENDOR, must be tagged and checked against “Original Material certificates” or “Certified True copies’ of originals, on receipt.

On receipt of the materials and equipment, VENDOR shall check against the shipping manifest and inspect the goods to ensure there is no missing and / or damaged items.

VENDOR shall provide adequate and proper storage for the materials and equipment. Materials and equipment which are sensitive to environmental conditions shall be stored in covered storage areas and / or air-conditioned rooms. Space and anti-condensation heaters of equipment shall be energized while the equipment is in storage. All valves shall be stored in covered storage areas not exposed to blasting and painting activities prior to installation at site. Proper maintenance and greasing of the valves internal shall be provided during construction phase.

Fabrication of different materials such as carbon steel, alloy low temperature steel, corrosion resistant alloys (CRA), etc., shall be carried out in segregated areas to prevent cross contamination and mix-up.

Material traceability shall be meet requirement in Appendix 1 of Exhibit VII – Directives For Quality.

9.0 SPARE PARTS AND SPECIAL TOOLS

9.1 SPECIAL TOOLS

The VENDOR shall design the Package to avoid the use of special tools or special handling equipment for the erection, maintenance and operation. All special tools, if required, to do installation and maintenance of the Package shall be provided. Special Tools shall mean any tool or fixture required to disassemble, assemble, or maintain Package equipment. Special tools shall be provided in long term storage special boxes, labelled and storable on shelves.

9.2 SPARE PARTS

Consumables and components that cannot achieve the full service life of the equipment and require periodic replacement shall be listed in design data submitted for Company approval. Consumables and replacement components shall be provided to allow completion of all package commissioning and use during: Topside integration; transportation; start-up.

VENDOR shall prepare SPIR in BUYER's SPIR Form (refer to Attachment) for the BUYER's approval.

10.0 PREPARATION FOR SHIPMENT

The VENDOR shall prepare the equipment for export shipment. VENDOR shall ensure that all Equipment, Materials and Spare Parts that it purchases hereunder are suitably packaged, crated, boxed or otherwise appropriately prepared for shipment and protected from the elements in accordance with Protection, Packing, Marking & Shipping Documentation Instructions provided by BUYER. All sea fastening and temporary supports used on the equipment for shipment shall be clearly identified. The BIDDER shall submit packing and preservation procedure prior to shipment for BUYER's review and approval

All packing and preservation in ITP shall be a Witness/HOLD point for BUYER inspection – review of VENDOR s report is not acceptable. Then, packing list shall be signed by BUYER compared to actual items including spare parts during inspection to check quantity and final appearance status based on packing list. VENDOR's check list and/or inspection report including pictures for the packing & preservation shall be developed and provided.

VENDOR shall be responsible to preserve the Equipment for the intended period of storage (Minimum 6 months). All the preservation activities required to be done by BUYER thereafter should be written down specifically and supplied with the Equipment. The VENDOR shall recommend preservation requirements for idle periods of non- operation up to 12 months for protection against rain, moisture, dirt, blasting grit etc.

VENDOR shall ensure that all equipment, materials, special tools and spare parts that VENDOR purchases hereunder are suitably packed, crated, preserved, boxed or otherwise appropriately prepared for shipment and protected from the elements.

Preservation and packing shall be proper for transportation and storage in a marine environment and protected against moisture and damage during transport, handling and lifting, in accordance with vendors and sub-vendors recommendations and specification. In any case, suitable preservation and protective measures shall be provided to prevent equipment deterioration prior to entering into service in the UNIT. The fumigation of the boxes shall be considered, where applicable.

All packing shall be clearly marked for shipping, including lifting points, weight, dimensions and center of gravity. All sea fastening and temporary supports used on the equipment for shipment shall be clearly identified.

VENDOR shall ensure that all loose valves, tubes and instruments are supplied with plastic caps. VENDOR shall also ensure that all electric panels and motors will be supplied with Volatile Corrosion Inhibitor (VCI) impregnated plastic protection or similar, and external plug for space heater connection.

VENDOR shall provide clear and comprehensive instructions on the exterior of all packages advising the necessary warning notices for unpacking, handling and installing the equipment on arrival at destination.

VENDOR shall submit the following shipping documents (Dispatch Dossier), containing at least:

1) Inspection Release Certificate (IRC), issued by SELLER or its representatives, which releases the equipment/material shipment;
2) Certificate of the Classification Society, if applicable;
3) Conformity Certificate;
4) Emergency card or "Material Safety Data Sheet", if applicable;
5) Requirements and instructions for lashing, protection, transportation, unloading and storage, whenever necessary;
6) General Arrangement and Data Sheet;
7) P&ID;
8) Packing List and Invoice
9) Lifting Scheme, if applicable
10) Packing, unpacking, preservation and storage procedures. Such documentation must cover all phases of assembly and testing on Unit.
11) Painting/coating repair procedures.
12) Fumigation Certificate, if executed;

During the transportation, all piping spools including ship loose spools shall have their ends sealed and internals properly preserved. VENDOR shall ensure that all loose valves, tubes and instruments are supplied with plastic caps. VENDOR shall also ensure that all electric panels and motors will be supplied with Volatile Corrosion Inhibitor (VCI) impregnated plastic protection or similar, and external plug for space heater connection.

Exposed machined surfaces, including weld bevels and flange facings, shall be covered with an approved preservative.

All equipment at outdoor locations shall be adequately protected against ingress of debris, dust, moisture, rains, etc. Covers that are temporarily removed for installation purposes shall be reinstated when work is suspended or craftsmen take a break. All open entries shall be effectively sealed at all times against ingress of water, dust and debris. Temporary air conditioning shall be provided for equipment which contain electronic parts / components, or sensitive to moisture / humidity.

VENDOR shall obtain the IRC from BUYER prior to shipment. Shipment cannot be proceeded without IRC, any approved punch and/or approved NCR. The IRC will be issued after BUYER’s satisfactory for final inspection including punch list, NCR and packing & preservation. In case any non-conformity issues out of tolerance is found against the approved drawings,

VENDOR shall take corrective actions before shipment else BIDDER shall do the corrections at BUYER yard at his cost.

When specified in the purchase order, spare parts for the equipment or item or comprising either part or complete equipment systems shall be inspected and tested to the same factory inspection and testing procedures as the equipment.

Documents, including unloading, unpacking and lifting instructions, a description of the reassembly steps required for installation and pre-commissioning and start up instructions shall be included with the package. The number of copies of documents to be provided shall be as stated in the Requisition.

The BUYER will be sole judge of adequacy of export preparation.

Prior to shipment, all instrumentation and other equipment vulnerable to damage shall be disassembled and boxed separately for shipment.

VENDOR shall prepare and implement a programme for preserving the equipment and components so that damage to equipment and materials is avoided and to ensure that the start-up of equipment and systems can take place without any problems. Modern preservation materials shall be used in line with BUYER's and COMPANY's experiences.

In addition, the following additional requirements shall apply:

1) All equipment skids to be delivered with tailor-made flameproof tarpaulin with openings for inspection and hook-up of pipes and cable trays. All openings shall be sealed with either adhesive lock or zip-fastener. The tarpaulin shall be installed inside the wood to enable the yard removing the wood without destroying the tarpaulin.
2) All stainless steel surfaces on vessels and tanks to be protected with suitable flame resistant materials to avoid contamination of the stainless steel surface during the construction/installation phase.
3) All control cabinets to be delivered with tailor-made flameproof tarpaulin with openings for inspection and hook-up. Openings shall be sealed with either adhesive lock or zip fastener. The tarpaulin shall be installed outside the heavy duty film.

The packing list shall be submitted in Microsoft Excel format (Native File). The packing list shall include detail information for each item/material such as Packing Box no. material description, material Size & weight, drawing & part no. packing type/method, shipping size & weight and VENDOR's identification no and tagged as a minimum with P.O number, Item number, Tag number / Part number and description of item.

The packing list shall be very exhaustive with details of all items shipped under different categories (Commissioning spares, Start-up spares, Special tools, etc). These items are to be boxed up category-wise. It is VENDOR's total responsibility to ship these items along with materials required for their installation (new gaskets, fasteners etc). Detail three-dimensional drawings for the re-assembly shall be enclosed with these packing lists. Each shipped loose item shall have tag plate (showing material description, assembly drawing no., and parts no) attached by wire to identity easily it during installation. VENDOR shall fill in the Template of HHI Standard Ship Loose Item List is included in this requisition for BUYER's approval. BIDDER shall submit GAD for each Ship Loose Item to be submitted for interface check (Size, Connection detail etc.).

VENDOR shall submit the shipping plan for BUYER approval two (2) months in advance to shipment. The plan shall identify all items, especially those are being shipped loose and date of

shipment.

11.0 DOCUMENT REQUIREMENT

11.1 VENDOR DOCUMENTATION AND SUBMISSION

All drawings and documents shall be presented in the English language and in general the ISO International System of Units (SI) shall be used.

All documents shall be submitted to in line with a required schedule for review & approval, as per VENDOR Document Requirements List (VDRL).

Penalty Milestones to BIDDER shall apply for the late submittal of all documentation. All documents as final (As Built) shall be completed and submitted within 8 Weeks of the equipment shipment from BIDDER's works.

1. User Manual (Operating and maintenance manual) - These documents shall be presented in English and Brazilian Portuguese
2. Copy of Manufacturing Record Book shall be submitted to BUYER prior to shipment of the equipment. MRB shall contain copy of all original documentation signed and/or stamped by the certifying and regulatory agency.
3. Manufacturing Record Book shall contain the following documentation, but not limited to;
   1) As built drawings annotated with pertinent information regarding welding, repairs, locations of specific sections, and relevant inspection comments
   2) Bolt tightening report
   3) Leak test report
   4) Non-conformance reports and concessions
   5) Manufacturer's mill certificates
   6) Mechanical Test certificates
   7) Records of dimensional control reports and other tests
   8) Records of witnessed tests
   9) NDT Test Record (incl. digitalized R.T. films per I-ET-3010.00-1000-970-P4X-002)
   10) Material and Equipment certificates
   11) Welding qualification certificates including welding parameter check sheet
   12) Index and map of radiographic and ultrasonic examinations
   13) Weld examination reports
   14) Protective coatings report
   15) Records of destructive and non-destructive tests
   16) Details of any repairs
   17) Heat treat records and charts
   18) Pressure test procedures and reports
   19) Marked-up documentation showing location of identified materials, identification of weld and welder
   20) Mechanical Completion and alignment test reports
   21) As-built weight & CG data
   22) NDE personal qualification certificate
   23) Painting operator/inspector qualification certificate
   24) Long term preservation procedure

4. Mechanical Completion Dossiers
5. Index Information
6. 2D CAD Files as well as other native files
7. Documentation for dispatch, receipt, storage and installation.
8. As-built Drawings
9. Purchase specification of all components for future purchasing during maintenance period.
10. A list of sub-vendors

Manufacturing Record Book (MRB) shall be submitted by the BIDDER as part of document submission. MRB shall consist of Design data book, Fabrication data book and Equipment data book.

The BIDDER shall submit a schedule of contract execution in the form of a bar chart covering engineering, procurement, manufacture, fabrication, assembly, inspection & testing within two weeks of award of contract. And, BIDDER shall submit monthly progress report including all activity for engineering, procurement, manufacture, fabrication, assembly, inspection & testing with area of concern, if any.

As part of any Award the BIDDER shall prepare a monthly progress/status report and submit it for BUYER's review. Progress reports shall indicate actual activities completed and note any items impacting on the project, such as change orders, late sub-BIDDER deliveries, test failures and production limitations.

During peak periods of manufacturing/testing, the BIDDER shall be required to submit bi-weekly reports.

BIDDER is responsible for maintaining the Vendor Master Document List (VMDL) in accordance with Vendor Document Requirements Listing (VDRL). BIDDER shall also include the list of all detailed and construction engineering documents required for the Work, including the subcontracted part of the work, required documents as per Project Quality Management Plan(PQMP) and the related time schedule for the production of such documents. VMDL shall be in Microsoft Excel format. The register shall be able to be provided, at any time, accurate and up-to-date information on the current status of all planned Vendor data submissions. The Project format of VMDL shall be strictly used to provide all information required.

For Approval, Final & As-Built: Electronic file upload to PIMS developed by OMEGA.

11.2 BIDDER'S DATA AND TECHNICAL DETAILS

The BIDDER's submissions shall provide sufficient data to allow the equipment to be evaluated against the total life of field costs to be incurred. In this respect, equipment capital cost, cost of spares and utility requirements shall be considered, as shall the cost of a proposed field life maintenance program.

The BIDDER shall submit full technical details of their standard equipment which best meets the requirements detailed herein. Aspects of the requirements that cannot be met shall be identified and the variances explained.

The BIDDER shall complete project data sheets, where these are listed with this specification, or API equipment data sheets, and submit them as part of the Submission.

11.3 FINAL HAND-OVER DOCUMENT

All documents, with the exception of e-mail and letters, shall be submitted under cover of a uniquely numbered Document Transmittal per Project Requirements. Final dossiers shall be available for final review within 1 month of dispatch of equipment.

VENDOR issue VENDOR Data Book (VDB) to be delivered to BUYER and CPY, one (1) printed copies, five (5) tagged pen drives copies and two (02) HD, SD or other electronic storage media, including all design and construction documentation on all materials, including from package units, purchased by BUYER, vendors, sub-vendors and subcontractors, broken down into five (5) parts and containing at least the following details:

1. The printed copies of data books shall be issued and delivered to CPY only after the electronic versions of Data Books be already in final version, previously analysed and approved by BUYER and accepted by CPY.
2. The first part (section 1) shall list all documents, certificate drawings, data sheets, technical specifications, performance curves and calculation memorandum, as follows:
   1) A list of documents
   2) A manufacture and inspection schedule
   3) Data sheets with dimensions and base loads for all equipment
   4) List of spare parts including the reference identification (containing its complete technical specification and original manufacturer part number), drawings, dimensions and weight, quantity per equipment, type of material and coating / surface treatment, if applicable; broken down into two (2) parts: The first part (all supplied spare parts for onshore and offshore commissioning and start-up) / The second part (two (2) years of operation spare part list)
   5) Cross sectional drawings or "exploded views" showing all components of the equipment and in a way it is possible to identify the recommended spare parts.
   6) The general arrangement of the equipment, with design and operation data regarding the maintenance area, connections, and skid limits, dimensions, weight and gravity centre of each equipment (for empty equipment, in operating and testing conditions), type of material and coating / surface treatment, description of parts and materials
   7) Construction drawings of equipment, the structure and piping
   8) Construction documents

Cross-sectional drawings, including a list of every item, materials and original manufacturer part numbers

12.0 APPENDICES

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APPENDIX #1

DATASHEETS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
1	I-FD-3010.2G-5330-660-KES-303	A	DATASHEET – OILY WATER SEPARATOR PACKAGE

APPENDIX #2

PACKAGE SPECIFICATIONS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
1	I-ET-3010.1M-5330-667-P4X-001	C	SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501)

APPENDIX #3

PROJECT SPECIFICATIONS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
~~1~~	~~I-ET-3010.00-1200-940-P4X-002~~	~~C~~	~~GENERAL TECHNICAL TERMS~~
~~2~~	~~I-ET-3010.00-5140-797-P4X-001~~	~~D~~	~~ELECTRICAL SYSTEM AUTOMATION ARCHITECTURE~~
~~3~~	~~I-ET-3010.00-5140-700-P4X-001~~	~~F~~	~~SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS~~
4	I-ET-3010.00-5140-700-P4X-002	E	SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS
5	I-ET-3010.00-5140-700-P4X-003	D	ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS
~~6~~	~~I-ET-3010.00-5140-700-P4X-005~~	~~A~~	~~REQUIREMENTS FOR HUMAN ENGINEERING DESIGN FOR ELECTRICAL SYSTEMS OF OFFSHORE UNITS~~
7	I-ET-3010.00-1200-956-P4X-002	F	GENERAL PAINTING
~~8~~	~~I-ET-3010.00-5143-700-P4X-001~~	~~D~~	~~ELECTRICAL SYSTEM PROTECTION CRITERIA~~
~~9~~	~~I-LI-3010.00-5140-700-P4X-001~~	~~B~~	~~ELECTRICAL EQUIPMENT DATA SHEET MODELS~~
~~10~~	~~I-LI-3010.00-5140-797-P4X-001~~	~~D~~	~~ELECTRICAL SYSTEM AUTOMATION INTERFACE SIGNALS LIST~~
11	I-ET-3010.1M-1200-300-P4X-002	A	NOISE CONTROL REQUIREMENTS FOR ACCOMMODATION / SHIPSIDE
~~12~~	~~DR-ENGP-M-I-1.3-R.5~~	~~0~~	~~SAFETY ENGINEERING~~
13	I-ET-3A36.00-1000-941-PPC-001	D	METOCEAN DATA
14	I-ET-3000.00-0000-940-P4X-002	A	SYMBOLS FOR PRODUCTION UNITS DESIGN
15	I-ET-3000.00-1200-940-P4X-001	C	TAGGING PROCEDURE FOR PRODUCTION UNITS DESIGN
16	DR-ENGP-I-1.15	0	COLOR CODING
~~17~~	~~I-ET-3010.00-5140-712-P4X-001~~	~~C~~	~~LOW-VOLTAGE INDUCTION MOTORS FOR OFFSHORE UNITS~~
18	I-ET-3010.00-1200-800-P4X-002	F	AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS
19	I-ET-3010.1M-1200-800-P4X-014	F	AUTOMATION INTERFACE OF PACKAGE UNITS PACKAGE INTERFACE
~~20~~	~~I-RL-3010.1M-1200-940-P4X-001~~	~~B~~	~~GENERAL SPECIFICATION FOR AVAILABLE UTILITIES~~
~~21~~	~~I-ET-3010.00-1200-251- P4X-001~~	~~B~~	~~REQUIREMENTS FOR BOLT MATERIALS~~
~~22~~	~~I-ET-3010.00-1200-200-P4X-116~~	~~A~~	~~REQUIREMENTS FOR BOLTED JOINTS ASSEMBLY AND MANAGEMENT~~
23	I-ET-3010.00-5400-947-P4X-002	0	SAFETY SIGNALING
24	I-ET-3010.1M-1200-200-P4X-002	A	PIPING SPECIFICATION FOR HULL
25	I-RL-3010.1M-1350-960-P4X-009	C	MOTION ANALYSIS

~~26~~	~~I-ET-3000.00-1350-940-P4X-013~~	~~F~~	~~TECHNICAL REQUIREMENTS FOR DIGITAL ENGINEERING~~
~~27~~	~~I-ET-3010.1M-1200-300-P57-001~~	~~0~~	~~COMPLIANCE WITH NR-12 REQUIREMENTS FOR MACHINES AND EQUIPMENT~~
~~28~~	~~I-ET-3010.1M-1200-500-PAH-001~~	~~A~~	~~COMPLIANCE WITH NR-13 REQUIREMENTS FOR PRESSURE VESSELS~~
~~29~~	~~EXHIBIT III w/ appendices~~	~~17~~	~~DIRECTIVES FOR ENGINEERING~~
~~30~~	~~I-ET-3010.00-1200-431-P4X-001~~	~~D~~	~~THERMAL INSULATION FOR MARITIME INSTALLATIONS~~
~~31~~	~~I-ET-3010.00-1000-970-P4X-002~~	~~0~~	~~REQUIREMENTS FOR NDT~~
~~32~~	~~I-ET-3010.00-1200-200-P4X-001~~	~~C~~	~~MINIMUM REQUIREMENTS FOR PIPING MECHANICAL DESIGN AND LAYOUT~~
~~33~~	~~I-ET-3010.00-5520-888-P4X-001~~	~~C~~	~~AUTOMATION PANELS~~
~~34~~	~~I-ET-3010.00-1200-800-P4X-013~~	~~G~~	~~GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS~~
~~35~~	~~I-ET-3010.00-1200-955-P4X-001~~	~~B~~	~~WELDING~~
~~36~~	~~I-ET-3010.00-1200-955-P4X-002~~	~~B~~	~~REQUIREMENTS FOR WELDING INSPECTION~~
~~37~~	~~I-ET-3010.00-1200-956-P4X-001~~	~~A~~	~~QUALIFICATION TESTS FOR PAINT SYSTEMS~~
~~38~~	~~I-ET-3010.00-0000-970-P4X-001~~	~~A~~	~~REQUIREMENTS FOR PROCEDURES AND PERSONNEL QUALIFICATION AND CERTIFICATION~~
~~39~~	~~EXHIBIT V (Applicable Sections)~~	~~8~~	~~DIRECTIVES FOR PROCUREMENT~~
~~40~~	~~EXHIBIT VII (Applicable Secions)~~	~~1~~	~~DIRECTIVES FOR QUALITY ASSURANCE SYSTEM~~
~~41~~	~~EXHIBIT VIII (Applicable Sections and Appendix II and III)~~	~~M~~	~~DIRECTIVES FOR COMMISSIONING~~
~~42~~	~~EXHIBIT XV (Applicable Sections)~~	~~2~~	~~DIRECTIVES FOR FPSO CLASSIFICATION~~
~~43~~	~~I-ET-3010.2G-9000-911-KSL-001~~	~~A~~	~~DOCUMENT CONTROL PROCEDURE~~
~~44~~	~~I-ET-3010.1M-1350-140-P4X-001~~	~~C~~	~~HULL STRUCTURAL REQUIREMENTS~~
~~45~~	~~I-ET-3010.00-1352-130-P4X-001~~	~~A~~	~~FLOOR GRATINGS, TRAY SYSTEMS AND GUARDRAILS MADE OF COMPOSITE MATERIALS~~
~~46~~	~~I-ET-3010.1M-1200-800-P4X-001~~	~~A~~	~~INSTRUMENTATION ADDITIONAL TECHNICAL REQUIREMENTS~~
~~47~~	~~I-ET-3010.00-1200-588-P4X-001~~	~~C~~	~~SAMPLE CONNECTIONS~~
~~48~~	~~I-ET-3010.1M-1200-800-P4X-005~~	~~F~~	~~FIELD INSTRUMENTATION~~
~~49~~	~~I-ET-3010.00-1200-800-P4X-01~~	~~A~~	~~CRITERIA FOR DETAILING DESIGN CAUSE & EFFECT MATRIX~~
~~50~~	~~I-ET-3010.00-1200-800-P4X-010~~	~~B~~	~~CRITERIA FOR ESTABLISHING CABLE CODES AND CABLE GLAND CODES~~
~~51~~	~~I-MD-3010.1M-1200-940-P4X-02~~	~~K~~	~~DESCRIPTIVE MEMORANDUM - AUTOMATION & CONTROL~~

APPENDIX #4

PROJECT DRAWINGS

S/N	DOCUMENT NUMBER	REV	DESCRIPTION
1	I-DE-3010.2G-1200-944-KES-301	0	P&ID GENERAL NOTES
2	I-DE-3010.2G-5330-944-KES-301	A	PID for BILGE, SLUDGE, BALLAST (AFT) AND GENERAL SERVICE SEAWATER SYSTEM
3	I-DE-3010.2G-6124-944-KES-301	A	P&ID FOR HULL SERVICE AND INSTRUMENT AIR DISTRIBUTION
4	I-DE-3010.2G-5115-944-KES-302	A	P&ID FOR FRESH HOT AND POTABLE WATER SYSTEM
5	I-DE-3010.00-5140-700-P4X-003	E	GROUNDING INSTALLATION TYPICAL DETAILS
6	I-DE-3010.2G-5140-700-KES-452	0	POWER INSTALLATION TYPICAL DETAILS
~~7~~	~~I-DE-3010.1M-1200-942-P4X-002~~	~~E~~	~~GENERAL ARRANGEMENT~~
8	I-DE-3010.2G-1351-942-KES-309	A	ENGINE ROOM LAYOUT
9	I-DE-3010.1M-5400-94A-P4X-001	A	AREA CLASSIFICATION - GENERAL
10	I-LI-3010.2G-5400-940-KES-301	0	AREA CLASSIFICATION DATA LIST
11	SHK-ST-DA-801-0151	B	STANDARD EQUIPMENT SEAT TYPE 3 & 4

APPENDIX #5

VENDOR DOCUMENT REQUIREMENT LIST (VDRL)

VENDOR DATA REQUIREMENT LIST - VDRL

REQUISITION DESCRIPTION : Oily Water Separator Package						Rev.:	0	
VDRL No. :						Date:	2-Sep-21	
VDRL Code	Data/Document Description	Accept	Critical	With Tender	For Approval		Final Data	
				No. Copies	No. Copies	Date Reqd.	No. Copies	IOM/ MDR
A	**CONTROL DOCUMENTS**							
A01	Vendor Document Schedule(VDS)		O	1E	1E(PIMS)	LOI+2W	2P+7E	MDR
A02	Fabrication/production Schedule			1E	1E(PIMS)	LOI+8W	2P+7E	MDR
A03	Progress reports(Note. 1)				1E(PIMS)	LOI+8W	2P+7E	MDR
A04	Vendor/Sub-Vendor Equipment List			1E	1E(PIMS)	LOI+8W	2P+7E	MDR
A05	Project Reference List			1E		NA	-	-
B	**INTERFACE & ARRANGEMENT DRAWINGS**							
B01	General Arrangement Drawings		O	1E	1E(PIMS)	LOI+4W	2P+7E	IOM
B02	Interface and Connection Schedule/Scope of Supply Drawing			1E	1E(PIMS)	LOI+8W	2P+7E	IOM
B03	Foundation Loading Diagram and Support Details				1E(PIMS)	LOI+8W	2P+7E	IOM
B04	Tie-In Schedule(TP Schedule)			1E	1E(PIMS)	LOI+8W	2P+7E	IOM
B99	Special Requirements					NA	-	-
C	**DESIGN & OPERATION DATA**							
C01	Piping and Instrument Diagrams(P&IDs)		O	1E	1E(PIMS)	LOI+4W	2P+7E	IOM
C02	Single Line Electrical Diagrams		O	1E	1E(PIMS)	LOI+8W	2P+7E	IOM
~~C03~~	~~Schematic Diagrams~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
C04	Utilities Schedule including Electrical Load List		O	1E	1E(PIMS)	LOI+8W	2P+7E	IOM
C05	Weight Data Sheets			1E	1E(PIMS)	LOI+8W	2P+7E	MDR
C06	Equipment Data Sheet			1E	1E(PIMS)	LOI+8W	2P+7E	IOM
C07	Noise Level Data Sheet			1E	1E(PIMS)	AS-2W	2P+7E	MDR
~~C08~~	~~Schedule of Electrical Equipment in Hazardous Areas~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~MDR~~
C09	Detailed Description of Operation/Specifications				1E(PIMS)	LOI+20W	2P+7E	IOM
C10	HVAC Schematic and Flow Diagrams					NA	-	-
~~C11~~	~~PFD's and Heat and Mass Balance~~				~~1E(PIMS)~~	~~LOI+4W~~	~~2P+7E~~	~~IOM~~
~~C12~~	~~Electrical Power Supplies Data Sheet~~			~~1E~~	~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
C13	Enclosure Ventilation Requirements					NA	-	-
C14	Piping Line List				1E(PIMS)	LOI+10W	2P+7E	IOM
C15	Valve List				1E(PIMS)	LOI+10W	2P+7E	IOM
~~C16~~	~~Cause & Effect Charts~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~C17~~	~~Visual screen Display Graphics~~				~~1E(PIMS)~~	~~LOI+20W~~	~~2P+7E~~	~~IOM~~
C18	Electrical & Instrument Wiring Diagram		O		1E(PIMS)	LOI+8W	2P+7E	IOM
C19	Bill of Materials(BOM)				1E(PIMS)	LOI+12W	2P+7E	IOM
~~C20~~	~~Instruments/Telecoms System Schematic Diagram~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
C21	Electric/Electronic/Pneumatic/Hydraulic Schematic				1E(PIMS)	LOI+10W	2P+7E	IOM
C22	Control and Shutdown Philosophy				1E(PIMS)	LOI+10W	2P+7E	IOM
D	**GENERAL DETAILS & SUB-ASSEMBLY DRAWINGS**							
D01	Cross-Sectional Drawing or Exploded View Dgrm with Parts List				1E(PIMS)	LOI+12W	2P+7E	IOM
~~D02~~	~~mechanical Seal Drawings~~				~~1E(PIMS)~~	~~LOI+12W~~	~~2P+7E~~	~~IOM~~
~~D03~~	~~Shaft Alignment Drawings~~				~~1E(PIMS)~~	~~LOI+12W~~	~~2P+7E~~	~~IOM~~
D04	Name Plate Drawings				1E(PIMS)	LOI+12W	2P+7E	IOM
D05	Sub Assembly Drawings				1E(PIMS)	AS-2W	2P+7E	IOM
~~D06~~	~~Detail Fabrication Drawings~~				~~1E(PIMS)~~	~~LOI+12W~~	~~2P+7E~~	~~IOM~~
~~D07~~	~~Installation & Dismantling Drawing~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~IOM~~
~~D08~~	~~Electrical/Instrument Layout Drawings~~				~~1E(PIMS)~~	~~LOI+14W~~	~~2P+7E~~	~~IOM~~
D09	Pressure Vessel / Tank Detail Drawings		O		1E(PIMS)	LOI+12W	2P+7E	IOM
D10	Skid/Module Structural Detail Drawings				1E(PIMS)	LOI+12W	2P+7E	IOM

VENDOR DATA REQUIREMENT LIST - VDRL

REQUISITION DESCRIPTION : Oily Water Separator Package | **Rev.:** 0

VDRL No. : | **Date:** 2-Sep-21

VDRL Code	Data/Document Description	Accept	Critical	With Tender	For Approval		Final Data	
				No. Copies	No. Copies	Date Reqd.	No. Copies	IOM/ MDR
~~D11~~	~~Skid/Module Fabrication Details~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~IOM~~
D12	Weld Location Plans				1E(PIMS)	LOI+12W	2P+7E	IOM
~~D13~~	~~Pipework Support Details~~				~~1E(PIMS)~~	~~LOI+14W~~	~~2P+7E~~	~~IOM~~
~~D99~~	~~Miscellaneous Drawings~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~IOM~~
E	**INSTRUMENT/ELECTRICAL INTERCONNECTIONS & DETAILS**							
E01	Electrical Connection Diagrams				1E(PIMS)	LOI+10W	2P+7E	IOM
E02	Instrument/Electrical Panel Detail Drawing				1E(PIMS)	LOI+10W	2P+7E	IOM
E03	Termination Diagram/Schedule				1E(PIMS)	LOI+10W	2P+7E	IOM
E04	Cable Schedule				1E(PIMS)	LOI+10W	2P+7E	IOM
~~E05~~	~~Instrument Termination and Hook Up Details(including Pneumatic)~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~E06~~	~~Loop Diagrams~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~E07~~	~~Instrument PLC Data communicator Protocol~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
E08	Instrument Index				1E(PIMS)	LOI+10W	2P+7E	IOM
E09	Instrument/Electrical Data Sheets				1E(PIMS)	LOI+10W	2P+7E	IOM
~~E10~~	~~Instrument/Electrical Logic Diagram~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
E11	Terminal Block Diagrams				1E(PIMS)	LOI+10W	2P+7E	IOM
E12	Alarm/Set Point Schedule				1E(PIMS)	LOI+10W	2P+7E	IOM
E13	I/O Schedule				1E(PIMS)	LOI+10W	2P+7E	IOM
~~E14~~	~~Serial Link Memory Maps~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~E15~~	~~Cable Block Diagram~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
F	**CALCULATION & PERFORMANCE DATA**							
F01	Calculations-Tank Mech Calcs				1E(PIMS)	LOI+10W	2P+7E	IOM
~~F02~~	~~Process/Utility Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~F03~~	~~Structural Steel Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~F04~~	~~Foundation Support Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
F05	System Head Loss Calculations					NA	-	-
F06	Lateral Critical Speed Calculations					NA	-	-
F07	Torsional Critical Speed Calculations					NA	-	-
F08	Bearing Life Calculations					NA		
F09	Thrust Bearing sizing Calculations					NA		
F10	Heat Emission Calculations					NA		
F11	Acoustic Analysis for Reciprocating Compressors					NA	-	-
F12	Hydraulic Calculations					NA	-	-
F13	Exchanger Thermal Rating Calculations					NA		
~~F14~~	~~Instrument Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
F15	Enclosure Ventilation System Calculations					NA	-	-
F16	Exhaust Duct Calculations					NA	-	-
F17	Coupling Selection Calculations					NA	-	-
F18	Lube and Seal Oil System Sizing Calculations					NA	-	-
F19	Anti Surge Valve Sizing					NA		
F20	Pulsation Dampener Design Calculations					NA		
F21	Rotor/Shaft system Unbalanced Response Analysis					NA	-	-
~~F22~~	~~Piping Stress Analysis~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
F23	Crane Failure Mode Analysis					NA	-	-
F24	End Valve Calculation					NA	-	-
~~F25~~	~~Relief Valve And Bursting Disc Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~

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VDRL No. : | **Date:** 2-Sep-21

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				No. Copies	No. Copies	Date Reqd.	No. Copies	IOM/ MDR
F26	Electrical Protection Curves					NA	-	-
F27	Current, Potential and Power Transformation Curves					NA	-	-
F28	Motor Performance Curves				1E(PIMS)	LOI+10W	2P+7E	IOM
F29	Combustion Gas Turbine Performance Curves					NA		
F30	Centrifugal Pump Performance curves					NA		
F31	Rotary Pump Curves				1E(PIMS)	LOI+10W	2P+7E	IOM
F32	Centrifugal Compressor Performance Curves					NA		
~~F33~~	~~Fan Performance Curves~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
F34	Engine Performance Curves					NA		
F35	General Performance Data				1E(PIMS)	LOI+12W	2P+7E	IOM
F36	Speed/Torque Starting Curves				1E(PIMS)	LOI+18W	2P+7E	IOM
F37	Reciprocating Pump Performance Curves					NA	-	-
F38	Lighting Performance Curves					NA	-	-
F39	Battery Charge/Discharge Curves					NA		
F40	Power System Analysis Data					NA		
~~F41~~	~~Reliability/Availability Data and Calculations~~				~~1E(PIMS)~~	~~LOI+20W~~	~~2P+7E~~	~~IOM~~
F42	Compressor Dynamic Simulations					NA		
F47	Electrical Relay Characteristics					NA		
F48	Flare/Relief System Calculations					NA	-	-
F49	Flare Dispersion Calculations					NA	-	-
F51	Fiscal Metering System Calculations					NA	-	-
F52	Non-Actuated Valve Torque Calculations					NA	-	-
F60	Thermal Growth Calculations					NA	-	-
F61	Fatigue Calculations(Well Conversion)					NA	-	-
F62	Fatigue Calculations(Riser System)					NA	-	-
~~F63~~	~~Acceptable Nozzle Load Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~F64~~	~~Center of Gravity Calculations~~				~~1E(PIMS)~~	~~LOI+10W~~	~~2P+7E~~	~~IOM~~
~~F65~~	~~Lifting Equipment Calculations~~				~~1E(PIMS)~~	~~LOI+12W~~	~~2P+7E~~	~~IOM~~
F66	Barge Transportation Calculations					NA	-	-
~~F67~~	~~System Performance Data~~				~~1E(PIMS)~~	~~LOI+12W~~	~~2P+7E~~	~~IOM~~
~~F68~~	~~Control Valve Calcs(in sizing, velocity, noise, actuator sizing, etc)~~				~~1E(PIMS)~~	~~LOI+12W~~	~~2P+7E~~	~~IOM~~
F69	Vortex Inducted Vibrations					NA	-	-
F70	Drains Systems Calculations					NA	-	-
F99	Rotating Stall Analysis					NA		
G	**HANDLING, INSTALLATION & PRECOMMISSIONING**							
G01	Lifting Drawing/Procedure & Calculation				1E(PIMS)	LOI+16W	2P+7E	IOM
G02	Unpacking and Preservation Procedure				1E(PIMS)	LOI+24W	2P+7E	IOM
G03	Handling and Shipping Procedures				1E(PIMS)	LOI+24W	2P+7E	IOM
G04	Pre-Commissioning/commissioning Procedure				1E(PIMS)	LOI+24W	2P+7E	IOM
H	**QHSE PROCEDURES**							
H01	Quality Manual				1E(PIMS)	LOI+10W	2P+7E	IOM
H02	Quality Plan				1E(PIMS)	LOI+10W	2P+7E	IOM
H03	Repair Procedures				1E(PIMS)	PO+16W	2P+7E	IOM
H04	Performance Testing and Acceptance Test Procedures				1E(PIMS)	LOI+24W	2P+7E	IOM
~~H05~~	~~Software Quality System~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~IOM~~
H06	Weighing Procedure				1E(PIMS)	LOI+16W	2P+7E	IOM

VENDOR DATA REQUIREMENT LIST - VDRL

REQUISITION DESCRIPTION : Oily Water Separator Package | **Rev.:** 0

VDRL No. : | **Date:** 2-Sep-21

VDRL Code	Data/Document Description	Accept	Critical	With Tender	For Approval		Final Data	
				No. Copies	No. Copies	Date Reqd.	No. Copies	IOM/ MDR
H07	Health, Safety & Environmental Plan					NA	-	-
H08	Health, Safety & Environmental Mgmt System					NA	-	-
H09	Supplier HS & Management System Interface					NA	-	-
H10	Hydrostatic and/or Pneumatic Test Procedures				1E(PIMS)	LOI+16W	2P+7E	IOM
H11	Non-Destructive Examination Procedures(NDE)				1E(PIMS)	LOI+12W	2P+7E	IOM
~~H12~~	~~Post Weld Heat Treatment Procedures~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~IOM~~
H13	Surface Preparation and Painting Procedure/Specification				1E(PIMS)	LOI+16W	2P+7E	IOM
H14	Non-Conformance Report (NCR) Procedure				1E(PIMS)	LOI+16W	2P+7E	IOM
~~H15~~	~~Positive Material Identification Procedure~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~IOM~~
H16	Material Safety Data Sheets				1E(PIMS)	AS-1W	2P+7E	IOM
J	**MAINTENANCE & SPARES DATA**							
J01	Lube Oil and Operating Fluids Schedule				1E(PIMS)	LOI+24W	2P+7E	IOM
J02	Recommended Start-up and Commissioning Spares List			1E	1E(PIMS)	LOI+24W	2P+7E	IOM
J03	Recommended Spares List for Operation			1E	1E(PIMS)	LOI+24W	2P+7E	IOM
~~J04~~	~~Erection Fasteners Summary List~~				~~1E(PIMS)~~	~~LOI+24W~~	~~2P+7E~~	~~IOM~~
~~J05~~	~~Maintenance Schedule and Guaranteed engine change out intervals~~			~~1E~~	~~1E(PIMS)~~	~~LOI+24W~~	~~2P+7E~~	~~IOM~~
K	**PERFORMANCE & DESIGN CERTIFICATION**							
~~K01~~	~~Performance Test Results~~				~~1E(PIMS)~~	~~Test+2W~~	~~2P+7E~~	~~MDR~~
K02	Factory Acceptance Test Report/Certificate(FAT)				1E(PIMS)	Test+2W	2P+7E	MDR
K03	Vibration Report				1E(PIMS)	Test+2W	2P+7E	MDR
K04	Noise Report				1E(PIMS)	Test+2W	2P+7E	MDR
K05	Weight Report				1E(PIMS)	Test+2W	2P+7E	MDR
K06	Dimensional Report				1E(PIMS)	Test+2W	2P+7E	MDR
~~K07~~	~~Fire Test Reports/Certificates~~				~~-~~	~~NA~~	~~-~~	~~-~~
~~K08~~	~~Equipment Hazardous Area Certificates & Schedule~~				~~1E(PIMS)~~	~~LOI+24W~~	~~2P+7E~~	~~MDR~~
~~K09~~	~~HAZOP Reports~~				~~WHENEVERAPPLIC~~		~~2P+7E~~	~~MDR~~
~~K10~~	~~Register of Safety Related Device~~				~~WHENEVERAPPLIC~~		~~2P+7E~~	~~MDR~~
K11	General Reports					WHENEVER APPLICABLE	2P+7E	MDR
K12	Certification Data Book				1E(PIMS)	AS+4W	2P+7E	MDR
~~K13~~	~~Weldability Data~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~MDR~~
K14	Certification Data Book Index				1E(PIMS)	LOI+24W	2P+7E	MDR
K15	Type Test Certification					AS+4W	2P+7E	MDR
K16	Pressure Test Certificates					Test+2W	2P+7E	MDR
K17	Design and Analysis Reports					NA	-	-
L	**TEST & INSPECTION REPORTS**							
L01	Material Test Certificate				1E(PIMS)	AS+4W	2P+7E	MDR
L02	Welder Performance Qualification Records/Certificates				1E(PIMS)	LOI+16W	2P+7E	MDR
L03	NDE Operator Qualifications				1E(PIMS)	PO+16W	2P+7E	MDR
~~L04~~	~~Production Test Results(Inc Welding)~~				~~1E(PIMS)~~	~~PO+16W~~	~~2P+7E~~	~~MDR~~
L05	NDE Records				1E(PIMS)	AS+4W	2P+7E	MDR
~~L06~~	~~Heat Treatment Records~~				~~1E(PIMS)~~	~~AS+4W~~	~~2P+7E~~	~~MDR~~
~~L07~~	~~Material Traceability Records~~				~~1E(PIMS)~~	~~AS+4W~~	~~2P+7E~~	~~MDR~~
L08	Name Plate Rubbing				1E(PIMS)	AS+4W	2P+7E	MDR
~~L09~~	~~Electrical Equipment Type Test~~				~~1E(PIMS)~~	~~AS+4W~~	~~2P+7E~~	~~MDR~~
~~L10~~	~~Routine Test Certificate Electrical Equipment~~				~~1E(PIMS)~~	~~AS+4W~~	~~2P+7E~~	~~MDR~~
L11	Measurement of Insulation, Resistance & Continuity					NA	-	-

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VDRL Code	Data/Document Description	Accept	Critical	With Tender	For Approval		Final Data	
				No. Copies	No. Copies	Date Reqd.	No. Copies	IOM/ MDR
L12	Painting Inspection Report				1E(PIMS)	AS+4W	2P+7E	MDR
L13	Inspection & Test Plan			E	1E(PIMS)	LOI+4W	2P+7E	MDR
~~L14~~	~~High Voltage Impulse Test~~				-	~~Test+2W~~	~~2P+7E~~	~~MDR~~
~~L15~~	~~POSTIVE MATERIAL IDENTIFICATION REPORTS (PMI)~~				~~1E(PIMS)~~	~~LOI+16W~~	~~2P+7E~~	~~MDR~~
M	**SHIPPING & PACKING DOCUMENTS**							
M01	Shipping and Packing Documents				1E(PIMS)	AS+1W	2P+7E	IOM
~~M02~~	~~Hazardous Material Shipping Certificate~~				~~1E(PIMS)~~	~~AS+1W~~	~~2P+7E~~	~~IOM~~
M03	Inspection Release Certificate				1E(PIMS)	AS+1W	2P+7E	IOM
N	**RENEWABLE TEST & CALIBRATION DATA**							
N01	Pressure Gauge Test Certificate				1E(PIMS)	Test+2W	2P+7E	MDR
N02	Instrument Test/Calibration Certificate				1E(PIMS)	Test+2W	2P+7E	MDR
~~N03~~	~~Lifting Equipment Test Certificates~~				~~1E(PIMS)~~	~~Test+2W~~	~~2P+7E~~	~~MDR~~
~~N04~~	~~Insulation Resistance Check~~				~~1E(PIMS)~~	~~Test+2W~~	~~2P+7E~~	~~MDR~~
~~N05~~	~~Vessel & Exchanger Code Data Reports~~				~~1E(PIMS)~~	~~AS-1W~~	~~2P+7E~~	~~MDR~~
N06	Certificate of Compliance				1E(PIMS)	AS-1W	2P+7E	MDR
~~N07~~	~~Company's Release Notes/Waivers~~				~~1E(PIMS)~~	~~AS-1W~~	~~2P+7E~~	~~MDR~~
~~N08~~	~~Certifying Authority Release Note/Waivers~~				~~1E(PIMS)~~	~~AS-1W~~	~~2P+7E~~	~~MDR~~
N09	Agreed Letter of Assurance (HSE)					NA	-	-
~~N10~~	~~Concession Records~~				-	~~NEVERAPPLIC~~	~~2P+7E~~	~~MDR~~
N11	Certificate of Conformity				1E(PIMS)	AS-1W	2P+7E	MDR
P	**WELDING DATA**							
P01	Weld Procedure Specifications				1E(PIMS)	LOI+12W	2P+7E	MDR
P02	Weld Procedure Qualification (WPQ) Records				1E(PIMS)	LOI+12W	2P+7E	MDR
P03	Weld Procedure Register				1E(PIMS)	LOI+12W	2P+7E	MDR
P04	Weld History Data (Weld Maps)				1E(PIMS)	LOI+12W	2P+7E	MDR
Q	**MATERIAL CERTS(SITE WELDED)**							
Q01	Material Certificates - Bulk Items					WHENEVER APPLICABLE	2P+7E	MDR
Q02	Material Certificates - Package Welded Interfaces					WHENEVER APPLICABLE	2P+7E	MDR
R	**MANUALS**							
R01	Despatch Dossiers					NA	-	-
R02	Installation , Operation and Maintenance Manual (IOM)					AS+4W	2P+7E	IOM
R03	Renewable Certification Data Book					AS+4W	2P+7E	MDR
R04	Manufacturing Records-Manufacturer Data Report(MDR)					AS+4W	2P+7E	MDR
R05	IOM Table of Contents				1E(PIMS)	LOI+24W	2P+7E	IOM
R06	MDR Table of Contents				1E(PIMS)	LOI+24W	2P+7E	MDR
Z	**MISCELLANEOUS**							
Z01	Inspection Data Sheets				1E(PIMS)	Test+2W	2P+7E	IOM
Z02	Spare Parts and Interchangeability Record(SPIR)				1E(PIMS)	LOI+24W	2P+7E	IOM
Z03	List of special tools furnished for maintenance				1E(PIMS)	LOI+24W	2P+7E	IOM

Note 1. Progress report shall be submitted as monthly basis during engineering stage and after delivery. This report shall be submitted bi-weekly during peak season(from the initiation of fabraction to FAT completion.

PO: After Purchase Order

AS: After Shipment

P : Paper Copy

E : Electric Copy(USB/CD/DVD/Portable Disc)

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W : Week(s)

IOM : Installation , Operation and Maintenance Manual

MDR: Manufacturing Records-Manufacturer Data Report

LOI: Letter of Intent to Vendor

APPENDIX #6

TECHNICAL CLARIFICATION SHEET

TECHNICAL CLARIFICATION SHEET FOR OILY WATER SEPARATOR

NO.	Disc.	Date	CLARIFICATION REQUEST	SPECIFICATION REQUIREMENT or REFERENCE DOC/No.	BID Clarificaiton Meeting (13-July-2021)	HHI's Reply	Bidder's Reply	STATUS	REMARK
1	Process	9-Jul-21	Maximum Suction Height is 4.5m in Bidder proposal. Purchaser understood additional feed pump is not required when elevation between bilge tank LT2 set and oily water sepator tank bottom is less than 4.5m. Bidder to confirm. The feed water suction availability.	OFE47862 rev0 OWSAN-5 TEC page 7 Maximum Suction Height	HHI will provide pressure drop calculation from bilge water settling tank to package tie-in. Detegasa will check and confirm on feed water suction availability based on HHI information for pressure drop.	HHI provided requested information and waiting for Detegasa response.	Proposal was updated accordingly.	Closed	
2	Process	9-Jul-21	Bidder to ensure package equipment design intergrity based on utility system data indicated in section 8.4 in the I-ET-3010.1M-5330-667-P4X-001_C_SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501).	OFE47862 rev0 OWSAN-5 TEC page 7 Air Consumption / Flushing Water Consumption	Detegas confirms and comply. Pressure regulator will be considered in scope of supply by Detegas inside skid. Pressure relief valve will also provided at compressed air supply line. For fresh water supply line, pressure relief valve at first oil separator will be protected for package.	Noted and closed		Closed	
3	Process	9-Jul-21	(quote) It is also possible to carry out the cleaning of the coalescent filter manually, selecting “Manual Flushing On” mode in the control panel, which produces flushing water intake and the discharge to the bilge or oily water tank through valve V7. (unquote) It seems that V7 will not used for manual back-flusing because it's a pressure safety valve. V8 should be used for manual back-flusing. Bidder to check.	OFE47862 rev0 OWSAN-5 TEC page 14 PROCESS DESCRIPTION	Detegas confirms that V7 will be used during manual flushing mode operation.	Noted and closed		Closed	
4	Process	9-Jul-21	(quote) If for any reason, the sample has hydrocarbon content higher than 15 ppm, the discharge valve to the sea V16 would close and the recirculation to bilges V17 would open. If this situation continues for more than 10 seconds, the equipment would be blocked and stopped automatically. If, on the contrary, the situation is re-established before this period of time, that is, the samples have a content of hydrocarbons lower than 15 ppm, the equipment would automatically change closing V17 recirculating valve and opening V16 overboard valve. (unquote) As per Sec.6.1 in I-ET-3010.1M-5330-667-P4X-001, Automatic stop of the SAO Bilge Water Pump and close of sea valve after the necessary length of time to block hydraulic hammer effect. Intention of requirement is to ensure package protection in case of inadvertent overboard discharge valve close. Bidder to explain how to control/protect package when overboard dhscharge valve close while package is in operation.	OFE47862 rev0 OWSAN-5 TEC page 15_7.PROCESS DESCRIPTION page 18_10. ALARMS AND SHUT DOWN	Detegasa advised PSVs in the skid will protect package itself and XV closing signal send to package control panel to stop feed pump in package. Additional Pressure Switch can be considered as alternative option. Deterges to provide optional price for Pressure Switch in next proposal.	Noted and closed		Closed	
5	Process	9-Jul-21	Pump data on page 16(140l/min@2.5bar) and pump curve(less than 20m @ 1500 rpm &140 l/min) on page 20 is not matched.(but, motor rate speed is 1155 rpm on page 21) Pump selection to be check whether selected model is suitable based on required flowrate and discharge pressure. Bidder to confirm dischage pressure @ package flowrate is suitable considering with discharge elevation (abt. 20m) and adsorption filter and system pressure drop.	OFE47862 rev0 OWSAN-5 TEC page 16 pump technical data page 20 pump curve	Detegasa will check and reply. HHI will provide required tie-in pressure or pressure drop at the discharge piping.	HHI provided requested information and waiting for Detegasa response.	Proposal was updated accordingly.	Closed	
6	Process	9-Jul-21	It was indicated that self prime from dry run up to 3 m meanwhile max. suction elevation is 4.5 in P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1). Bidder to clarify max. suction lift elevation and consider to provide priming provision if required.	OFE47862 rev0 OWSAN-5 TEC page 20 pump curve	3m of dry running is self-priming requirement of pump suction line. Oily water separator and pump suction line to be filled before start-up through fresh water filling line by manual. It will be described in operation manual.	Noted and closed		Closed	
7	Process	9-Jul-21	Bidder to inform any package performance restriction for unit trim condition as per I-RL-3010.1M-1350-960-P4X-009.(or I-RL-3010.2G-1350-960-KES-326)	OFE47862 rev0 OWSAN-5 TEC	Detegasa will check and reply comparing with motion value considered in design based on IMO 107.	Detegasa to update on response	20 -Jul: The package is designed according to MEPC 107(49) requisitions, which highly exceed the design values of the FPSO. Please see the design criteria of MEPC 107(49) *Inclination test:* *Equipment should operate satisfactorily at angles of inclination up to 22.5º in any plane from the normal operating position;*	Closed	
8	Process	9-Jul-21	Bidder to inform the scope of system sizing report.(Vessel,Pump,PSV,CV etc.)	OFE47862 rev0 OWSAN-5 TEC	Detegasa will provide vessel calculation only. Others are part of vendor standards design. However, Detegasa will provide datasheets for valves and instruments as part of vendor data.	Noted and closed		Closed	

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9	Process	9-Jul-21	Bidder to inform max. operating temperature & design temperature of package(expecially 1st oily water separator) when operating temperature is higher than 60 degC, personnel protection shall be provided as per I-ET-3010.00-1200-431-P4X-001.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	Detegasa informed operating temperature set will be 45degC and design temperature will be 60degC except electrical parts in panel.	Noted and closed		Closed	
10	Process	9-Jul-21	Bidder to confirm the requirement of drain nozzel with valve on 1st oily water separator.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	Detegasa informed additional drain valve will be considered in pump suction line. It will be included in next proposal. Regarding the comment on drain point for Adsorption separator, "I" in the GAD, Detegasa informed current GAD is preliminary drawing and skid drip pan will be extended to cover drain point.	Noted and closed		Closed	
11	Process	9-Jul-21	Bidder to provide check valve on flush water supply line avoid to contaminate oil water into fresh water line and confirm the requirement of strainer. Purchaser doesn't have check valve and strainer on the line.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	HHI requested to include strainer and check valve on flushing fresh water inlet line in Detegasa scope of supply as base scope. Detegasa confirmed and proposal will be updated accordingly.	Noted and closed		Closed	
12	Process	9-Jul-21	Bidder to review set pressure of V22. It seems too narrow between operating pressure and set pressure.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	Detegasa will evaluate and reconsider pump and 2nd stage tank sizing. HHI will provide latest pressure drop calculation and layout. Detegasa informed this change will not impact on IMO type approval.	HHI provided requested information and waiting for Detegasa response.	Proposal was updated accordingly.	Closed	
13	Process	9-Jul-21	Oily Water separator need to ensure vacuum condition. Bidder to confirm.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	Detegasa confirms	Noted and closed		Closed	
14	Process	9-Jul-21	Bidder to check pump protection at blockage condition. PSV to be provided at pump downstream.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	Detegasa will consider PSV at pump downstream. Proposal will be updated accordingly.	Noted and closed		Closed	
15	Process	9-Jul-21	Purchaser wish to know the purpose of V20. Is it re-circulation required as per IMO MEPC 107(49) 6.1.1?	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ1)	Detegasa confirms that it is recirculation required as per IMO MEPC 107.	Noted and closed		Closed	
16	Process	9-Jul-21	Vendor to advice maximum line size of "L" considering 4.5m max height.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ2)	HHI will provide piping layout. Based on HHI layout information, Detegasa will check and advise.	HHI provided requested information and waiting for Detegasa response.	Proposal was updated accordingly.	Closed	
17	Process	9-Jul-21	Vendor to advice the function of V2 valve.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ2)	V2 valve is manual recirculation to adjust flow to ensure correct vacuum condition on 1st stage tank.	Noted and closed		Closed	
18	Process	9-Jul-21	Vendor to advice the requirement of heater, RS1 and advice requirment of temperature protection for Heater itselt.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ2)	Heater is to keep the temperature oil on top of 1st stage of tank. In order to remove the oil easily during flashing operation. Electrical heater has thermal temperature switch for electrical heater protection.	Noted and closed		Closed	
19	Process	9-Jul-21	Vendor to advice current line size of G,F is enough to prevent clogging problem of sludge.	P&ID_1801-UNIC-44-P-XB-00001_01_Z1 (OWSAN-20423-F01I RZ2)	Detegasa informs that there was no critical problem based on current design. Recommend to keep current line size.	Noted and closed		Closed	
20	Mechanical	9-Jul-21	Interface piping flanges are not matching with project's pining specification, B14H. (FF Vs RF)	-	Detegasa confirms on availability on FF flanges based on project requests.	Noted and closed		Closed	
21	Mechanical	9-Jul-21	Bidder to comply with the below instruction for selection of sub vendor list. 1) Super Critical category items within Package Unit Packager(s) shall select the Sub-vendor(s) as pre Appendix I Project Vendor List in Exhibit V - Directives for Procurement. 2) Critical category items within Package Unit Packager(s) shall select the Sub-vendor(s) as pre Appendix I Project Vendor List in Exhibit V - Directives for Procurement. In case Packager wants to present a Sub-Vendor(s) for a critical equipment different("alternate Sub-Vendor") from the Appendix I Project Vendor List in Exhibit V, Packager(s) will follow clause 4.7.4 in Exhibit V - Directives for Procurement. 3) For equipment/materials(including bulk materials) NOT covered by the Appendix I Project Vendor List in Exhibit V within Package Unit Packager(s) shall select his own standard Sub-Vendor(s) with proven pre-qualifications for the sub-supplied items. But, Packager(s) shall submit a Pre-Qualification documents for final approval of sub-vendor as requested in section 4.7.3 in Exhibit V. a) A list of similar installations b) Technical capability c) Manufacturing capability d) ISO 9001 certification or equivalent e) Type approval certificates f) Classification Society certificates	Exhibit V	1) Detegasa confirms to follow vendor list for Pressure relief valve. 2) Detegasa proposed own design Oil Monitor, DYNAMIC DOM-107. HHI requested to add this proposed vendor for oil in water analyzer in the deviation list for official approval by Petrobras with detail technical justification. 3) Detegasa shall provide Pre-Qualification documents for items which are not covered in Exhibit V during execution stage, if requested. Detegasa to provide sub supplier list with updating pressure relief valve, 1st/2nd Tanks and information for country.	Detegasa to provide sub supplier list with updating pressure relief valve, 1st/2nd Tanks and information for country. Detegas to update deviation list for oil monitor.	20 Jul: Deviation for the oil monitor updated in the deviation list	Closed	
22	Mechanical	9-Jul-21	Bidder to add table shown weight and size for major components in the GA drawing in order to check material handling of those item with Module layout.	Bidder's GA Drawing	Detegasa will provide required information in the GA drawing during execution stage after order.	Noted and closed		Closed	

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23	Mechanical	9-Jul-21	Bidder to indicate access & maintenance area required volume around skid in the GA drawing.	Bidder's GA Drawing	Detegasa will provide required information in the GA drawing during execution stage after order.	Noted and closed		Closed	
24	Mechanical	9-Jul-21	Bidder to confirm and comply on Quality Requirement attached as Appendix 15 in Material Requisition. If there are deviation and/or clarification, it shall be listed and updated in deviation & exception form, Appendix 8 in the MR. Bidder to confirm.	Quality Requirement attached as Appendix 15 in Material Requisition.	HHI confirms that Detegasa to consider only documents listed in I-ET-3010.1M-5330-667-P4X-001(SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501)). The appendix 15 will be updated accordingly.	Detegas to provide updated deviation list.	Deviation list was updated	Closed	
25	Mechanical	9-Jul-21	The Project assigned American Bureau of Shipping (ABS) as Classification Society for this project. Bidder shall comply with required ABS rules. The required ABS rules are not limited in Main Package. Bidder shall manage the ABS requirements for sub-equipment /component as well. Bidder shall directly communicate with ABS if required. - Class No. / Hull No.(YY283791/Q554) can be referred for communication with ABS for this project. - Notation ✠A1, Floating Production, Storage and Offloading System (Ship-Type), (S) Buzios Field Santos Basin Brazil, FL(30), 205X, ✠AMCC, EXP, CRC(OC), HAB++(OS), UWILD, HELIDK, OHCM - ABS VEC coordinator: Ms Sirin Safie (Tel: +65 6371 2653, ssafie@eagle.org) - VENDOR / SUB-vendor to allow CS (ABS) to share Vendor Comment List with HHI (Hull Contractor) and his Consortium lead KSL. - VENDOR to enable BUYER oversight access in Eagle Construct (ABS). VENDOR to note that ABS VEC	MR	Detegasa confirmed.	Noted and closed		Closed	
26	Mechanical	9-Jul-21	The Operation Manuals shall be delivered in English and Brazilian Portuguese Language including sub order items.	MR	Detegasa confirmed.	Noted and closed		Closed	
27	Mechanical	9-Jul-21	Bidder to provide on information of proposed sub vendor for fabrication and assembly.	-	Detegasa informed Skid assembly and FAT will be by Detagasa own factory in Spain.	Noted and closed		Closed	
28	Mechanical	9-Jul-21	Bidder shall confirm and comply with allowable nozzle load, Attachment_30-01.	-	Detegasa confirmed to follow nozzle load value but any calculation for piping stress is not part of Detegasa scope of work. HHI agreed but confirmed nozzle load values to be indicated in GA drawing during execution stage.	Noted and closed		Closed	
29	Mechanical	9-Jul-21	Bidder to advise boundary of Type Approval for IMO MEPC 107(49). Bidder to advise limitation of compliance project's requirements(especially compliance of all applicable documents in technical specification for oily water separator package(I-ET-3010.1M-5330-667-P4X-001) due to maintain the effect of Type Approval Certificate.	MR	Detegasa confirmed to maintain Type Approval for IMO MEPC 107(49). The critical points impact to IMO type approval, if any will be added in deviation list.	Noted and closed		Closed	
30	Mechanical	9-Jul-21	Flange isolation kits shall be provided where applicable as per ANNEX E in I-ET-3010.1M-1200-200-P4X-002_K_HULL PIPING SPECIFICATION. Bidder to confirm.	I-ET-3010.1M-1200-200-P4X-002_K_HULL PIPING SPECIFICATION	Detegasa confirmed.	Noted and closed		Closed	
31	Mechanical	9-Jul-21	Structural isolation kits shall be provided where applicable. (e.g. Between Stainless Steel Equipment's Support and Carbon Steel Skid's beam). Bidder to confirm.	-	Detegasa confirmed.	Noted and closed		Closed	
32	Mechanical	9-Jul-21	** Fabrication Site Conditions Supplier shall ensure the suitability of the package or equipment for the fabrication site condition as follows in every respect from design to operation in addition to the offshore site condition: • Location : South Korea • Duration : Maximum 1.5 year • Climate : 4 seasons, 100% relative humidity, Salt laden, marine coastal environment * Ambient temperature : 38.6 deg. C Max. , -16.7 deg. C Min.	-	Detegasa recommends to storage the equipment indoor before installation. Detegasa expect that this equipment will not be operated with below 0 deg.C and there is no concern for this clarification point.	Noted and closed		Closed	
33	Mechanical	9-Jul-21	Bidder to check and confirm on the revised Piping Specification For Hull. Refer to attachment #36-1.	-	Dedegasa will review Piping Specification for Hull which was provided in this TC No.36. Detegasa will add deviation points in the deviation list, if required.	Noted and closed		Closed	

TECHNICAL CLARIFICATION SHEET FOR OILY WATER SEPARATOR

NO.	Disc.	Date	CLARIFICATION REQUEST	SPECIFICATION REQUIREMENT or REFERENCE DOC/No.	BID Clarificaiton Meeting (13-July-2021)	HHI's Reply	Bidder's Reply	STATUS	REMARK
34	Mechanical	9-Jul-21	- ALL STEEL SERRATED BAR GRATING AND STAIR TREADS SHALL HAVE A MESH 30x100 mm: HEIGHT 30 MM; MINIMUM BEARING BRS THICKNESS 4.76 mm; CROSS BARS DIAMETER 8 mm. GRATING PANELS WITH EDGE BARS WELDED TO BEARING BARS AROUND PERIMETER AND PENETRATIONS; - ALL SERRATED WELDED STEEL GRATING AND STAIR THREADS SHALL BE HOT DIP GALVANIZED BY THE FABRICATOR ACCORDING TO I-ET-3010.00-1200-956-P4X-006 - GENERAL PAINTING; - ALL GRATING PANELS AND STAIR THREADS SHALL BE FASTENED TO SUPPORT STEEL EITHER BY POWER DRIVEN THREADED STUDS OR BY SADDLE CLIPS C/W BOLTS, NUTS, LOCK WASHER. STUDS SHALL USE AISI 316 STAINLESS STEEL. SADDLE CLIPS SHALL BE HOT DIP GALVANIZED. AT LEAST 4 FASTENERS PER SQUARE METER OR PER PIECE. DRAIN HOLES/GUILLIES MAY NOT BE FIXED. STAIR THREADS FASTENED WITH BOLTS AND ITS SURFACE COATING ACCORDING TO BOLT MATERIALS SPECIFICATION GRATING FIXING BY GRATING FABRICATOR; - STEEL GRATING SHALL BE DESIGNED FOR A MINIMUM 5kN/M^2 DISTRIBUTED LOAD AND FOR A CONCENTRATED LOAD OF 5kN APPLIED IN 100x100 mm AREA (NOT SIMULTANEOUSLY). STEEL GRATINGS IN AREAS WITH LOADS HIGHER THAN 5kN/M2 SHALL BE RIFIED TO THIS SPECIFIC LOAD CONDITIONS. STAIR THREAD SHALL BE DESIGNED FOR A CONCENTRATED LOAD OF 1.33 kN FOR EACH 600 mm OR FRACTION OF STAIR WIDTH; - MINIMUM GRATING SUPPORTING LENGTH 25 mm, CONSIDER INSTALLATION CLEARANCES TO DETERMINE THIS LENGTH; - MINIIMUM CLEARANCE BETWEEN GRATINGS IS 10 mm; - MEALLIC REMOVABLE FLOOR GRATING SHALL BE EQUALIZED (GROUNDED) TO STRUCTURE POTENTIAL ACCORDING TO REQUIREMENTS OF I-ET-3010.00-5140-700-P4X-001 - SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS AND I-DE-3010.00-5140-700-P4X-003 - GROUNDING INSTALLATION TYPICAL DETAILS; - DETAILED DESIGN SHALL ISSUE A REPORT WITH MEASUREMENT OF RESISTANCE BETWEEN REMOVABLE GRATING AND HULL REFERENCE GROUNDING PLATES, ACCORDING TO LIMITS DEFINED BY I-ET-3010.00-5140-700-P4X-001 - SPECIFICATION FOR ELETRICAL DESIGN FOR	-	There are No grating on the skid. In principle, access/maintenance of skid will be from outside(around) skid.	Noted and closed		Closed	
35	Mechanical	9-Jul-21	VENDOR's standard mechanical completion check sheets are to be detailed for the overall package, as well each individual component (e.g. Pump, Motor, Filters, Instruments, Control panel, etc.) as pre section 8.7 in the MR. The mechanical completion check sheets shall be included in FAT procedure which will be submitted as part of vendor document during execution stage for HHI review and approval. Bidder to confirm.	MR	Detegasa will check and revert	Detegasa to update on response	Detegasa confirm.	Closed	
36	Mechanical	9-Jul-21	Piping flow marking(stripes for identification) shall be provided as per Section 11 in DR-ENGP-I-1.15-R.4_0_COLOR CODING. The details for piping flow marking shall be included in Painting procedure which will be submitted as part of vendor document during execution stage for HHI review and approval.	DR-ENGP-I-1.15-R.4_0_COLOR CODING	Detegasa will check and revert	Detegasa to update on response	Detegasa confirm.	Closed	
37	Mechanical	9-Jul-21	Critical vendor documents which are directly related to HHI's interface engineering shall be submitted as requested in below. Appendix 5 in MR will be updated accordingly. Bidder to confirm on this submission schedule. 1) Vendor Document Schedule(VDS): LOI + 2 weeks 2) GA Drawing: LOI + 4 weeks (including weight, COG, piping tie-in location, E&I tie-in location, detail of structural interface, table for sub-equipment showing size and weight, required maintenance space around skid and etc.) 3) P&ID: LOI + 4 weeks 4) Utilities Schedule including Elec. Load list : LOI + 4 weeks 5) Single Line Diagrams: LOI + 8 weeks 6) Schematic Diagrams: LOI + 8 weeks	Appendix 5 in MR	Detegasa confirmed.	Noted and closed		Closed	
38	Mechanical	9-Jul-21	Bidder to submit utility consumption list including elec. Load list.	-	Detegasa will provide updated utility consumption list including elec. Load via separate offer document.	Detegasa to send utility consumption list.	20-Jul: Noted and confirmed	Closed	
39	Mechanical	9-Jul-21	Bidder to comply with vendor for oil in water analyser, Advanced Sensor.	Exhibit V	Detegasa will add this point in the deviation list.	Noted and closed. Further discussion will be clarified via deviation list.		Closed	
40	Mechanical	9-Jul-21	Bidder to comply with I-ET-3010.1M-5330-667-P4X-001_C_SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501)	I-ET-3010.1M-5330-667-P4X-001_C_SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501)	Detegasa will review and clarify on deviation points, if any via deviation and clarification list.	Noted and closed. Further discussion will be clarified via deviation list.		Closed	
41	Automation	9-Jul-21	Control Narrative and Cause & Effect chart shall be provided as part of vendor data.	General	Detegasa does not usually supply this document. We provide a troubleshooting guideline. If it is finally requested, Detegasa can supply that info, but it will have an impact in the price of the package	Noted but Detegasa to provide control/logic description as vendor data.	Detegasa confirm.	Closed	
42	Automation	9-Jul-21	Test report to include all instrument calibration certification / tubing leak test / instrument mechanical check sheet.	General	Detegasa confirms	Noted and closed		Closed	

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43	Automation	9-Jul-21	HART type field instrument device shall be applied.	GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS(I-ET-3010.00-1200-800-P4X-013) 4.2 Electronic Instrumentation 4.2.1 4 to 20 mA, with addition of digital communication using HART protocol, certified by HART FOUNDATION.	There are no field instruments in our package, consequently, we consider that it is not applicable	Noted and closed		Closed	
46	Automation	9-Jul-21	Package I/O list should be provided.	General	To be provided during project execution	Noted and closed		Closed	
47	Automation	9-Jul-21	IP Grade Requirement	AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS(I-ET-3010.00-1200-800-P4X-002) 5.1.4 All instruments and their accessories shall be at least IP-56 (IEC 60529) and according to Classification Society rules. 5.11.5 Junction boxes for instrumentation cables in classified or open areas shall have a minimum Ingress Protection rating of IP-56 and shall be "Ex e" or "Ex d" certified and certified for at least Zone 2. 5.13.6.1 Indoor panels and accessories installed in air conditioned room or ventilated room (indoor panels) shall be supplied with IP-22 protection degree, according to IEC 60529, as a minimum. 5.13.7.1 Outdoor panels and accessories shall be supplied with IP-56 protection degree, according to IEC 60529, as a minimum. Hazardous area requirements shall also be taken into account. See also item 4.1.	Detegasa comfirms	Noted and closed		Closed	
48	Automation	9-Jul-21	Junction Boxes Segregation Requirement	According to AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS(I-ET-3010.00-1200-800-P4X-002) 5.11.4 Signals to CSS related to PCS, PSD, HCS, HSD and FGS sub-systems, must be segregated in different terminal strips and multicables (where applicable), when installed inside the same junction box. The junction boxes shall be segregated into 3 types: • TYPE 1: for signals using flame-resistant cables (PSD/HSD/FGS); • TYPE 2: for signals related to safety interlocks (PSD/HSD); • TYPE 3: for signals related to process control (PCS/HCS).	No junction boxes in our scope of supply	Noted and closed		Closed	
49	Automation	9-Jul-21	Enclosure Material Requirment	According to GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS(I-ET-3010.00-1200-800-P4X-013) 6.3.2, Boxes or enclosures of electronic field instruments shall be made of ASTM A351 Gr. CF8M stainless steel.	No field instruments in our package	Noted and closed		Closed	
50	Automation	9-Jul-21	According to P1 package catagory requirements, package system shall consist and set the function.	AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS(I-ET-3010.00-1200-800-P4X-002) 4. PACKAGE INSTRUMENTATION AND CONTROL	Noted	Noted and closed		Closed	
51	Automation	9-Jul-21	Control Panel Color Requirement	COLOR CODING (DR-ENGP-I-1.15-R.4) 17.2. Control Panels [RC]: Light Cream (MUNSELL notation 2.5 Y 9/4).	Detegasa confirms	Noted and closed		Closed	
52	Automation	9-Jul-21	Instrument Color Requirement	COLOR CODING (DR-ENGP-I-1.15-R.4) 17.1. Instruments: follow the piping or equipment color. * See note bellow. Instrumentos: * Note: For new parts, it may be kept the original maker painting, except for control valves. GENERAL PAINTING(I-ET-3010.00-1200-956-P4X-002)	Detegasa confims	Noted and closed		Closed	

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53	Automation	9-Jul-21	Redundant power for package shall be provided.	Redundant power for package shall be provided.	To be discussed	HHI will more detail for this request. The available power supplied by the Vendor to be used by the Automation panels is defined in I-ET-3010.00-5140-700-P4X-003 – ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS. It is part of the panel SUPPLIER's scope to convert and redundantly distribute the different power supplies inside panels, including, where necessary, redundant AC/DC or DC/DC stabilized power supply units for cabinet's internal distribution of 24 VDC. (22-July) As mentioned in Annex 1 of I-ET-3010.00-5140-700-P4X-003, 1 feeder of 220V(1ph) will be supplied to OWS control panel by HHI. Bidder shall install 2 sets of AC/DC stabilized power supply units for cabinets' internal distribution of 24VDC as mentioned in I-ET-3010.00-5520-888-P4X-001, para 6.4. (26-July) With regard to power supplies, it's confirmed that "440/480V 60Hz 3ph", 220V 1ph CC(control), 220V 1ph AC(space heater for control panel) will be provided. Bidder to install 2 sets of AC/DC stabilized power supply units for control. Motor space heater is not requirement.	20 -Jul: The package will be fed by one main power supply to the main control system (440 V 60 Hz 3 Ph) and one auxiliary electrical supply for the space heater of the cabinet. Electrical istribution and step down by means of transformers will be carried out internally in the package. in that sense we do not really understand the requisition about power redundancy, more precisely, redundant distribution of power supplies. Please clarify further this point (22-July): We understand that our package will be fed by three independent power supplies: - 440/480 V 60 Hz 3 Ph - 220 V 1 Ph CC (control) - 220 V 1 Ph AC (space heater for main control panel) * We understand that space heater is NOT required for the motor of the pump **(26-July): Noted and confirmed**	Closed	
54	Automation	9-Jul-21	SPARE PARTS:	According to AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS(I-ET-3010.00-1200-800-P4X-002) 5.1.5 Continuity of spare parts supply and maintenance shall be given for a 5 year period for instrument.	Detegasa confirms	Noted and closed		Closed	
55	Automation	9-Jul-21	Cable Tray Material Requirement	GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS(I-ET-3010.00-1200-800-P4X-013) 17.4.1 Cable trays shall be made of AISI 316 stainless steel. Cable trays made of galvanized steel may be used in sheltered areas (rooms, etc.).	Detegasa comfirms	Noted and closed		Closed	
56	Automation	9-Jul-21	Cable tray also shall be segregated by signal type.	General	Detegasa confirms	Noted and closed		Closed	
57	Automation	9-Jul-21	Instruments in type P1 packages are not directly and permanently connected to AMS(Asset Management System), although the DTM/DD of package instruments for P1 Packages shall be delievered and configured in AMS, as defined in I-MD-3010.1M-1200-800-P4X-001_D, in this way, allowing the handheld use for these instruments. As stated in I-MD-3010.1M-1200-940-P4X-025, a descriptive memorandum (MD) has precedence over a technical specification (ET), so the clause 9.14.4.11 of I-MD-3010.1M-1200-800-P4X-001_D shall be complied.	I-MD-3010.1M-1200-800-P4X-001 I-MD-3010.1M-1200-940-P4X-025	To be discussed	HHI will more detail for this request. Hart protocol type of instruments for deliverying DTM/DD to be provided.	20- Jul: Noted and confimed if instruments provide signals to AMS. If those signals are only for internal use in the package, HART protocol will not be considered	Closed	
58	Automation	9-Jul-21	PLC and HMI shall be selected from below list. 1) GE 2) SIEMENS 3) Rockwell Because Standardization for PLC and HMI is requested in P-78 Project. Final selection of supplier for PLC and HMI will be informed during Execution Stage.	-	Our scope includes a SIEMENS PLC. In case of changes, it might have an impact in the price	Noted and closed. HMI also to be included in base scope as per response on TC#45.		Closed	
59	Electrical	9-Jul-21	Bidder's Technical Specification 5.2 Electrical and control design basis The equipment includes a main control panel to be installed in safe zone, which contains the motor starters and the main power connection.	Motor starters will be provided by Buyer. Control panel only shall be provided by Bidder. Interface cable between Motor starters and control panel shall be provided by Buyer.	The electrical connection between the starter and the motor is within the limits of our package	Noted. Further clarification shall be done during detail design if required.		Closed	
60	Electrical	9-Jul-21	Oily Water Separator General Arrangement Electrical Supply : 690Vac, 3ph, 60Hz	Motor shall be 440Vac, 3ph, 60Hz.	Detegasa confirms	Noted.		Closed	

APPENDIX #7
PROJECT STANDARD FORMAT

1. ATTACHMENT #1. SPIR
2. ATTACHMENT #2. SPECIAL TOOL REGISTER
3. ATTACHMENT #3. LUBRICANT SCHEDULE
4. ATTACHMENT #4. RAM & FMEA INPUT FORM
5. ATTACHMENT #5. NOT USED
6. ATTACHMENT #6. NOT USED
7. ATTACHMENT #7. NOT USED
8. ATTACHMENT #8. NOZZLE CHECK SHEET
9. ATTACHMENT #9. WEIGHT DATASHEET(A) AND CERTIFICATE(B)
10. ATTACHMENT #10. WELDING PARAMETER CHECK SHEET
11. ATTACHMENT #11. INSPECTION & TEST PLAN
12. ATTACHMENT #12. INSPECTION NOTIFICATION
13. ATTACHMENT #13. INSPECTION RELEASE NOTE
14. ATTACHMENT #14. PUNCH LIST FORMAT
15. ATTACHMENT #15. PACKING LIST
16. ATTACHMENT #16. DOCUMENT TEMPLATE
17. ATTACHMENT #17. DRAWING TEMPLATE
18. ATTACHMENT #18. INSTRUCTION FOR SKID VISUAL AND DIMENSION INSPECTION
19. ATTACHMENT #19. EX REGISTER FORMAT
20. ATTACHMENT #20. DESIGN DOCUMENT ANALYSIS (DDA)
21. ATTACHMENT #21. TECHNICAL QUERY FORM (TQF)
22. ATTACHMENT #22. VENDOR DOCUMENT ANALYSIS (VDA)
23. ATTACHMENT #23. NR 12 ANNEX II REGISTER FORM

APPENDIX #8

DEVIATION & EXCEPTION LIST

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TECHNICAL EXCEPTIONS/ DEVIATIONS

Project NO:

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1	I-ET-3010.1M-5330-667-P4X-001 OILY WATER SEPARATOR PACKAGE (SAO-5330501)	3.0 REFERENCE DOCUMENTS	D	Only the the following documents have been considered as applicable: 1) I-ET-3A36.00-1000-941-PPC-001 METOCEAN DATA 2) I-ET-3000.00-0000-940-P4X-002 SYMBOLS FOR PRODUCTION UNITS DESIGN 3) I-ET-3000.00-1200-940-P4X-001 TAGGING PROCEDURE FOR PRODUCTION UNITS DESIGN 4) I-ET-3010.00-1200-956-P4X-002 GENERAL PAINTING 5) DR-ENGP-I-1.15 COLOR CODING 6) I-ET-3010.00-5140-700-P4X-002 SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS 7) I-ET-3010.00-5140-700-P4X-003 ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS 8) I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS 9) I-ET-3010.1M-1200-800-P4X-014 AUTOMATION INTERFACE OF PACKAGE UNITS PACKAGE INTERFACE 10) I-ET-3010.1M-1200-200-P4X-002 HULL PIPING SPECIFICATION 11) I-ET-3010.00-5400-947-P4X-002 SAFETY SIGNALLING 12) I-T-3010.1M-1200-300-P4X-002 NOISE CONTROL REQUIREMENTS FOR ACCOMMODATION / SHIPSIDE 13) I-DE-3010.1M-1200-942-P4X-002 GENERAL ARRANGEMENT 14) I-DE-3010.1M-5400-94A-P4X-001 AREA CLASSIFICATION GENERAL; I-RL-3010.1M-1350-960-P4X-009 MOTION ANALYSIS 15) I-DE-3010.1M-5330-944-P4X-001 BILGE, SLUDGE, BALLAST (ENGINE ROOM) AND GENERAL SERVICE SEA WATER. 16) I-ET-3010.1M-5330-667-P4X-001 SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501) 17) Contents of Material Requisition Any other document not listed above (including any subsequently referenced document) is considered as not applicable and not included in the scope of supply. (16-July) Detegasa confirm and compliance international codes and Brazilian Codes which are listed in Section 2.1 and 2.2 in I-ET-3010.1M-5330-667-P4X-001 SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501).	The documents listed in this deviation list is completely aligned with reference documents in I-ET-3010.1M-5330-667-P4X-001 SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501). This clarification is acceptable for HHI. However, Detegasa also to confirm the international codes and Brazilian Codes which are listed in Section 2.1 and 2.2 in I-ET-3010.1M-5330-667-P4X-001 SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501). (16-July) Noted and acceptable to HHI.	I-ET-3010.1M-5330-667-P4X-001 SPECIFICATION FOR OILY WATER SEPARATOR PACKAGE (SAO-5330501) Section 2.1 and 2.2 shall be applied.	Closed	**Closed**
2	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	4.SCOPE OF WORK Control Panel (PN-SAO-5336501)	C	Control panel installed on the common skid of the OWS	Noted and Closed.		Closed	Closed
3	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	4.SCOPE OF WORK INMETRO compliance Certification	D	INMETRO certification is not included in our scope of supply	Noted and Closed. OWS is installed in safe area.		Closed	Closed
4	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	4. SCOPE OF WORK Documentation as per VDRL, this material requisition and project specifications	D	Along with the technical proposal, we submit our proposal for VDRL (16-July) Detegasa confirm the documents required by OILY WATER SEPARATOR PACKAGE(I-ET-3010.1M-5330-667-P4X-001) at least.	Noted. Bidder shall comply at least with the documents required by OILY WATER SEPARATOR PACKAGE(I-ET-3010.1M-5330-667-P4X-001) (16-July) Noted and Closed.		Closed	Closed

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5	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	The following meeting shall be included in BIDDER's base scope of work 1) Kick Off Meeting at BUYER's Office (1 time, 2 working days) 2) Pre-Inspection Meeting at BIDDER's Premises (1 working day) 3) Clarification Meeting at BUYER's Office (2 times, 2 working days each) 4) HAZOP Meeting at BUYER's Office (1 time, 2 working days) 5) Technical Clarification Meeting at BIDDER's Premises or Conference Call (anytime when required)	D	Due to the Covid-19 Pandemic, we propose to hold those meeting via videocall. If it is not acceptable, then we will provide our standard services rates for onsite meetings in Korea.	Noted. Bidder shall consider meeting via videocall.		Closed	Closed
6	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	PDMS 3D model	C	We will provide a 3D model in STEP format. (16-July) Detegasa confirm to support and provide required information of attribute, if required during execution stage.	Noted but bidder shall provide supports to covert CAE 3D tools by HHI during execution supports. Refer to I-ET-3000.00-1350-940-P4X-013_F_TECHNICAL REQUIREMENTS FOR DIGITAL ENG. All required attribute values to be provided by vendor for proper converting from STEP format to CAE 3D tools. (16-July) Noted and Closed.		Closed	Closed
7	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	4. SCOPEO OF WORK Supervision service for installation, pre/commissioning & startup ; Required dates' amount for listed activities and lump sum pricing with daily rates shall be proposed by Bidder. - During Installation at Fabrication Site - During Pre-commissioning and Commissioning at Fabrication Site and Offshore Field Training of for BUYER Operating and Maintenance Teams ; Minimum Eight (8) hours	C	Please see our proposal in terms of commissioning and training for this package, which is included in the scope of supply (16-July) Detegasa confirm that training and commissioning will be reflected as optional scope in updated proposal.	1) Bidder shall provide Training service as **optional scope**. Bidder shall provide 3 optional proposals for each 3 locations of Korea(On-shore), Singapore(On-shore), Brazil(Off-shore). 2) Below requirements are still valid for each option. - Training Program for each piece of equipment shall cover maintenance and operation aspects, since the public to attend the training shall be presented by two (2) separated groups of fifteen (15) technicians, being one group for maintenance staff and other for operation staff. Attendees shall be awarded certificates of training, upon completion of the course. - The training shall be performed in Brazilian Portuguese language or in English with translation to Brazilian Portuguese language. - Training shall be minimum sixteen (8) hours per each group. 3) Bidder shall provide Supervision service for installation, pre/commissioning & startup as **optional scope.**		Closed	Closed
8	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	5.4 ELECTRICAL The application of electrical equipment, instruments and accessories on hazardous areas shall follow the requirements of IEC series 60079, 61892-6 and 61892-7, as well as have their conformity certificates according to INMETRO Portaria 179, May 18th 2010 and INMETRO Portaria 89, Feb 23rd 2012 and approved by Classification Society. Electrical equipment and accessories installed in external safe or hazardous areas, which shall be kept operating during emergency shutdown ESD-3P or ESD-3T shall be certified for installation in hazardous areas Zone 2 Group IIA temperature T3 (IEC 61892-7).	C	We understand that our packege in located indoors and in safe area. Consequently, Ex requirements are not applicable for the package	Detegasa understading is correct.		Closed	Closed

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9	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	5.5 INSTRUMENTATION Electronic instruments shall meet the requirements of IEC 61000-6-1/2 regarding electromagnetic compatibility. All instruments, junction box, panels, materials and equipment proper to be used in hazardous areas, shall have conformity certificates complying with PORTARIA INMETRO Nº 179, DE 18/maio/2010, and its annexes, changed by PORTARIA INMETRO Nº 89, de 23/fereveiro/2012, and shall be approved by Classification Society. Electrical equipment installed in external areas, that shall be kept operating during emergency shutdown ESD-3P and ESD-3T shall be certified for installation in hazardous areas Zone 2 Group IIA temperature T3.	C	Same comments as in the previous point	Detegasa understading is correct.		Closed	Closed
10	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	6.0 BIDDER´S RESPONSABILITIES BIDDER shall participate in Hazard and Operability sessions (HAZOP) for his supplied process and utility system equipment in accordance with Project HAZOP Strategy. BIDDER shall bear ALL costs associated with participants and implement the outcome into equipment design.	C	Decisions associated to HAZARD meetings held in posterior dates upon agreeing the scope of supply and price for the package, might have an impact in the price (extracost) and delivery time of the package.	Noted and Closed.		Closed	Closed
11	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	61.APPROVED VENDOR LIST	D	Our proposed brands for main equipment are shown in section 11 of our technical proposal (16-July) Detegasa confirm to this instruction. Only Safety Valve and TOG(Oil content in water) Analisyer are applicable for our package. Detegasa confirm to AVL for Safety Valve. Detegasa will make deviation for TOG Analisyer.	Bidder to comply with the below instruction for selection of sub vendor list. 1) Super Critical category items within Package Unit Packager(s) shall select the Sub-vendor(s) as pre Appendix I Project Vendor List in Exhibit V - Directives for Procurement. 2) Critical category items within Package Unit Packager(s) shall select the Sub-vendor(s) as pre Appendix I Project Vendor List in Exhibit V - Directives for Procurement. In case Packager wants to present a Sub-Vendor(s) for a critical equipment different("alternate Sub-Vendor") from the Appendix I Project Vendor List in Exhibit V, Packager(s) will follow clause 4.7.4 in Exhibit V - Directives for Procurement. 3) For equipment/materials(including bulk materials) NOT covered by the Appendix I Project Vendor List in Exhibit V within Package Unit Packager(s) shall select his own standard Sub-Vendor(s) with proven pre-qualifications for the sub-supplied items. But, Packager(s) shall submit a Pre-Qualification documents for final approval of sub-vendor as requested in section 4.7.3 in Exhibit V during execution stage, if requested from comments of sub-vendor list which will be submitted as part of vendor data. a) A list of similar installations b) Technical capability c) Manufacturing capability d) ISO 9001 certification or equivalent e) Type approval certificates f) Classification Society certificates (16-July) Noted and closed. Regarding the deviation for TOG Analisyer will be clarified via related item in this list.		Closed	Closed
12	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	6.3 WEIGHT CONTROL	D	Our standard weight control and procedure will be applied. See the attachment.	Attached Detegasa proposed weighing procedure can be Acceptable but this procedure shall be provided as part of vendor data during execution stage. T:₩03.DESIGN₩ hanical₩12_TBE R		Closed	Closed
13	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	7.0 NOISE AND VIBRATION LIMITATION	C	Noise and vibration measures will be carried out during FAT. Noise and vibration report will be provided along with the equipment	Noted and Closed.		Closed	Closed

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14	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	8. INSPECTION AND TESTING	C	According to our standards in section 5.4 of our technical proposal 20-Jul: Noted and confirmed	Bidder to comply with Section 6 along with Table 1 and Table 2 in EXHIBIT V as minimum.		Closed	Closed
15	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	8.1 INSPECTION AND TEST PLAN	C	We will provide an ITP for your revision during project execution, in case of order	Noted and Closed.		Closed	Closed
16	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	8.2 INSPECTION AUTHORITY	C	In our proposal we include all DNV_GL activites required by Class to get the approval for the package	Noted. Please note that the Project assigned American Bureau of Shipping (ABS) as Classification Society for this project.		Closed	Closed
17	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	8.5 WELDING AND WELDING INSPECTION	D	According to our standards in section 5.4 of our technical proposal and NR-13 and ASME applicable standards for our package	Acceptable but Welding Book(WPS/PQR/Welder Qualification) shall be submitted as part of vendor data druing execution stage.		Closed	Closed
18	I-RM-3A36.06-5330-667-HHI-301 MATERIAL REQUISITION FOR OILY WATER SEPARATOR	11 DOCUMENT REQUIREMENT Manufacturing Record Book	D	The following documentation are not included in the scope of supply: - Manufacturer's mill certificates - Index and map of radiographic and ultrasonic examinations (not applicable for the package) - Heat treat records and charts (non-applicable for the package) (16-July) Detegasa confirm to provide mill certificates except commercial items such as bolt/nut, motor, and such components.	Bidder to comply on providing Manufacturer's mill certificates. (16-July) Noted and closed.		Closed	Closed
19	I-ET-3010.1M-5330-667-P4X-001 OILY WATER SEPARATOR PACKAGE (SAO-5330501)	4.1 QA/QC SYSTEM PACKAGER shall submit his Quality Assurance / Quality Control handbook to HULL SUPPLIER for information	C	Quality manual is only available in spanish. If translation is required it may has an impact in price. (16-July) Noted and confirm.	Quality manual shall be submitted as part of vendor data with English. (16-July) Noted and closed.		Closed	Closed
20	I-ET-3010.1M-5330-667-P4X-001 OILY WATER SEPARATOR PACKAGE (SAO-5330501)	6.1 SCOPE OF SUPPLY For foreign made equipment, the standard manufacturing parts (couplings, mechanical type seals, anti-friction bearings) shall be purchased from Manufacturers with representative branches located in Brazil, with service parts and maintenance workshops	D	Considered not applicable to our package (16-July) Detegasa confirm to local supplier in Brazil for pump spare parts.	Detegasa to confirm availability of spare parts for pump from Brazil. (16-July) Noted and closed.	Avalability of spare parts from Brazil to be confirmed.	Detegasa confirmed that local supplier in Brazil for pump spare parts.	**Closed**
21	I-ET-3010.1M-5330-667-P4X-001 OILY WATER SEPARATOR PACKAGE (SAO-5330501)	8.8 PRESERVATION, PACKING AND TRANSPORTATION The preparation shall make the equipment suitable for 12 months outdoor storage in a coastal tropical climate from the time of Shipment	D	The equipment shall be stored indoor. If special package for outdoor storage must be provide, it shall be included as an extracost. 20-Jul: Noted and confirmed Just one remark in reference to the painting specification, more precisely about the Inspections and tests: DELTA TG tests are NOT included in our scope of supply, since we consider that they are not applicable for our package	HHI confirm that equipment will be installed indoor before instllation but Detegasa to consider the preservation requirements in Material Requisition and I-ET-3010.00-1200-956-P4X-002(GENERAL PAINTING). (21-July) In terms of preservation, this item can be closed.		Closed	Closed
22	I-ET-3010.1M-5330-667-P4X-001 OILY WATER SEPARATOR PACKAGE (SAO-5330501)	10. TRAINING	C	Included as optional scope, not in the base scope of supply	Noted and Closed.		Closed	Closed

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23	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	5.1 TEMPORARY PAINTING (SHOP PRIMER) 5.1.2.2 The integrity of holding primer shall be evaluated by visual inspection and a pull-off test. A minimum value of 5MPa is required	D	Pull off or adherence tests are not included in the base scope of supply. (16-July) Cost will be added in final proposal.	Detegasa to include it in the base scope. (16-July) Noted and closed.		Closed	Closed
24	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	5.7.11.1 All welds in the line/equipment shall be full penetration welds	C	Considered not applicable to our package	Noted and Closed.		Closed	Closed
25	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	6 COATINGS SYSTEM QUALIFICATION Performance (pre-qualification) testing shall be carried out by the coating manufacturer according to this specification	C	Coating system is qualified by the paint manufacturer according internatinoal standards. Specific qualification for this projects are considered not applicable for our package and not included in the scope of supply. 16/08/2021: we follow the international standards ISO 12944-5 C5M Exterior High, ISO 8501-3:2006 and ISO 8501-1:2007	Acceptable to HHI. Pre-Qualification testing by coating manufactuer can be excluded for this particular skid.	Bidder shall inform the international standards to be applicable. [08/27/2021] HHI confirmed by e-mail on 08/25/2021 that this item could be close according the following DETEGASA's response: "Regarding the item no.26 in the deviation list what we communicated in below e-mail, Detegasa confirmed they can withdraw this deviation based on final clarification with their paining manufacturer. This item can be closed." Note: The no of deviation is wrong on the e-mail, the correct is no 25.	Detegasa to advise international standards mentioned in bidder's deviation list and summary of comparison between Detegasa proposed standards and table 4 in the section 6 in I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING.	Closed
26	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	8.3 PAINTERS QUALIFICATION	C	Considered not applicable to our package 16/08/2021: FROSION or NACE level 2 inpector will attend inpections according to ITP, to be discussed and approved after PO.	Acceptable to HHI. Attached Detegasa's Paintrers qualification carried out by FROSIO inspector can be acceptable to HHI,. T:₩03.DESIGN₩ hanical₩12_TBE R	Seller to confirm the attendance of the I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING with the painters qualification received. [08/23/21] Item can be closed.	Painting inspector is under bidder's responsibility however, Seller's qualified inspector who is requested to FROSIO or NACE Level 2 in general will attend inspection according to vendor ITP which will be set and approved after PO.	Closed
27	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	12.4 PIPING INTERNAL COATING APPLICATION	C	Pipes in the package are made of CRES316 and internal coating is considered not applicable.	Acceptable in conjunction with the item no. 31 and no.61.	Noted.	Closed	**Closed**
28	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	13.4 TESTS ON COMPLETION OF COATING WORKS	D	Only the following tests are considered: DFT and visual inspection. Other tests not included in the scope of supply. (16-July) Cost will be added in final proposal.	Detegasa to include it in the base scope. (16-July) Noted and closed.		Closed	Closed
29	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	15 DOCUMENTATION 17 WARRANTY	C	Considered not applicable to our package	Acceptable to HHI. Almost required documents and warranty are not directly related to package.		Closed	Closed
30	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	15 DOCUMENTATION 17 WARRANTY	C	Considered not applicable to our package	Acceptable to HHI. Almost required documents and warranty are not directly related to package.		Closed	Closed
31	I-ET-3010.00-1200-956-P4X-002_F_GENERAL PAINTING	ANNEX A.	C	For carbon steel components the system number 2 shall be used. For stainless steel components the system number 4 shall be used.	Noted and Closed. Please note that the coating for Austenitic stainless steel with service temperature below 50°C can be excluded based on section 4.1 in I-ET-3010.00-1200-956-P4X-002 (GENERAL PAINTING).		Closed	Closed
32	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	3.1 ENVIRONMENTAL CONDITIONS Printed circuit-boards shall additionally comply with the following requirements…	C	Considered not applicable to our package. Motors shall be tropicalized. However, the electric/electronic equipment shall be as per Detegasa's standard, suitable for temperatures up to 55ºC and humidity up to 90%. (16-July) Cost will be added in final proposal but HHI to confirm on required max. temperature for electrical device outside panel. 20-Jul: Noted and confirmed	Detegasa to comply with Environmental condtion. In terms of temperatures, Detegasa to comply with Table 1 in I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS. (16-July) HHI will check and advise. (19-July) HHI clarified that intention of max. temperature for electrical device outside panel is for device on panel itself.		Closed	Closed

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33	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	3.6.3.1 Paint system for external coating of all electrical equipment, material and supports shall be proper for offshore installation and pre-qualified according to I-ET-3010.00-1200-956-P4X-002 - GENERAL PAINTING. 3.6.3.2 Painting system for internal coating of all electrical equipment and material shall be proper for offshore installation according to manufacturer requirements, when not specified by PETROBRAS.	D	Coating for control panels and electrical components shall be as per OEM standard with the exception of the color code, that shall follow Petrobras requirements. 16/08/2021: for control panel, ISO 9227:2017, ISO 12944-6, ISO 12944-5 For other electrical components, they are not available at this stage, to be provided in a later stage, during project execution.	Acceptable subject to Petrobras approval.	Deviation to be clarified.	Based on conference call with Detegasa on 2-Sep-2021, Detegasa shall following conclusion; 1. Painting for E&I item : Detegasa to comply with the project specification 2. Junction box on Electrical Heater : Detegasa to change proposal to meet project requirement. Junction box will be non-metallic material(Polyamide).	Closed
34	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	3.6.5. 2 All equipment with mechanical protection degree IP44 or higher shall have test certificates issued by an authorized and properly identified laboratory.	C	Individual components shall have OEM certificates for IP. However, please note that IP tests for assemblies as control panels are not considered in the scope of supply. 20-Jul: Noted and confirmed	Acceptable but Detegasa to confirm on OEM certificates for IP is still valid after assembly.		Closed	Closed
35	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	3.10 EMC AND RFI REQUIREMENTS	D	Electric or electronic equipment shall be compliant with 2014/30/EU for EMC and in general, all the components shall be compliant with IEC61000-6-2 and IEC61000-6-4 or equivalent. Other requirements are considered not applicable for this package.	Acceptable subject to Petrobras approval.	Deviation to be clarified. [08/23/21] Clarification accepted. Item can be closed.	Seller understands that this is clarification from bidder. Bidder confirmed that ompliant with 2014/30/EU for EMC and in general, all the components shall be compliant with IEC61000-6-2 and IEC61000-6-4 or equivalent. Other general sentences in section 3.10 of I-ET-3010.00-5140-700-P4X-002 are not applicable for this package.	Closed
36	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	4.7 TRANSFORMERS	C	Considered not applicable to the single phase small transformer for control tension included in the control panel of our package.	Acceptable subject to Petrobras approval.	Deviation to be clarified. [08/23/21] As informed by Bidder, the requirement is not applicable. Item can be closed.	Seller understands that this is clarification from bidder. Referred section is not applicable for this package.	Closed
37	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	4.8.3.3.13 The following criteria shall be applied: a) Panels shall be built with steel sheets AISI 316L;	D	The control panel of the equipment is made of coated carbon steel. If not acceptable, please advise and we'll revise our proposal.	Plates constructed in painted carbon steel are accepted for indoor panels based on I-ET-3010.00-5520-888-P4X-001(AUTOMATION PANELS)		Closed	Closed
38	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	4.8.3.5.1 Panel shall have busbars of electrolytic copper in A.C. systems	D	The control panel of the equipment is only using eath bars. Power distribution is carried out by internal wiring and not busbars.	Acceptable.		Closed	Closed
39	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	4.8.3.17 HEATING RESISTORS	D	Considered not required for our package (16-July) Cost will be added in final proposal.	Detegasa to include it in the base scope. (16-July) Noted and closed.		Closed	Closed
40	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	5.1.6 Unless otherwise stated, all threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B1.20.1.	D	Our equipment also uses cilindrical (metric or BSP) threads. (16-July) Noted. Metric will be considered.	Metric is acceptable but, BSP is not acceptable. (16-July) Noted and closed.		Closed	Closed

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41	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	5.3 CABLE TRAYS	D	The cable trays external to the control panel are made of SS316L but the cable trays internal to the control panel may be made of non metalic or plastic material. Tests for these non metalic materials are not available. Structural calculations for external cable trays are considered not required due to the small size of the equipment. (16-July) Yes, our intention of cable trays internal of panel is cable duct. We noted and confirm on cable overfill verification.	If Detegasa's intention of cable trays internal is means cable duct inside panel, this clarification is acceptable. The submission of structural calculations can be waived but Detegasa to confirm that cable overfill shall be verified by Detegasa including consideration of free space on cable tray for spare cables. (16-July) Noted and closed.		Closed	Closed
42	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	5.11 TERMINAL LUGS AND TERMINAL BLOCKS FOR CABLES	C	Isolated ferrules for internal control panel cable connections are understood as acceptable.	Acceptable to HHI		Seller understands that Buyer agree on clarification between HHI and Bidder. This item can be closed.	**Closed**
43	I-ET-3010.00-5140-700-P4X-002, SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS	5.13.3.4 In control cables where there is traffic of analog signals and in control cables with interface/interconnection with PLC, control cables shall follow the specification of cables for instruments, with twisted pair, individual shielding for each pair and general shielding under the outer cover.	D	2 core cables with common copper braid/foil shall be also acceptable for control cables. (16-July) Confirm and will be reflected in final proposal.	Not acceptable. The twisted pair, individual shielding cable must be used. (16-July) Noted and closed.		Closed	Closed
44	I-ET-3010.00-5140-700-P4X-003, ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS	6.5.15.5. Panels shall be installed in areas free of vibration and far from heavy maintenance areas. It shall be avoided the installation of panels on the same skid of the driven/controlled equipment. maintenance areas. It shall be avoided the installation of panels on the same skid of the driven/controlled equipment.	C	Considered not applicable to our package	Noted and closed.		Closed	Closed
45	I-ET-3010.00-5140-700-P4X-003, ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS	9.2. For Packages with power panel, Packager shall issue a protection coordination study including graphical plotting for all protective devices of the panel	C	Considered not applicable to our package	Noted and closed.		Closed	Closed
46	I-ET-3010.00-5140-700-P4X-003, ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS	For items 9.5 and 9.6, software shall be furnished in their most recent versions at purchase time, accompanied by their corresponding licensing, access passwords, source code, installation media(s) and manuals, as well as with one year of technical support and maintenance.	D	Detegasa shall provide the files with the PLC logic and parameterization but it's not including the commercial licenses (from Siemens or PLC/HMI OEM) to install/modify the software. 20-Jul: We will include the sofware licence in the scope of supply	Not acceptable. Detegasa to send us more detail of available options for HHI consideration. (21-July) Noted and Closed.		Closed	Closed

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47	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	4.2 Package Classification	C	We understand that our package falls under P1 type	Detegasa understading is correct.		Closed	Closed
48	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	4.3.2.6 All logic diagrams, cause and effect matrix, P&IDs, Instrument List and I/O List, depicting all Package logic, shall be supplied along with Package.	C	This information will be supplied according to our standards	Noted but Detegasa to follow up project template, tagging and symbol as per MR requirements.		Closed	Closed
49	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	4.3.3 P1 Package Requirements	C	Signals to be according to I-DE-3010.1M-5330-944-P4X-001 PIPING AND INSTRUMENT DIAGRAM BILGE, SLUDGE, BALLAST (AFT) AND GENERAL SERVICE SEAWATER SYSTEM	Noted and finalizing of interface signal will be fixed as part of vendor data during execution stage.		Closed	Closed
50	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	4.3.3.10 For analog signals interchanged between PACKAGE UCP and CSS, galvanic isolators shall be foreseen. The isolator shall be located in the corresponding CSS controller.	C	No analog signals to CSS in our package. Consequently, we undertand that all signal shall be hardwired by voltage free dry contacts.	Detegasa understading is correct.		Closed	Closed
51	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	5.1.2 All instruments, junction box, panels, materials and equipment proper to be used in hazardous areas, shall have conformity certificates complying with PORTARIA INMETRO Nº 179, DE 18/maio/2010, and its annexes, changed by PORTARIA INMETRO N º 89, de 23/fereveiro/2012, and shall be approved by Classification Society.	C	We understand that our package is located indoors and in safe area, consequently this requirement is not applicable. Our instruments will comply with IEC regulations	Detegasa understading is correct.		Closed	Closed
52	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	5.5 Control Valves	C	There are not control valves in our package	Noted and Closed.		Closed	Closed
53	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	5.6 Shutdown Valves (SDV)	C	There are not shutdown valves in our package	Noted and Closed.		Closed	Closed
54	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	5.7 Blowdown Valves (BDV)	C	There are not blowdown valves in our package	Noted and Closed.		Closed	Closed

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55	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	5.15 Asset Management System (AMS)	C	We understand that our package is P1 type, consequently AMS is not applicable	Detegasa understading is correct.		Closed	Closed
56	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	6.2 UCP Human Machine Interface (HMI) 6.2.1 For P1 type Packages, the operator interface shall have at least: • Push-buttons and leds signaling; • Small electronic graphic units and keyboard; • Video screen and keyboard.	D	The UCP HMI of our package only have Push-buttoms and led signaling. There are not screens in our scope of supply. (16-July) Noted and consider to add HMI with SIEMENS model. Cost will be added in final proposal.	HHI understands that current considered Push-buttons and singals are not HMI. HMI shall be included in Detegasa scope of supply. But HHI accept to exclude keyboard in the design for this package. HMI shall be selected from below list. 1) GE 2) SIEMENS 3) Rockwell		Closed	Closed
57	I-ET-3010.00-1200-800-P4X-002 AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS	7 TEST REQUIREMENTS	C	Tests will be conducted according to the FAT protocol agreed by both parties	Noted and Closed.		Closed	Closed
58	I-ET-3010.1M-1200-800-P4X-014 AUTOMATION INTERFACE OF PACKAGE UNITS	4.1 SIGNALS The minimal interface signals between PACKAGE UNITs and CSS are according annex spreadsheet, below. Other signals may be necessary during Detail Engineering Design.	C	Signals to be according to I-DE-3010.1M-5330-944-P4X-001 PIPING AND INSTRUMENT DIAGRAM BILGE, SLUDGE, BALLAST (AFT) AND GENERAL SERVICE SEAWATER SYSTEM	Noted and finalizing of interface signal will be fixed as part of vendor data during execution stage.		Closed	Closed
59	I-ET-3010.1M-1200-800-P4X-014 AUTOMATION INTERFACE OF PACKAGE UNITS	6 FIELD INSTRUMENTS INTERFACE	C	All skid instruments shall be connected to the main control panel in Skid. The interface junction box is not applicable for our package.	Acceptable to HHI. This package has small q'ty of instruments and skid size also samll.	Noted.	Closed	**Closed**
60	I-ET-3010.1M-1200-800-P4X-014 AUTOMATION INTERFACE OF PACKAGE UNITS	7 NETWORK INTERFACE 7.1 GENERAL	C	We undertand that all signal shall be hardwired by voltage free dry contacts. Consequently, no LAN communication in our package	Noted and Closed.		Closed	Closed
61	I-ET-3010.1M-1200-200-P4X-002 PIPING SPECIFICATION FOR HULL	5. PIPING SPECIFICATION X SERVICE	D	For the piping of our package we propose code piping and valves made of 316L, instead of B14H. Those components will be according to our standards described in the section 5.1 and 5.3 of our technical proposal You can also see more info in section 11 of our technical proposal, where you can find the data sheet of the main components	Acceptable for 316L material for both piping and valves without application of Piping Specification, B14H. 316L is suitable for bilge water service and proposed application is bidder's proven standard application.	Noted.	Closed	**Closed**
62	APPENDIX 13 APPROVED VENDOR LIST		D	We have submitted the information about the brands of the main components for your approval. One important remark is the Bilge alarm or Oil analyser. We propose our standard model, DOM-107, designed and manufactured by our company, certified according to MEPC 107 (49) by ABS on behalf of European Administration.	Acceptable. DOM-107 is bidder's own design and certified according to IMO MEPC 107 (49) as attached. T:₩03.DESIGN₩ hanical₩12_TBE R	Noted.	Closed	**Closed**

APPENDIX #9

APPROVED VENDOR LIST

APPENDIX #10

E&I TECHNICAL COMPLIANCE CHECK SHEET

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
1. Document				**P-78**	**P-78**					
1	Certification	1	Material certification to be provided.	Packager shall provide complete documentation according to Brazilian Labour and Employment Ministry NR-10. This documentation shall be issued and signed by legally qualified personnel. The documentation shall include the items listed below and all items foreseen in NR-10: a) Conformity Technical Appraisal regarding applicable standards for all internal equipment and installations, including tests, inspections, and measurement reports; b) Project Technical Report, including internal protection, installation, safety grounding, etc.; c) Manuals with safety and health procedures and instructions regarding electrical equipment and installations; d) List with electrical and electronic equipment installed in hazardous areas, including for each equipment, the classification Ex, nameplate data, number of Conformity Certificate and name of entity that issued the certificate; e) Conformity Certificate for all electrical and electronic equipment proper for installation in hazardous areas, according to INMETRO Portaria nº 179, May 18th 2010 and Portaria nº 89, Feb 23rd 2012.	I-ET-3010.00-5140-700-P4X-003	YES		Noted	ONLY FOR APPLICABLE REQUIREMENTS TO OUR PACKAGE	Closed
		2	Explosion proof certification to be provided.	The application of electrical equipment, instruments and accessories on hazardous areas shall follow the requirements of IEC series 60079, 61892-6 and 61892-7, as well as have their conformity certificates according to INMETRO Portaria 179, May 18th 2010 and INMETRO Portaria 89, Feb 23rd 2012 and approved by Classification Society. Electrical equipment and accessories installed in external safe or hazardous areas, which shall be kept operating during emergency shutdown ESD-3P or ESD-3T shall be certified for installation in hazardous areas Zone 2 Group IIA temperature T3 (IEC 61892-7). All hazardous areas equipment type certificates shall be delivered together with respective equipment. All instruments that operate with electric power (transmitters, analyzers, switches, valve positioners and solenoids, fire detectors, gas detectors, sight gas detector beam emitters, manual call points - MCPs etc.) shall be certified to operate in zone 2. The type of protection to be used is Ex dc IIA T3 in wide area and zone 2 and Ex db IIA T3 when installed in zone 1. The cases where another type of protection is acceptable are: a) Manual call points, MCP; b) Instruments installed in zone 0 (zero) shall have Ex ia IIA T3 certification. Other types of protection with EPL Ga are not accepted. c) For analyzers, where the only available technology is pressurized type, an air reservoir, adjacent and exclusive to the analyzer, shall be provided. It shall allow a minimum autonomy equal to the operating time of the CSS equipment defined in DR-ENGP-M-I-1.3 – SAFETY ENGINEERING If the protection type used is for zone 1, Ex pX, all circuits (DC power, AC power, discrete signal, analog signal etc.) shall pass through the disconnecting device. [INTERDISCIPLINARY INTERFACE] [PRO, MEC, HSE] d) Protection type increased safety, Ex e, for junction boxes. NOTE: Any other type of protection shall be submitted to PETROBRAS for evaluation.	I-ET-3010.00-5140-700-P4X-003 I-ET-3010.00-1200-800-P4X-013		NO	Noted	IT IS NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA	Closed
		3	Instrument calibration certification to be provided.	Instrument calibration certification to be provided. 1) Valid period shall meet the whole project schedule(Project handover) plus min 2 years. 2) Original of the calibration sheet shall be delivered before delivery.	N/A (HHI Standard)	YES		Noted		Closed
		4	WPS/PQR to be provided.	Weld certificates shall be provided attesting procedures, qualification of the performer and penetrating liquid testing of the weld in accordance with the technical specification document of the project entitled PIPING SPECIFICATION. This is general requirement for all instrument which needs welding.	I-ET-3010.00-1200-800-P4X-013	YES		Noted	NDE According to our standards	Closed
		5	SIL certification to be provided Reliability and Availability	SIL certification to be provided for Safety devices.	N/A (HHI Standard)	YES		Noted	ONLY FOR A SAFETY RELAY IN THE MAIN CONTROL PANEL	Closed
2	Instrument input data	1	IO list and index -IO list/ Index to be provided via project format.	All document and drawings shall be in PDF format and provided with its respective native files.	N/A (HHI Standard)	YES		Noted		Closed
		2	Mapping list / Serial IO list -Mapping list to be provided via project format.	All document and drawings shall be in PDF format and provided with its respective native files.	N/A (HHI Standard)	YES		Noted		Closed
		3	Cable schedule -Cable schedule to be provided via project format.	All document and drawings shall be in PDF format and provided with its respective native files.	N/A (HHI Standard)	YES		Noted		Closed
		4	Wiring/termination DWG -Cable termination to be provided.	All document and drawings shall be in PDF format and provided with its respective native files.	N/A (HHI Standard)	YES		Noted		Closed
		5	Graphic layout -Graphic layout to be provided if applicable.	To be provided as per requirement	N/A (HHI Standard)	YES		Noted		Closed
		6	Cause and Effect chart /diagram -ESD/FGS cause and effect chart to be provided as per project requirement	To be provided as per requirement. Specially, even though fire gas system is not a part of PKG system, FGS C&E shall be provided when it shall be protected by fire fighting system such as CO2 or watermist.	I-ET-3010.00-1200-800-P4X-012		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		7	System architectures -System architectures to be provided.	To be provided as per requirement	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		8	Functional logic diagram -Functional Logic Diagrams, Function Descriptions and Control Narratives to be provided.	To be provided as per requirement	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		9	Instrument data sheet -Instrument data sheet to be provided via project format.	Data sheet to be provided as per ISA format All document and drawings shall be in PDF format and provided with its respective native files.	ISA	YES		Noted		Closed
		10	Instrument loop DWG -Instrument loop DWG to be provided. -Instrument loop index shall be provided as per project requirement.	All document and drawings shall be in PDF format and provided with its respective native files.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		1	Load list -Power consumption calculation shall be provided as per project requirement.	Electrical Loads List The following information shall be provided, as a minimum: Project Template will be provided later. - Load Identification (TAG); - Load Identification (Description); - Rated voltage (V); - Rated power (kW for motor and kVA for other loads); - Rated current (A); - Starting time (s) and starting current (p.u.); - Speed (rpm); - Load required power (bkW); - Motor plus load consumptions (kVA); - Service factor; - Efficiency (%) at 25, 50, 75 and 100% of load; - Power factor at 25, 50, 75 and 100% of load; - Maximum locked-rotor time (s).	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		2	Heat dissipation calculation sheet - Heat dissipation data shall be provided.	Heat dissipation calculation at maximum and normal operational conditions (W) shall be provided.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		3	Motor data sheet -Requirement of maximum number of start of motor/ restart inhibit time at cold and hot condition shall be clearly indicated on motor datasheet by vendor, if required. Motor data sheet including minimum below data in case MV motor shall be provided for power system calculation. - Motor thermal over load curve (Hot/Cold) for Stator at 100% and 80% voltage - Motor locked rotor withstand curve (Hot/Cold) for Stator at 100% and 80% voltage - Motor Acceleration curve (current vs time) at 100% and 80% voltage - Motor Acceleration thermal limit curve (Refer attached example as per IEEE C37.96-2012) - Motor Load Torque curve at 100% and 80% voltage - Motor and Load Inertia	As per I-LI-3010.00-5140-700-P4X-001 and I-ET-3010.00-5140-712-P4X-001(or I-ET-3010.00-5140-712-P4X-002)	I-LI-3010.00-5140-700-P4X-001 I-ET-3010.00-5140-712-P4X-001 (for LV Motor) I-ET-3010.00-5140-712-P4X-002 (for MV Motor)	YES		Noted	WE WILL PROVIDE THE INFORMATION AVAILABLE FROM THE SUB-SUPPLIER	Closed
		4	THD harmonic spectrum data report -THD value shall comply with project requirement - To be provided.	The voltage distortions and the notch depth shall remain within the following limits, as stated in IEEE 519: a) Total Harmonic Distortion (THD) – 5%; b) Individual Harmonic Distortion – 3%; c) Notch depth – 20%.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		5	Schematic diagram / Single line diagram -To be provided for electric equipment	Schematic / Single line diagram diagram to be submitted for all electric equipment.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
3	Elec. input data	6	Equipment layout - Component, equipment location with tag - Gland plate with dimension and cable - Filed cable interface point for MCT and Tie-in cable tray. - Earth connection point - Recommended minimum clearance distance (side, front, rear and top) of equipment which is reqiured for proper operation, test and maintenance shall be indicated on general arrangement drawing. - The door opening direction, Front & Rear direction and dimension for panels shall be indicated on general arrangement drawing. - Weight data to be provided.	Equipment layout shall have the plans and its sections & details which depict, in the vertical plane, all sections and views indicated on the Equipment Layout Plan, including complete identification and dimensions of equipment. - Component, equipment location with tag - Gland plate with dimension and cable - Filed cable interface point for MCT and Tie-in cable tray. - Earth connection point - Recommended minimum clearance distance (side, front, rear and top) of equipment which is reqiured for proper operation, test and maintenance shall be indicated on general arrangement drawing. - The door opening direction, Front & Rear direction and dimension for panels shall be indicated on general arrangement drawing. - Weight data to be provided.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		7	Connection diagram -Connection DWG to be provided for interface cable	Packager shall present interface cables list and interconnection diagrams, clearly indicating the cables supplier (Packager, or Module Supplier, or Bidder).	I-ET-3010.00-5140-700-P4X-003	YES		Noted		Closed
		8	Electrical diagram and schematic dwg -To include BOM with original manufacturer part number.	Electrical diagram and schematic dwg shall indicate BOM with original manufacturer part number.	N/A (HHI Standard)	YES		Noted		Closed
		9	Serial mapping address data for electric system interface -It shall be submitted if required.	If any communication between eletric system and PKG, serial mapping list to be provided.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		10	The detail list for all ship loose items shall be provided. The installation details like hook up and layout drawings for these items to be also included along with this list.	The detail list for all ship loose items shall be provided. The installation details like hook up and layout drawings for these items to be also included along with this list.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		11	3D modeling for entire package shall be done by the vendor and native files shall be provided to HHI. This 3D model shall include the layout for E&I items like panel, instruments, JB and trays within the package. All the clearances and dimensions shall be clear in the model.	3D modeling for entire package shall be done by the vendor and native files shall be provided to HHI. This 3D model shall include the layout for E&I items like panel, instruments, JB and trays within the package. All the clearances and dimensions shall be clear in the model.	N/A (HHI Standard)	YES		Noted	3D MODEL OF THE PACKAGE IN STEP FORMAT	Closed
		12	The MCT cable assignment -MCT cable assigment to be provided including yard cables.	MCT cable assigment to be provided including yard cables.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		13	Certificate of Conformity for Ex Electrical Apparatus - To be provided.	2.23. Certificate of Conformity for Ex Electrical Apparatus 2.23.1. The Certificate of Conformity related to a given Ex electrical apparatus shall be issued in a dedicated design document, coded in accordance with the requirements of NI-1710 and made available through a searchable PDF file. 2.23.2. A given Certificate of Conformity may be assigned to identical Ex electrical apparatus from the same subsystem/contractor. In this case, only one design document need to be issued.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA	Closed
		14	Ex Electrical Apparatus List	Packager shall furnish a table with the data presented in Table below for Ex Electrical Apparatus. Equipment Tag. No. / Component identification Equipment Description / Characteristics Unit Location / Module Equipment Manufacturer Catalogue / Type Number IP Code Type of Ex Protection Gas Group EPL (Equipment Protection Level) Temperature Class Approval Body Certificate Type Certificate Number Zone of Use Accessories Certificate of Conformity	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA	Closed
7	Reliability data for SIL	1	PKG vendor shall submit the reliability data for SIL verification via the project format.	To be provided.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
8	Cyber security work sheet	1	Cyber security work sheet shall be submitted via project format if applicable.	*DETAILED ENGINEERING DESIGN DOCUMENTS specifed on I-MD-3010.1M-5520-800-P4X-003 _section 9 shall be supplied for review/approval. 9.2 NETWORK ONE-LINE DRAWING 9.3 DETAILED NETWORK INTERCONNECTION DIAGRAM 9.4 IP LIST 9.5 ASSET LIST 9.6 Other documents A cybersecurity robustness testing shall also be performed. This test includes vulnerability analysis, penetration testing, and submitting the system to unusual network loads to ensure that it maintains its correct behavior.	I-MD-3010.1M-5520-800-P4X-003		NO	Not acceptable. eventhough no network connection with CSS, still remain other cybersecurity points such as hardening, password.	NOT APPLICABLE FOR OUR PACKAGE 14/07/2021: The bilge alarm in protected by a passwword, in order to avoid any changes in the configuration. PLC is also protected by a password. HHI20210716] Noted	Closed
9	E&I Tie in document	1	1. E&I tie in document shall be provided as per E&I Tie document guidance. 2) E&I tie in document shall indicate the all information related to HHI yard cable interface. 3. The information related cable installed by vendor shall exclude from this document. 4. If any discrepancy between this doc and others, E&I tie in document shall be pripority for HHI yard interface finalization. 5. The contents of table for this document shall be - Tie in Cable block diagram [Project format] - Tin in Cable schedule [Project format] - Tie in E&I layout - Tie in panel layout - Tie in installation detail for ship loose items	Refer to E&I tie in document guidance.	E&I tie in document guidance.	YES		Noted	WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
		2	Tie in Cable block diagram - It shall include all cables which are installead by HHI at yard. - Cable installed by vendor within skid shall excluded in this document. - The project format shall be used. - All required data on format shall be filled out.	This document shall be submitted by employing project tie in document template / format. This template requests to fill out the following data and it is mandatory to fill out all required data as per project requirement/numbering procedure. A. Cable tag numbering B. Cable type C. IO tag / type for instrument D. Load description for electric E. Power rating / consumption F. From / To equipment tag number/description G. Cable entry size	E&I tie in document guidance.	YES		Noted	WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
		3	Tie in Cable schedule/ Tubing schedule - It shall include all cable/tubing list which is installead by HHI at yard. - Cable/tubing list installed by vendor within skid shall excluded in this document. - Cable scheudle shall be generated based on tie in cable block diagram.	Tie in cable schedule shall be submitted after receiving 1st comment on cable block diagram from Bidder. It is Packager's responsibility not only to incorporate all comment /data on cable block diagram to this document but also to exclude cable list installed by VENDOR from this DWG. In addition, it is required to submit it via tie in cable schedule format.	E&I tie in document guidance.			Noted	WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
		4	Tie in Cable wiring/termination - It shall incldue all wiring/termination under HHI scope of work. -Termination information under vendor scope shall exclude in this document.	Tie in termination DWG shall include following information as minimum. A. Cable tag numbering B. Cable type C. IO tag / type for instrument D. Power load / Power signal description for Electric E. From / To equipment tag number F. Terminal block number / Terminal block strip number G. Yard cable screen/shield termination H. Diode / resistance for line monitoring if applicable	E&I tie in document guidance.			Noted	WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
		5	Tie in E&I layout -It shall indicate all cable tray/tubing tray that HHI yard cable/tubing to be laid. -It shall indicate all equipment that HHI yard cable to be terminated. -Cable benidng radius shall be visualized based on mximum yard cable OD be laid on each cable tray. -List for yard cable laid on each cable tray shall be shown. -Tray size information to be shown	This document is key data to determine yard cable routing. Following information shall be shown on package skid layout. A. The location of E&I equipment. B. Note: E&I equipment tag number shall be matched with the data on tie in cable block diagram C. Cable tray with dimensional data between skid edge tie in location and E&I equipment. D. Note: Only cable tray on which laid yard cable need to be shown. E. Skid edge tie in location / points F. Cable list with assumed in skid length / tubing list laid on each cable tray G. Cable tray spare space data after installing yard cables.	E&I tie in document guidance.			Noted	WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
		6	Tie in panel layout -Termination point for each yard cable to be shown in this DWG. - Foot print / fixing hole size / panel dimension to be indicated. - Door opening to be shown. - Gland plate hole arrangement layout	It is critical document for the control room arrangement. It shall show the information of panel foot print with fixing holes detail, door opening, gland plate holes arrangement and termination points for each yard cables.	E&I tie in document guidance.			Noted	WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
		7	Tie in hook up / Installation detail -It shall include the installation detail for all field instrument/detectors/electric devices that are supplied by vendor but are installed by HHI.	Tie in E&I installation detail shall indicate the mounting method for all loosed E&I equipment which is installed by CONTRACTOR(Bidder) at yard including required bill of material. . It is crucial to indicate the mounting holes size on mounting bracket/plate as well as recommended mounting bolt sizes with torque value.	E&I tie in document guidance.		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
10	Project Document List	1	Documentation required in I-ET-3010.00-5140-700-P4X-003 clause 9 shall be submitted for approval.	Documentation required in I-ET-3010.00-5140-700-P4X-003 clause 9 shall be submitted for approval as a minimum.	I-ET-3010.00-5140-700-P4X-003				ACCORDING TO VDRL AGREED BY BOTH PARTIES	Closed
2. General requirement										
1	Painting	1	All E&I devices shall be painted/coated as per project painting requirement.	1. Electrical Paint system for external coating of all electrical equipment, material and supports shall be proper for offshore installation and pre-qualified according to I-ET-3010.00-1200-956-P4X-002 - GENERAL PAINTING. Painting system for internal coating of all electrical equipment and material shall be proper for offshore installation according to manufacturer requirements, when not specified by PETROBRAS. Unless otherwise stated in equipment documentation, the last coat colour for electrical equipment shall be Light Green Munsell 5G8/4. Inner components mounting plate, internal doors faces shall be Safety Orange Munsell 2.5YR6/14. Note: Colours shall comply with NR-37. For carbon steel parts of lighting fixtures and junction boxes, the last coat colour shall be White Munsell N9.5. For stainless steel lighting fixtures no coat colour is required. For FRP materials, no coat colour is required, unless otherwise required by a standard, regulation or PETROBRAS. For fire fighting associated equipment, the last coat colour shall be Safety Red Munsell 5R4/14. Stainless steel surfaces shall be painted if exposed do saline atmosphere and temperatures above the limits indicated, as defined in I-ET-3010.00-1200-956-P4X-002 - GENERAL PAINTING: ● 316L - with service temperature over 50°C shall be coated; ● Duplex SS - service temperature over 80°C shall be coated; ● Superduplex SS - service temperature over 90°C shall be coated. 2. Instrument Instrument shall follow the piping or equipment color.(DR-ENGP-I-1.15) Instrument panel color shall be Light cream. All instrument shall be painted as per I-ET-3010.00-1200-956-P4X-002.	1. I-ET-3010.00-5140-700-P4X-002 2. I-ET-3010.00-1200-956-P4X-002 DR-ENGP-I-1.15	YES		Noted		Closed
2	VCI Coating	1	VCI coating shall be applied as per project requirement.	Refer to AUTOMATION PANELS, I-ET-3010.00-5520-888-P4X-001 . In order to prevent corrosion, VCI shall be used adequately, where applicable, as part of preparation for shipment and storage instead of desiccants such as silica gel. The latter shall be used only in cases where VCI is not applicable. Both VCI and desiccants shall not be used together for protecting the same compartment.	I-ET-3010.00-5520-888-P4X-001		NO	vendor to provide	NOT APPLICABLE FOR OUR PACKAGE 14/07/2021: noted, to be discussed in the meeting HHI20210716] Noted	Closed
3	Cable glands	1	Cable gland type shall be selected as per project requirement.	Cable gland shall follow IEC 62444. Cable glands for installation at non threaded holes of removable plates or steel sheet enclosures shall have cylindrical thread with locknut. Unless otherwise stated, all threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B1.20.1. Cylindrical threaded joints shall be acceptable for lighting fixtures for fluorescent lamps, for control boxes for push-buttons and signalling, for floodlights and for power socket-outlets. Cylindrical threaded joints for other applications shall be submitted to PETROBRAS approval. For equipment installed in hazardous areas, the threads shall comply with the requirements of IEC 60079-0. Cable glands for armoured cables shall be metal type and the metallic pair shall not create an electrolytic corrosion in case of dissimilar metallic material. Stainless steel and copper armour shall use Stainless Steel Cable glands. Barrier type Cable gland shall be supplied for Exd devices when IEC-60079 requests it.	I-ET-3010.00-5140-700-P4X-002	YES		Noted		Closed
		2	Material shall be selected as per project requirement.	Cable gland materials shall be selected in order to avoid electrolytic corrosion caused by contact of dissimilar materials. Cable glands material shall follow the requirements of Table 18: *Table 18 - Cable Gland Material* Enclosure Material: Stainless Steel — Cable Gland Material: Stainless Steel AISI-316L Enclosure Material: Cast Iron — Cable Gland Material: Aluminium Enclosure Material: Aluminium — Cable Gland Material: Aluminium Enclosure Material: FRP (Fiber Reinforced Plastic) — Cable Gland Material: Aluminium; Nylon (1); Naval Bronze (2); Stainless Steel AISI-316L Enclosure Material: Stainless Steel, Naval Bronze — Cable Gland Material: Naval Bronze (2) Notes: 1) Nylon cable glands shall be accepted only up to maximum size 1", with metallic plate (made of Stainless Steel AISI-316L, Aluminium or Naval Bronze(2)) for grounding, with internal locknut and if they are certified as Ex e or Ex n. 2) Naval Bronze shall be classic marine, high-strength and corrosion-resistant alloy C-462 or C-464.	I-ET-3010.00-5140-700-P4X-002	YES		Noted		Closed
		3	Ex certification shall be provided as per project requirement	Conformity Certificate for all electrical and electronic equipment proper for installation in hazardous areas, according to INMETRO Portaria nº 179, May 18th 2010 and Portaria nº 89, Feb 23rd 2012.	I-ET-3010.00-5140-700-P4X-002		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA	Closed
		4	IP washers shall be installed as per cable gland manufacturers' recommendation and project requirement.Fabric or rubber type IP washer is prohibited.	To be provided.	N/A (HHI Standard)	YES		Noted		Closed
		5	Serrated washers shall be provided as per project requirement.	To be provided.	N/A (HHI Standard)	YES		Noted		Closed
		6	Cable gland for HHI yard cable interface shall be provided by PKG vendor. Cable detail information to be provided by HHI during detailed engineering.	To be provided.	N/A (HHI Standard)	YES		Noted		Closed
		7	All spare cable entries shall be plugged with Ex certified cable plug.	To be provided.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA	Closed
		8	Cable gland shroud shall be provided as per project requirement. Long type shroud to be selected in case cable gland fitting (reducer, adoptor) is applied to cover all part of gland and accessory	N/A	N/A			Noted	N/A	Closed
		9	Cable Gland Code List	Enforce that CONTRACTOR's sub-suppliers and PACKAGED UNIT's SUPPLIERS also create such codes and emit a similar document.	I-ET-3010.00-1200-800-P4X-010 Refer to Form-10			Bidder to check	14/07/2021: NOTED	Closed
4	INST. Hook up	1	Hook-up drawing to be provided by PKG vendor.	To be provided by VENDOR	API RP 551		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		2	Tubing/tube fitting material, size, type, selected vendor to be in line with project requirement.	Tubing shall be made of ASTM A 269 Gr. TP 316L stainless steel with a minimum Molybdenum content of 2.5%. The material of the associated connections and valves shall be made of AISI 316 stainless steel, not requiring a minimum Molybdenum content of 2.5%. Note: In cases where the atmosphere (microclimate) is unsuitable for tubing in the material described in item I-ET-3010.00-1200-800-P4X-013 _ 19.2.1, the use of tubing, fittings and associated valves in the 6Mo stainless steel material (ASTM A269 S31254) may be considered. Each batch of tubing shall have a Positive Material Identification certificate in accordance with API RP 578. Connections shall have certificate of conformity with ASTM F1387. The certificates shall comprise at least the Standard Qualification Tests specified on 19.1.6.2 _I-ET-3010.00-1200-800-P4X-013. (A) and the following Supplementary Tests (S): The wall diameters and thicknesses which may be used are presented in Table 19.2-I_I-ET-3010.00-1200-800-P4X-013 Tube fitting shall be supplied by manufacturer who will be determined by CONTRACTOR during Detailed Engineering. Tentative approved supplier is SWAGELOK, PARKER, HY-LOK. So, the highset cost to be considered for bidding.	I-ET-3010.00-1200-800-P4X-013	YES		Noted	ACCORDING TO OUR STANDARDS	Closed
		3	All entry size and type for tubing connection shall be clearly specified on hook-up DWG.	All entry size and type for tubing connection shall be clearly specified on hook-up DWG.	N/A (HHI Standard)	YES		Noted		Closed
		4	If yard tubing to be connected to PKG skid or devices by HHI, it shall be included on hook-up DWG.	If yard tubing to be connected to PKG skid or devices by HHI, it shall be included on hook-up DWG.	N/A (HHI Standard)	YES		Noted		Closed
		5	Instrument manifold and isolation valve shall be provided as per project requirement.	Instrument manifold and isolation valve shall be provided as per project requirement.	I-ET-3010.00-1200-800-P4X-013	YES		Noted		Closed
		6	Vent/drain line with cap for instrument venting shall be provided as per project requirement	Vent/drain line with cap for instrument venting shall be provided as per project requirement	N/A (HHI Standard)				N/A	Closed
		7	The slope for tubing line shall be inline with project requirement as well as API PR 551.	When measuring liquids, impulse lines shall have a downward slope of 1:10 from the impulse tap up to the manifold block of the instrument. When measuring gas, the impulse lines shall have a rising slope of 1:10 from the impulse tap up to the manifold block of the instrument.	I-ET-3010.00-1200-800-P4X-013	YES		Noted	ACCORDING TO OUR STANDARD DESIGN FOR PROPER OPERATION	Closed
		8	Open end for tubing line shall be provided with cap, bug screen, flame arrestor and etc as per project requirement.	Open end for tubing line shall be provided with cap(calibration connection/drain), bug screen(safety gas venting), flame arrestor(flamable gas venting).	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed

Technical Compliance Sheet

PROJECT :	P-78
RFQ No.:	HQ-54
EQUIPMENT NAME :	OILY WATER SEPARATOR
INSPECTED ON :	
INSPECTED BY :	

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
		9	Tubing clamp/ tray shall be provided as per project requirement.	When routing more than one tubing in parallel, appropriate insulation shall be provided to avoid contact between them. To avoid corrosion points, this insulation cannot create regions that can accumulate water Support systems that can create crevice corrosion points shall not be used, such as strip-type or clamp-type supports. Tubing support systems shall be designed to minimize the points of contact between the tubing and the support to avoid crevice corrosion points. The metal-to-metal contact shall be eliminated using insulating material for the parts in contact with the tubing. Parallel tubing routing shall be done in covered trays and provision shall be made to avoid water accumulation.	I-ET-3010.00-1200-800-P4X-013	YES		Noted		Closed
		10	Sealant material shall meet project requirement.	All thread connection shall be sealed with sealant material.	N/A (HHI Standard)	YES		Noted		Closed
		11	Leak test report or mechanical completion check sheet for instrument hook up shall be provided.	Leak test report and mechanical completion check sheet shall be provided.	N/A (HHI Standard)	YES		Noted		Closed
		12	Tubing Support	The length of unsupported tubing to final destination (such as control valve or transmitter) for single tubes shall not exceed 0.5m.	N/A (HHI Standard)	YES		Noted		Closed
5	Labeling on Elec. Equipment	1	All E&I devices such as eletrical panels, juction boxs, field instruments and etc shall be provided with rigid tag plate as per project requirement	Identification nameplates for equipment with specific Technical Specification (e.g. induction motors, generators, MCCs and switchgears) shall follow the requirements of that specification. All other equipment shall have 02 (two) identification nameplates. The first plate shall have the equipment tag number, in black acrylic engraved with white letter for equipment installed indoors and in stainless steel AISI-316 for equipment installed outdoors. The other plate shall be in stainless steel AISI-316 with the following information, when applicable, in Portuguese language: Petróleo Brasileiro S.A. – PETROBRAS; Fornecedor (vendor name); Número de Série (equipment serial number); Ano de fabricação (year of manufacture); Número do pedido de compra (purchase order number); Other equipment specific data; Fabricante (manufacturer name); Tipo (type); Padrão de desempenho (performance standard); Valores nominais para: potência aparente de saída, tensões, frequência, correntes (rated values for: output apparent power, voltage(s), frequency, currents); Classificação térmica da isolação (thermal classification of insulation); Grau de proteção IP do invólucro e caixa de terminais (IP code of enclosure and termination box); Grupo de ligação de bobinas (vector group of windings); Máxima temperatura admissível para o meio refrigerante (maximum permissible cooling medium temperature); Valores das impedâncias de curto-circuito (short-circuit impedances values); Tipo do meio refrigerante (type of cooling medium); Massa total (total mass). Cable trays and channels shall have body identification. AISI 316 stainless steel identification plate with at least, the information specified on I-ET-3010.00-1200-800-P4X-013. Besides the identification plate described in item 6.1.1, a stainless steel plate with a minimum size of 50 mm x 20 mm x 0.5 mm (Width x Height x Thickness) shall be provided with instrument respective TAG engraved as per I-ET-3010.00-1200-800-P4X-013.	I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-1200-800-P4X-013	YES		Noted		Closed
		2	All E&I devices shall be provided with location tag plate as per project requirement	For instrument at field [To be considered for bidding] UV resistant weather proof laminated engraved traffolyte location tag plate to be supplied on instrument support so that when the instrument is replaced, the nameplates shall remain on the support.	N/A (HHI Standard)		NO	Bidder to comply on this requirement except UV resistant weather proof requirement.	NOT APPLICABLE FOR OUR PACKAGE 14/07/2021: there are not field instruments in our package. However, we will follow your requirement for the instruments within our package HHI 20210716] Noted	Closed
		3	Tag plates for electrical components such as power supply, IO card, terminal strips, relay, Earthing bus bar and etc inside panel / junction box shall be provided as per project requirement. Fuses shall be labeled with their current ratings. If the color coded wiring to fuses connected to the main circuit is not visible from the front of the fuses, the fuses shall also be labeled with their phase color.	Laminated engraved traffolyte name plate to be provided in order to identify each component. Nameplates shall be provided for all devices, accessories, equipment, switches, etc., installed on the front plate or inside the panels, adjacent to each component, so that when the components is replaced, the nameplates shall remain in the panel. Identification shall be in Brazilian Portuguese. The grounding bars shall be clearly and permanently identified with nameplates in Brazilian Portuguese. The nameplates shall be made of black acrylic material and white lettering. The identifications to be engraved shall be "ATERRAMENTO DE SINAL" for instrument grounding, "ATERRAMENTO DE SEGURANÇA" for safety grounding and "ATERRAMENTO DE SEGURANÇA INTRÍNSECA" for intrinsic safety grounding. Identification of panels and internal equipment shall follow Brazilian standard regulation NR-12 - SEGURANÇA NO TRABALHO EM MÁQUINAS E EQUIPAMENTOS.	I-ET-3010.00-5520-888-P4X-001	YES		Noted		Closed
		4	Tubing tag plate shall be provided as per project requirement	Tubing tag plate shall be provided.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
		5	Wire marker shall be provided as per wire marker numbering procedure.	Wiring shall be identified by heat-shrinkable ferrules or plastic rings, at each end, with numbers allocated by the Manufacturer. All wiring drawings shall clearly identify the appropriate reference numbers. For bidding, highest cost between heat-shrinkable ferrules and plastic rings shall be considered. During detailed engineering, type of wire marker shall be approved by CONTRACTOR. All panel wiring shall be identified at both ends by plastic rings with the terminal block number. Wire numbering procedure to be confirmed during detailed engineering.	I-ET-3010.00-1200-800-P4X-002	YES		Noted		Closed
		6	Multi cable transit(MCT) shall be provided with tag plate.	1) Tag Plate : Each side of frame 2) Material : AISI-316L stainless steel 3) Tagging procedure as per clause 3.10.4 of I-ET-3010.00-5140-700-P4X-001	I-ET-3010.00-5140-700-P4X-001	YES		Noted	IF APPLICABLE	Closed
		7	Cable tag plate shall be provided as per project requirement	All cables shall be clearly identified by means of plastic rings for identification or other method that supports the operation conditions, with the design number, at least at the following points: a) at panels, boxes and equipment entrance; b) at both sides of MCTs and any bulkhead crossing point incomings and outgoings; c) at the ends of conduits feeding single equipment.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		8	All inscriptions on nameplates shall be in the English language and Language required in Project.	Both English language and Brazilian Portuguese language to be provided for applicable items as per project specification.	I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-1200-800-P4X-013	YES		Noted		Closed
		9	Danger and warning labels shall comply with IEC 60417 and ISO 3864.	All electrical equipment, flour mounted, panels, or similar in construction to a panel, regardless of the area where it is installed, shall have the warnings as required by NR-10 and NR-12. Warnings shall follow the standard labels as required in ABNT NBR 13434-2 for electrical panels risk of shock as informed in I-ET-3010.00-5400-947-P4X-002 - SAFETY SIGNALLING. Power Panels shall have warning labels indicating the protective clothing risk category that shall be used for technical intervention. Power Panels shall have warning labels indicating that any technical intervention in the panels shall be executed only by authorized people. All Ex equipment which need to be operated during ESD condition shall be marked for easy identification with "ESD" label, including Ex equipment installed in safe area (outside hazardous area).	I-ET-3010.00-5140-700-P4X-002	YES		Noted		Closed
		10	Nameplates shall be permanently fixed with the screws, rivets, or nuts and bolts. Paper labels fixed with adhesives are not acceptable.	Nameplates shall be permanently fixed with the screws, rivets, or nuts and bolts. Paper labels fixed with adhesives are not acceptable.	N/A (HHI Standard)	YES		Noted		Closed
		11	All items of equipment shall have suitably inscribed identification labels indicating tag number and function. In addition, control stations and similar equipment shall carry normal operating function labels i.e. "stop", "start", "on", "off", etc.	All items of equipment shall have suitably inscribed identification labels indicating tag number and function. For more details, refer to I-ET-3010.00-5140-700-P4X-005 REQUIREMENTS FOR HUMAN ENGINEERING DESIGN FOR ELECTRICAL SYSTEMS OF OFFSHORE UNITS.	I-ET-3010.00-5140-700-P4X-005	YES		Noted		Closed
		12	Withdrawable parts shall be equipped with an engraved identification nameplate.	Withdrawable parts shall be equipped with an engraved identification nameplate.	N/A (HHI Standard)	YES		Noted		Closed
		13	Labels shall not be fixed to components which may be replaced or removed, unless a second label is fitted internally to ensure correct replacement.	Labels shall not be fixed to components which may be replaced or removed, unless a second label is fitted internally to ensure correct replacement.	N/A (HHI Standard)	YES		Noted		Closed
		14	Certified enclosures are not to be drilled or altered in any way for the addition of labels. If labels are found to be missing on certified equipment, then the Supplier shall refer to their sub-Supplier or manufacturer for corrective action.	Certified enclosures are not to be drilled or altered in any way for the addition of labels. If labels are found to be missing on certified equipment, then the Supplier shall refer to their sub-Supplier or manufacturer for corrective action.	N/A (HHI Standard)	YES		Noted		Closed
		1	Earthing bar, boss, plate, stud shall be provided for yard earthing cable. Ex) Junction box, vessel, skid, motor, panel, field devices……	Requirement in section 3.13 of I-ET-3010.00-5140-700-P4X-001 SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS shall be fulfilled. AISI 316L Grounding bar and grounding copper cable inside package shall be provided by Packager. For details, refer to I-DE-3010.00-5140-700-P4X-003 GROUNDING INSTALLATION TYPICAL DETAILS.	I-ET-3010.00-5140-700-P4X-001 I-DE-3010.00-5140-700-P4X-003	YES		Noted		Closed
		2	Equipment door (JB, cabinet and etc) hinges shall be heavy duty type to support the door when opened. Earthing shall be accomplished by means of an external/internal earth stud, which can be connected to the terminal mounting rail or compound mounting plate.	Vertical sections shall have hinged front door. Doors shall be bonded to the panel structure through flexible copper cable. Hinged doors shall have an open position lock device.	I-ET-3010.00-5140-700-P4X-002	YES		Noted	IF APPLICABLE	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
6	Earthing	3	All non-current carrying metallic parts which may develop to electrical potential from any electrical component shall be earthed by a separate insulated earth conductor of suitable cross section.	All non-current carrying metallic parts which may develop to electrical potential from any electrical component shall be earthed by a separate insulated earth conductor of suitable cross section.	I-ET-3010.00-5140-700-P4X-002	YES		Noted		Closed
		4	The following items shall be earthed to the skid frame: - Motor frames and generator frames - Metallic armoring, cable troughs, wireways and tray, equipment racks, and busways. - The armor or cladding on all cables shall be earthed at both ends.	Metallic structures, equipment, cables and accessories shall be grounded in accordance with the requirements of IEC 61892-6 and Classification Society. For FPSO and FSO units, the requirements of IEC 60092-201 and IEC 60092-502 shall also be complied with for grounding of equipment, cables and accessories. In hazardous areas, the requirements of IEC 61892-7 shall be additionally complied with. The grounding installation shall equalize all extraneous conductive metallic parts (conductive part not forming a part of the electrical installation and liable to propagate a potential, including ground potential) at the same potential to ensure that danger to operator is minimized. All grounding connections shall be made with self-locking screws (connectors). Welded terminals or exothermic connections are permitted only between earth bosses and structure/skid.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		5	Metalic pipe flange shall be earthed to ensure electrical continuity to avoid any static current. Pipe earhthing detail shall be included in offical vendor data for approval.	Bonding jumpers shall be installed across metal piping flange connections provided with non-conductive gaskets between them, regardlessof the kind of fluid that flows inside the piping.However, if RTJ (Ring Type Joint) metal gaskets or equivalent are used, the interconnection of flangeswith bonding becomes unnecessary.In either case (need of bonding jumpers or use of RTJ metal gaskets), the whole structure shall be in compliance with the requirements below. Unless otherwise stated in this specification, metallic structures (equipment, vessels, tanks, piping, etc.) for use in fluid handling applications,when directly welded or bolted to the main plataform structure, need no additionalconnections to earth,provided the resistance to ground, as measured with a DC multimeter at any point of the said metallic structures, does not exceed 10 Ω. Measurements shall be carried out at the final stage of the delivery, aiming at keeping the resistance to ground below 106 Ω over the lifetime of the offshore unit, according to IEC/TS 60079-32-1.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		6	Earth bar inside panel and JB shall accommodate all earth connetion with interface cable fully. This earth bar hole q'ty and arrangement dwg shall be included in GA dwg.	Earth bar inside panel and JB shall accommodate all earth connetion with interface cable fully. This earth bar hole q'ty and arrangement dwg shall be included in GA dwg.	N/A (HHI Standard)	YES		Noted		Closed
		7	Earth bar type shall be bus bar type with drilled hole. Special type bus bar is not acceptable.	Earth bar type shall be Stainless steel AISI 316L flat bus bar type with drilled hole and installed directly using weld. Special type bus bar is not acceptable.	N/A (HHI Standard)	YES		Noted	ACCORDING TO OUR STANDARDS	Closed
		8	Cable tray bonding shall be provided as per project requirement.	Cable tray bonding shall be provided as per project requirement, refer to I-DE-3010.00-5140-700-P4X-003_E_GROUNDING INSTALLATION TYPICAL DETAILS.	I-DE-3010.00-5140-700-P4X-003	YES		Noted	ACCORDING TO OUR STANDARDS	Closed
		9	Earthing for unit control panel /Elec. cabinet 1. Earthing shall be provided as per project requirement. 2) Proper earthing bus bar (Qty, bolt size....) shall be provided for yard earthing cables. 3. Earthing bus bar material/size shall be in line with project requirement 4. Maximum 2 earth wire connections are allowd per one earth bus bar bolt. 5. Dedicated earthing bus bar for each earthing system such as IE, ISE and SE shall be provided as per project requirement. 6. Earthing method for instrument cable shield shall be provided as per project requirement.	1. All panels shall be supplied with internal grounding busbars and all panel metallic parts, which are not intended for current conduction, shall be interconnected to the grounding bar, including movable parts. For the details, refer to I-ET-3010.00-5140-700-P4X-002_SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS clause 4.8.3.6 & 4.8.3.21. 2. Proper earthing bus bar (Qty, bolt size....) shall be provided for yard earthing cables which to be confirmed by vendor drawing. 3. Size of earthing bus bar shall be designed to accomodate total number of earthing cables to be connected. 4. Only one earth wire connection is allowd per one earth bus bar bolt. 5. Dedicated earthing bus bar for each earthing system such as IE, ISE and SE shall be provided as per project requirement. 6. Earthing method for instrument cable shield shall be provided as per project requirement	I-ET-3010.00-5140-700-P4X-002	YES		Noted	ACCORDING TO OUR STANDARDS	Closed
		10	VENDOR shall provide Earthing boss/plate which is suitable for M10. Otherwise, required size shall be specifed on relevant document.	VENDOR shall provide Earthing boss/plate which is suitable for M10. Otherwise, required size shall be specifed on relevant document.	N/A (HHI Standard)	YES		Noted		Closed
		11	Tubing Earthing	Hydraulic pipes and tubings shall hasve electrical continuity and shall be grounded according to classification society rules.	Claasification Society	YES		Noted		Closed
7	Cable tray	1	Tray segregation shall be provided as per project requirement	Refer to P-78 CABLE TRAY SEGREGATION TABLE in folder 7. Other project key requirement.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		2	Tray tie in location for yard cables to be provided with PDMS model and tray layout. Special attention for yard cable bending radius to be provided for the design of tie in point .	Tray tie in location for yard cables to be provided with PDMS model and tray layout. Special attention for yard cable bending radius to be provided for the design of tie in point	N/A (HHI Standard)			Noted	3D MODEL OF THE PACKAGE TO BE PROVIDED IN STEP FORMAT	Closed
		3	Cable bending radius shall be reviewed at each cable entry, tie in points and bending routing. .	Cable bending radius shall be reviewed at each cable entry, tie in points and bending routing. .	N/A (HHI Standard)	YES		Noted		Closed
		4	Cable tray for interface cable shall be supplied / installed at the skid edge by PKG vendor. All internal cable route in package between skid edge and package equipment shall be prepared by vendor.	Cable tray for interface cable shall be supplied / installed at the skid edge by PKG vendor. All internal cable route in package between skid edge and package equipment shall be prepared by vendor and identified in vendor drawing.	N/A (HHI Standard)	YES		Noted		Closed
		5	Cable tray type, material, painting shall be inline with project requirement.	1) Cable trays for outdoor use shall be made from heavy duty stainless steel AISI 316L. 2) Cable trays for indoor use shall be either stainless steel AISI-316L or carbon steel painted according to clause 3.6.3 of I-ET-3010.00-5140-700-P4X-002. 3) All screws, nuts and washers shall be made of bichromatized steel or AISI-316L stainless steel. For application in cable trays and cable trays supports only AISI-316L stainless steel shall be used. 4) Requirement for cable tray specified in clause 5.3 of I-ET-3010.00-5140-700-P4X-002 to be fulfilled.	I-ET-3010.00-5140-700-P4X-002	YES		Bidder to comply with I-ET-3010.00-5140-700-P4X-002.	ACCORDING TO OUR STANDARDS 14/07/2021: Cable trays and bolting to be made SS316L	Closed
		6	Minimum of spare capacity shall be provided in cable tray as per project requirement.	Cable trays, channels and ladders shall be designed with a reserve section of at least 20% of the total cross-section.	I-ET-3010.00-5140-700-P4X-001	YES		Noted		Closed
		7	Cable tray overfill to be verified by VENDOR. Cable tray overfilll shall be calculated considering future space requirement as well as yard cables. Yard cables are cables which is installed at HHI yard/site by CONTRACTOR .	Trays, channels and ladders sizing calculation report and list shall be submitted for approval, refer to the details in clause 2.13 of I-ET-3010.00-5140-700-P4X-001 SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		8	Cable tray routing shall be designed to maintain the mechanical continuity. If any deviation, it shall be approved via official documentation. Supports for cable tray shall be provided to ensure to withstand the maximum cable load including future load.	Cable tray routing shall be designed to maintain the mechanical & electrical continuity. If any deviation, it shall be approved via official documentation. Supports for cable tray shall be provided to ensure to withstand the maximum cable load including future load.	N/A (HHI Standard)	YES		Noted		Closed
		9	Cable tray cover shall be provided as per project requirement. And the cover shall withstand wind speed required in project operating environment condition.	1. Cables within trays that transit decks and floors shall be protected from mechanical damage by installation of tray covers between grade level and up to 2m above grade. 2. Cable tray cover same material as the tray shall be provided, where required. i.e from exposure to sun, heat and mechanical damage. 3. Cable trays for voltage 6.6kV and above shall be fully covered reagradless areas.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NOT APPLICABLE, PACKAGE LOCATED INDOORS	Closed
		10	If tubing line is installed between equipment within skid and skid outside, tubing within skid should be completed by vendor and prepared interface point edge of skid. Interface size and connection detail to be specified on hook up DWG.	If tubing line is installed between equipment within skid and skid outside, tubing within skid should be completed by vendor and prepared interface point edge of skid. Interface size and connection detail to be specified on hook up DWG.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		11	Other cable tray requirement.	Cable tray accessories must be such as to permit the execution of the minimum radius of curvature of the biggest cable installed thereon. Spacing and characteristics of the tray supports shall be determined so that they shall not be subjected to permanent distortion (allowing for 120% of installed cables), and also taking into account an 80kg individual point supplementary load applied at any point. Supports holding up the cable trays and conduits shall have a maximum spacing of 3 meters between them. A minimum clearance of 300mm shall be kept between cable trays and heated piping or engine exhaust lines in case of parallel runs and 150mm in case of a cross-over.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
8	Junction box	1	Material, painting and type for junction box shall be in line with project requirement.	Junction boxes enclosures shall be made of stainless steel AISI -316L and "Ex e" or "Ex de". Junction boxes shall be provided with terminal blocks suitable for interconnection. All metallic junction boxes shall comply with painting criteria of clause 3.6.3 of I-ET-3010.00-5140-700-P4X-002.	I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-1200-800-P4X-002		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		2	It shall comply ot IP/Explosion poof requirement.	All outdoor JBs enclosures shall be IP-56 minimum ingress protection degree, according to IEC 60529.	I-ET-3010.1M-1200-800-P4X-005		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		3	Jumper bar for cable shield shall be provided as per project requirement. It shall include jumper bar for yard multi cable interface.	Jumper bar to be provided for shield termination of yard multi cables.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		4	Dedicated junction box for each system/signal (Ex. Analogue, digital, electric..) shall be provided as per project requirement.	Signals to CSS related to PSD/HSD and FGS/HFGS sub-systems shall be segregated in different terminal strips and multicables (where applicable), when installed inside the same junction box. Cables from field instruments shall converge to junction boxes, segregated into 3 (three) types: i. for signals using fire-resistant cables; ii. for signals related to safety interlocks; iii. for signals related to process control. Junction boxes interconnecting fire-resistant cables and multicables shall have passive protection to provide the same fire resistance as the cables. The terminal strips shall also be of a fire-resistant material, such as ceramic. The junction box shall be internally divided into two regions, one for discrete signals and one for analog signals. Intrinsic safety signals shall be conditioned in a separate region inside the junction box, segregated from the other regions by a separating septum and identified as intrinsically safe signals. The terminal strips shall be blue. "Ex e" or "Ex d" certification and certified to work according to the area classification where it is installed. In open areas, the minimum ertification shall be for zone 2 (EPL Gc).	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		5	Top cable entry shall be probibitted. Bottom cable entry for multi cable is preferred. Side cable entry depends on project requirement.	Cables or multicables connecting instruments to the junction box may only be attached to the side or bottom of the junction box. Multicables that connect the junction box to a panel or to another junction box shall be connected to the bottom of the junction box. Entry of cables through the top of the junction box is not allowed.	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		6	Cable entry hole/gland shall be provided according to yard cable information.	Cable entry hole/gland shall be supplied by vendor according to yard cable information to be provided during detailed engineering. JBs shall be ample sized, with minimum of 20% spare (Terminals, cable entries).	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		7	Removable gland plate shall be provided as per project requirement.	All enclosures shall have removable cable gland plates, If MCT is not required on project requirement. For instrument JB, In the lower part of the junction boxes, the passage of cables and multicables shall be done using a removable blind plate. No removable blind plate shall be used at the side entrance.	I-ET-3010.1M-1200-800-P4X-005		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		8	Minimum spare capacity of terminal block shall be provided as per project requirement. Terminal blocks shall be provided to terminate all yard multicores cables.	The terminal strips used inside the junction boxes shall be firmly attached and suitable for use in vibrating conditions to avoid poor contact and consequent spark.	I-ET-3010.00-1200-800-P4X-013		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		9	Door hinge, cable duct shall be provided as per project requirement if applicable.	Door hinge, cable duct shall be provided as per project requirement if applicable.	N/A (HHI Standard)		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		10	Spare cable entries shall be provided as per project requirement. All spare entries shall be plugged with Ex certified cable plug.	JBs shall be ample sized, with minimum of 20% spare (Terminals, cable entries) MCTs shall be supplied with a spare of 10%. All spare entry shall be pluged with certified plug.	I-ET-3010.1M-1200-800-P4X-005		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		11	In case of power Junction box, voltage warning labels shall be installed on door cover.	Refer to clause 3.7 of I-ET-3010.00-5140-700-P4X-002	I-ET-3010.00-5140-700-P4X-002		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		12	Space for yard cable installation with bending radius should be reserved.	Space for yard cable installation with bending radius should be reserved.	N/A (HHI Standard)		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		13	Cable entry thread type shall be in line with project requirement.	For all other cases, the threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B 1.20.1.	I-ET-3010.1M-1200-800-P4X-005		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		14	Earth continuity plate shall be provided in cable entries when non-metallic junction boxes are used.	Terminals strips, connectors and bolts used inside the JBs, located in Zone 2 shall be adequate for use when vibrations are present and avoiding sparks due to bad contact. Instrument Where required, the JBs shall have 1 (one) ground bar for grounding the armoring of cables. This bar shall be internally wired to the ground terminal of the box and it shall be provided with sufficient screws for terminating armor grounding wires. Each gland or gland plate shall be electrically bonded to its relevant equipment ground bar or terminal /junction box grounding stud. For discrete instrument's cables, shield drain wires from instruments shall be interconnected in the junction box, nevertheless shall only be grounded on a dedicated ground bar for shield drain wires on the control panel. For Analogic instrument's cable shield drain wires shall be connected to its respective multicable pair/triad/quad individual shield drain wire and then connected to a dedicated grounding bar shield drain wires on the control panel. Shield drain shall be electrically continuous from the instruments until the control panel. he general shield drain wire on multicables shall be connected to dedicated grounding bar shield drain wires on the control panel and shall be left unconnected at junction box according to IEC-60079-14. Each outdoor junction box shall have a ground bolt at the outside for bonding to the skid structure. This bolt shall terminate at the inside to provide a grounding means inside the junction box to the safety ground bar.	I-ET-3010.1M-1200-800-P4X-005		NO	Noted	NO JUNCTION BOXES IN OUR PACKAGE	Closed
		1	General 1) Scale unit shall be employed with project requirement. 2) Mounting accessaries shall be provided as per project requirement. 3) If any dissimilar metal contact during installation is expected, insulation to be provided to avoid galvanized corrosion. 4) Accessibility shall be provided as per project requirement.	The following units for the main variables shall be used: • Temperature: C • Flow rate Liquids: m³/h • Flow rate Water Vapor t/h • Flow rate Gas: m³/h (@ 20 °C and 101.325 kPa abs) • Pressure: bar-g or kPa-g • Vacuum and Low Pressures: bar-a or kPa-a • Level: % of the range, m or mm Field insturment shall be supplied with mounting bracket and insulation shall be provided to void galvanized corrosion.	I-ET-3010.00-1200-800-P4X-002		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE 21/07/2021: noted HHI20210716] Noted. However, Vendor to follow MR requirement for the instruments within vendor package	Closed
		2	Pressure transmitters 1) Housing 2) SIL 3) Material (wetted part Vs non wetted part) 4) Cablibration range 5) Accuracy / Repeatability / Temperature Effect 6) Instrument manifold if applicable 7) IP/ Ex. Proof. 8) Local display 9) Protocol (Hart, FF, PB..) 10) Asset management interface configulation 11) Capillary tube protection if applicable 12) IS/Non IS 13) Flush ring with nipple & valve if applicable. 14) Cable Entry Size	1) ASTM A351 Gr. CF8M stainless steel 2) SIL 2 for Safety (Bidding) 3) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material. 4) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point The working range of the pressure instruments shall be chosen so that the maximum value is 75% of the range and the minimum is 15% of the range. 5) The accuracy class shall be 0.5 FS (full scale), or better. -The total uncertainty of the instrument shall not exceed 0.5%, including process influences (temperature and static pressure) and combined effects of hysteresis and repeatability; - The dead band shall not exceed 0.25% of the measuring range; - A change in ambient temperature of up to 50ºC shall not affect the output signal by more than 1% of the measuring range. 6) Required for 2 valves or 5 valves 7) Min IP 56 / General Ex d 8) Required 9) Hart 10) Yes 11) Required (tray cover, thermal insulation) 12) Ex d (if Ex ia is not required for specific application on the SPEC.. Ex. Zone 0) 13) Required for diaphram seal type 14) 1/2" NPT	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE 21/07/2021: No pressure transmitter in our package, consequently, not applicable HHI2021071 6] Noted.	Closed

Technical Compliance Sheet

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EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
9	Field Instrument	3	Local indicators (gauges) 1) Material (wetted part Vs non wetted part) 2) Cablibration range 3) Accuracy 4 Instrument manifold if applicable 5) IP 6) Glass size / type / Filled liquid / dial 7) Over pressure protector, over temperature protector, snubber(dampener) 8) Capillary tube protection if applicable 9) Flush ring with nipple & valve if applicable. 10) Measuring scale to be provided for all indicator as per project unit.	1) AISI 316 stainless steel 2) Adequate for operating with working pressure equal to the upper limit of the nominal range between 1/4 and 3/4 of the nominal range 3) ± 1,0% (full scale) 4) 3-way manifold shall be provided for impulse line installation or alternately, close-coupled AISI 316 stainless steel 2-valve manifold according API 551. 5) Min IP56 6) a nominal diameter of 114 mm or 100 mm 7) Yes if required 8) Yes 9) Yes for diaphram seal type 10) Double scale indication (kPa and bar) for pressure gauge	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE 14/07/2021: To be discussed in the meeting HHI20210716] Noted. However, Vendor to follow MR requirement for the instruments within vendor package	Closed
		4	Flow meters 1) SIL 2) Material (wetted part Vs non wetted part) 3) Cablibration range/ Flow calculation / Turndown ratio 4) Accuracy 5) Instrument manifold if applicable 6) IP/ Ex. Proof. 7) Local indicator 8) Protocol (Hart, FF, PB..) 9) Asset management interface configulation 10) Upstream / downstream straight length requirement. 11) Installation manual / recommendation from manufacturer. 12) Capillary tube protection if applicable 13) IS/Non IS 14) External power requirement 15) Flush ring with nipple & valve if applicable. 16) Cable Entry	1) SIL 2 for Safety (Bidding) 2) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material. 3) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point 4) Refer to I-ET-3010.1M-1200-800-P4X-005 & I-ET-3010.00-1200-800-P4X-013 5) 5 valves for DP flowmeter 6) IP56 / General Ex d 7) Required 8) i) pulses up to 10kHz; ii) totalized in the instrument transmitter itself and transmitted via non-proprietary field network. 9) Yes 10) As per manufacturer recommendation 11) To comply manufacturer recommendation 12) Required for diaphram seal type 13) Ex d (if Ex ia is not required for specific application on the SPEC.. Ex. Zone 0) 14) Vendor to specify 15) Yes for diaphram seal type 16) NPT	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE 21/07/2021: No flow meters in our package, consequently, not applicable HHI20210716] Noted.	Closed
		5	Level transmitter including chamber 1) SIL 2) Material (wetted part Vs non wetted part) 3) Cablibration range 4) Accuracy 5) Instrument manifold if applicable 6) IP/ Ex. Proof. 7) Local indicator 8) Protocol (Hart, FF, PB..) 9) Asset management interface configulation 10) Installation manual / recommendation from manufacturer. 11) Capillary tube protection if applicable 12) Glass size / type 13) Vent/Drain valve for chamber 14) Flush ring with nipple & valve if applicable.	1) SIL 2 for Safety (Bidding) 2) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material. 3) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point 4) Refer to I-ET-3010.1M-1200-800-P4X-005 & I-ET-3010.00-1200-800-P4X-013 5) 5 valves for DP flowmeter 6) IP56 / General Ex d 7) Required 8) Hart. 9) Yes 10) To comply manufacturer recommendation 11) Required for diaphram seal type 12) Use glass sections of dimensions 7, 8 or 9 with a maximum height limited to 5 (five) units of size 9 13) Level gauges shall have drain and vent connections with valves and caps included in the instrument 14) Yes for diaphram seal type	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE 14/07/2021: No llevel transmitters in our package, consequently, not applicanble HHI20210716] Noted.	Closed
		6	Solenoid valves 1) Rating 2) IS/Non IS 3) Dual solenoid requirement	1) Limited to 5W 2) NON IS (Exd) 3) As per P&ID legned.	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE HHI20210716] Noted. However, Vendor to follow MR requirement for the instruments within vendor package	Closed
		7	Temperature transmitter/Gauge with Termowell 1) SIL 2) Material (wetted part Vs non wetted part) 3) Cablibration range 4) Accuracy 5) Glass size / type 6) IP/ Ex. Proof. 7) Local indicator 8) Protocol (Hart, FF, PB..) 9) Asset management interface configulation 10) Mininum distance requirement between temp. gauges and temp. transmitters. 11) Installation manual / recommendation from manufacturer. 12) Stress calculation 13) Thermowell	1) SIL 2 for Safety (Bidding) 2) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material. 3) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point The indication range shall be such that the normal operating temperature is between 30 and 70% of this range, and the minimum and maximum temp. (For temp. Gauge) 4) ± 0.5 °C. 5) Display with minimum nominal diameter of 100 mm for gauge 6) IP56 / General Ex d 7) Required 8) Hart 9) Yes 10) As per P&ID requirement if applicable 11) To comply manufacturer recommendation 12) To be provided 13) For flanged wells, materials and construction shall respect the following order to maintain compatibility with connection to piping or equipment: welded to flange with full penetration weld in accordance with API RP 551 and ASME PTC 19.3 TW, the material of flange and the thermowell being AISI 316L stainless steel. When the AISI 316L stainless steel flange is not suitable for process conditions or line specification, the thermowell shall be made from a single AISI 316 stainless steel piece, without any welds. If this is still not suitable for process conditions, another material, equal to flange and body, with corrosion resistance and superior mechanical strength such as Duplex and Superduplex shall be selected.	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser.	NO FILED INSTRUMENTS IN OUR PACKAGE 21/07/2021: no temperature transmitter in our package, consequently, not applicable HHI20210716] Noted.	Closed
10	FGS detector	1	Fusible lines / Heat sensitive cable 1. If fusible line/Heat sensitive cable is required for PKG skid, fusible lines/Heat sensitive cables shall be supplied/installed by PKG vendor. 2) If fusible lines/Heat sensitive cable can't be installed by vendor, technical deviation shall be submitted for approval. 3. When fusible lines/Heat sensitive cable is not supplied and installed by vendor based on agreement, the support of fusible line/Heat sensitive cable shall be supplied on skid by PKG vendor to avoid hot work damage at yard. Ex) Fusible line is applied to vessels, PKG vendor shall install required support on the vessel skid	The Fusible Plug system shall conform to DR-ENGP-M-I-1.3 – SAFETY ENGINEERING. • High-risk areas are protected by fusible plugs or by fusible plugs and flame detectors, which, in presence of fire, bleed the deluge valve actuator and flood the area with seawater. For further information, refer to I-ET-3010.1M-1200-800-P4X-005 - FIELD INSTRUMENTATION and I-FD-3010.1M-5400-947-P4X-001 - SAFETY DATA SHEET.	I-ET-3010.00-1200-800-P4X-013 DR-ENGP-M-I-1.3 I-MD-3010.1M-5520-800-P4X-001		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
		2	VESDA(Very Early Smoke Detection Apparatus) 1. If VESDA is required for electrical panel supplied by PKG vendor, VESDA pipe support within panel shall be supplied and installed by PKG vendor.	N/A	N/A		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
		3	Flame detectors 1. Flame detectors type shall be selected as per project requirement. 2) Calibration tools shall be supplied by vendor.	1) Multi-Spectrum type (IR3) detectors with microprocessor technologies Weather protection shall be provided for all detectors installed in open area. 2) A calibration kit, one per detector type, shall be provided	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
		4	Smoke detectors 1. Smoke detectors type shall be selected as per project requirement. 2) Calibration tools/tubing shall be supplied by vendor. 3.For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage.	Usiung smoke detectors shall be reviewed before application for PKG. 1) Point optical detectors shall be used for smoke detection. 2) A calibration kit shall be provided, one per detector type	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
		5	IR point gas detectors 1. IR point gas detectors type shall be selected as per project requirement. 2) Calibration tools/tubing shall be supplied by vendor.	1) Combustible gas detectors (point or open path type IR) shall be used to monitor flammable gas (CH4) leaks. 2) A calibration kit shall be provided, one per detector type	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
		6	Heat detectors 1. Heat detectors type shall be selected as per project requirement. 2) Calibration tools/tubing shall be supplied by vendor. 3. For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage.	Usiung heat detectors shall be reviewed before application for PKG. 1) Heat detectors shall be "raise of temperature rate" (Thermovelocimetric) type or fixed temperature type according to their location 2) A calibration kit shall be provided, one per detector type	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-1200-800-P4X-013		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
		7	Shutdown manual stations(push button)/ manual call points 1. Shutdown manual stations(push button)/ manual call points type shall be selected as per project requirement. 2)For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage.	5.5.3.10 There are various AMIs, also called Manual Call Points installed in suitable zone locations, around the field. Basically, these devices switch an electric contact upon the intentional lifting of their lid and push button manual actuation. Normally, the AMI switches are part of the AFDS but, in special cases, some AMI may be connected to a digital input card of a Remote I/O Panel of the FGS System. In this case, the connection shall have line monitoring. 5.3.5.1 Field ESD shutdown manual push-buttons shall have latching action and require local reset. 7.9.1 If required by Classification Society, MANUAL ESD STATIONS shall be provided in open areas, and their quantities and locations shall be defined during the Project's Detail Engineering Design Phase.	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-5140-700-P4X-002 I-MD-3010.1M-5520-800-P4X-001 I-ET-3010.00-5520-861-P4X-001		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
		8	Beacon / horn 1.Beacon / horn type shall be selected as per project requirement. 2)For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage.	Beacon / horn 1.Beacon / horn type shall be selected as per project requirement. 2)For line monitoring purpose in cases of non fail safe, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage.	I-ET-3010.1M-5510-767-PPT-001		NO	Noted.	NO F&G DETECTORS IN OUR PACKAGE	Closed
11	Cable	1	Cable type(fire resistant, flame retardant…), color and etc shall meet project cable specification.	1. For Electrical Cables - Refer to clause 5.13 of I-ET-3010.00-5140-700-P4X-002 SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS. - Armoured cable shall be applied except the wires in the panels. - For cables to be installed only inside the accommodation areas, all materials used on their formation shall be halogen free. - All cables shall be at least flame retardant. - Cables for circuits that shall operate under fire conditions (systems for firefighting, fire and gas detection, alarm, public address, shut-down, emergency switch-off, emergency lighting, etc.), cables that feed essential and emergency services that are installed in hazardous areas, and cables crossing machinery space category A, as defined by SOLAS, shall be certified for circuit integrity under fire conditions, according to IEC 60331. These cables and related brackets may also be painted with specific products intended to keep their integrity for the expected fire and time conditions. The test reports of these products and the application procedures shall be subjected to PETROBRAS approval. 2. For Automation and Instrument cables - Refer to clause 16 of I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS. - Cable for intrinsically safe circuits shall be armoured. (Both indoor and outdoor)	I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-1200-800-P4X-013	YES		Noted		Closed
		2	All cable shall be binded with cable tie as per project requirement	All cable shall be binded with cable tie and/or cable cleat as per project requirement	N/A (HHI Standard)	YES		Noted		Closed
		3	Cable coils/ loops for field devices shall be provided to avoid water dripping and fucture location adjusting.	Cable coils/ loops for field devices shall be provided to avoid water dripping and fucture location adjusting.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE, NO FIELD DEVICES	Closed
		4	Cables shall laid on cable tray. Cables running without cable tray shall be approved before implementation. (For whole skid including edge interface.)	Cables shall laid on cable tray. Cables running without cable tray shall be approved before implementation. (For whole skid including edge interface.) It is not permitted to install cables in flat bars; Armoured cables shall be installed in cable trays or perforated channels;	N/A (HHI Standard)	YES		Noted		Closed
		5	Wire color shall be in line with project requirement.	Cable core color shall be in line with project requirement which to be advised during detailed engineering.	For instrument, I-ET-3010.00-1200-800-P4X-013 Refer to attachment#3	YES		Noted		Closed
		6	Armour material shall comply to project requirement.	Galvanized steel threads braid protected by anti-corrosive cover for multi-core signal or power cables. Copper or other non-magnetic metal threads braid protected by anti-corrosive cover for single-core cables.	I-ET-3010.00-5140-700-P4X-002	YES		Bidder to comply with I-ET-3010.00-5140-700-P4X-002.	ACCORDING TO OUR STANDARD CABLES 14/07/2021: noted	Closed
		7	If fiber optic cable is required, the selection of multimode VS singemode shall be approved during detail engineering based on project specification.	If fiber optic cable is required, the selection of multimode VS singemode shall be approved during detail engineering based on project specification.	N/A (HHI Standard)	YES		Noted	IF APPLICABLE	Closed
		8	Minimum of spare capacity shall be provided in multicore cables as per project requirement.	Multi pair / triple / core cables shall have at least 20% spare at Mechanical Acceptance.	TBD	YES		Noted		Closed
		9	Cable Code List	Enforce that CONTRACTOR's sub-suppliers and PACKAGED UNIT's SUPPLIERS also create such codes and emit a similar document.	I-ET-3010.00-1200-800-P4X-010 Refer to Form-11	YES		Noted		Closed
	Lighting & Small Power	1	Where specified on the Purchaser's Electrical Requirements for Packaged Equipment Data Sheet, the Supplier shall provide lighting fixtures and/ or socket outlets. Lighting fixtures and socket outlets when required shall be provided in accordance with project Lighting and Small Power Systems requirement.	Lighting fixtures and socket outlets to be supplied shall fulfill the requirement specified in I-ET-3010.00-5140-700-P4X-002 SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS. And lighting and small power incl. socket outlet shall be designed in accordance with I-ET-3010.00-5140-700-P4X-001 SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS.	I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-5140-700-P4X-001		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		2	Lighting type shall be selected as per project requirement depending on application. Ex) Indoor, Outdoor areas and inside of cabinet.	1. Ligihting type shall be selected as per project requirement depending on application. Ex) Indoor, Outdoor areas and inside of cabinet. 2. Extra safeguarding against falling down if the screwed connections loosen, shall be provided, refer to I-DE-3010.00-5140-700-P4X-001_LIGHTING INSTALLATION TYPICAL DETAILS. 3. Floodlight and searchlight installed at hazardous areas or open areas shall have plug-socket set for removal and maintenance at floor level. These plug-socket sets shall be Ex-e type according hazardous area classification.	I-ET-3010.00-5140-700-P4X-002 I-DE-3010.00-5140-700-P4X-001		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		3	A maintenance factor shall be used in design calculations according to project requirement. and the lumen output of lamps shall be "average through life" value. LED shall have dispersive metal reflectors and shall be fitted with anti-vibration mountings.	N/A	N/A		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		4	Lighting fixtures shall be solidly fixed and not suspended by means of chains or conduits, etc. with the exception of stanchion mounted fixtures. They shall be mounted such that routine maintenance can be carried out safely and without the use of temporary scaffolding.	Lighting fixtures shall be solidly fixed and not suspended by means of chains or conduits, etc. with the exception of stanchion mounted fixtures. They shall be mounted such that routine maintenance can be carried out safely and without the use of temporary scaffolding.	N/A (HHI Standard)		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		5	Light fixtures shall be mounted at sufficient elevation and directed so as to avoid stroboscopic effects.	Light fixtures shall be mounted at sufficient elevation and directed so as to avoid stroboscopic effects.	N/A (HHI Standard)		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		6	A maximum of 12 lighting fixtures per lighting loop shall be applied. The arrangement shall be such that failure of one lighting loop will not result in complete darkness of one area. Therefore as a minimum two lighting loops per area shall be applied.	The lighting circuits shall be distributed in order not to exceed 16A or to feed no more than 24 lighting points (each lamp is deemed to be a point, while each convenience outlet comprises 2 points). A minimum of 2 (two) lighting circuits shall be provided on each area or compartment, being one normal and one essential.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		7	Lighting equipment shall be installed in such a manner that it will not obstruct or impair the clearance of passageways for personnel or equipment and such that shadows are not cast on valves, controllers, control panels, etc	Lighting equipment shall be installed in such a manner that it will not obstruct or impair the clearance of passageways for personnel or equipment and such that shadows are not cast on valves, controllers, control panels, etc	N/A (HHI Standard)		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		8	The maximum number of convenience outlets on one supply shall comply to project requirement	The lighting circuits shall be distributed in order not to exceed 16A or to feed no more than 24 lighting points (each lamp is deemed to be a point, while each convenience outlet comprises 2 points).	I-ET-3010.00-5140-700-P4X-001		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		9	All maintenance sockets circuits shall not be mixed with lighting circuits if it is not allowd on project specification.	Socket-outlets and lamps shall be fed by different circuits.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		10	Socket/ plug type shall comply with project requirement. And Male plug also to be provided in case of outdoor and indurstical type of socket.	Socket/ plug type shall comply with project requirement, refer to clause 5.7 of I-ET-3010.00-5140-700-P4X-002. And Male plug also to be provided in case of outdoor and indurstical type of socket.	I-ET-3010.00-5140-700-P4X-002		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		11	Interface Junction box for skid lighitng and socket to be provided by VENDOR. Wiring DWG/layout for above to be provided.	Interface Junction box for skid lighitng and socket to be provided by VENDOR. Wiring DWG/layout for above to be provided.	I-ET-3010.00-5140-700-P4X-003		NO	Noted	LIGHTING ARE NOT INCLUDED IN OUR SCOPE	Closed
		1	The VENDOR shall provide all the required control stations such as push buttons, selector switches, indicating lamps, to be mounted on the package as per project requirement.	The VENDOR shall provide all the required control stations such as push buttons, selector switches, indicating lamps, to be mounted on the package as per project requirement.	N/A (HHI Standard)			Bidder to check	14/07/2021: noted	Closed
		2	The local control stations shall be installed on suitable steel supports near the respective pump/skids. The motor tag number and description shall be clearly identified close to the start/stop button. The contact block shall be hermeticaly sealed individually explosion protected single pole units with 1 NO or 1NC spare contacts. The location should be such that it would not hinder access for normal maintenance and would not be liable to impact damage.	The local control stations shall be installed on suitable steel supports near the respective pump/skids. The motor tag number and description shall be clearly identified close to the start/stop button. The contact block shall be hermetically sealed individually explosion protected single pole units with 1 NO or 1NC spare contacts. The location should be such that it would not hinder access for normal maintenance and would not be liable to impact damage.	N/A (HHI Standard)	YES		Noted		Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out)		HHI 1st Comment	Remark	Status
						Yes	No			
13	Local Control Station (Push-button)	3	The local control station shall be/have: - Minimum IP rating as per project requirement. - Explosion proof as per project requirement. In addition, it is resistant to saliferous weather conditions, corrosion and ambient environment as specified. - 'Start' push button: color, contact type(NO/NC), button type to be in line with project requirement. - 'Stop' push button: color, contact type(NO/NC), button type to be in line with project requirement. - Protection cover for Emergenty push button if required. - Bottom cable entry with ISO metric threads and provision for external earthing. Provision for two-cable entry and the spare entry shall be closed with dummy stopper plug. All unused entries shall have Ex'd' certified nickelplated brass plug to meet the requirement certification. - Sunshade for LCS - Label with LCS tag number, motor tag and description - LCS shall be provided from nominated vendor. Nominated vendor will be confirmed during detail engineering.	The local control station shall be/have: 1. Control boxes for push-buttons and signalling installed in hazardous areas and for external equipment kept in operation during ESD condition, shall be Ex de type. 2. Control boxes for push-buttons and signalling shall be made of non-metallic materials according to clause 5.2 of I-ET-3010.00-5140-700-P4X-002. 3. Push-buttons for ON (START) function shall be without release (return after push). Push-buttons for OFF (STOP) function shall be with release (retain after push). 4. Push-buttons for OFF (STOP) function shall have means for locking with padlock in OFF (STOP) positions'. 5. All field push-button shall have a load tag and a load identification or push-button function identification. Identification plate shall have be in black acrylic engraved with white letter for equipment installed indoors or in stainless steel AISI-316L for equipment installed outdoors. 6. Push-buttons for ON (START) function shall have a clear protective cover in order to avoid involuntary operation, as defined in NR-12. 7. Unless otherwise stated, all threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B1.20.1. Cylindrical threaded joints shall be acceptable for lighting fixtures for fluorescent lamps, for control boxes for push-buttons and signalling, for floodlights and for power socket-outlets. Cylindrical threaded joints for other applications shall be submitted to PETROBRAS approval. For equipment installed in hazardous areas, the threads shall comply with the requirements of IEC 60079-0. 8. All screws, nuts and washers shall be made of bichromatized steel or AISI-316L stainless steel.	I-ET-3010.00-5140-700-P4X-002	YES		Noted	PACKAGE LOCATED IN SAFE AREA INDORS, SO Ex AND OUTDOOR REQUIRMENTS ARE NOT APPLICABLE	Closed
14	Electrical Heat tracing	1	Electrical heat tracing shall meet the requirements of project specification for Electric Heat Tracing	Electrical heat tracing shall meet the requirements of project specification for Electric Heat Tracing	N/A (HHI Standard)		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
		2	Material from electrical heat tracing vendor selected in Project shall be proivided in order to ensure electrical heat tracing sytem control.	Material from electrical heat tracing vendor selected in Project shall be proivided in order to ensure electrical heat tracing sytem control.	N/A (HHI Standard)		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
		3	Eletrical heat tracing equipment (heat tracing cable, Power kit, end kit etc) including power cable from inerface junction box installation shall be completed on PKG skid. External Power interface Junction box shall arranged on boundray of skid for Field cable termination.	Eletrical heat tracing equipment (heat tracing cable, Power kit, end kit etc) including power cable from inerface junction box installation shall be completed on PKG skid. External Power interface Junction box shall arranged on boundray of skid for Field cable termination.	N/A (HHI Standard)		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
		4	Following document for Electrical heat tracing wiring shall be provided, but not limited to the following. - Electrical heat tracing caluculation note - Electrical heat tracing Installation detail dwg - Wiring diagram - Electrical heat tracing fulling pipe layout dwg	Following document for Electrical heat tracing wiring shall be provided, but not limited to the following. - Electrical heat tracing caluculation note - Electrical heat tracing Installation detail dwg - Wiring diagram - Electrical heat tracing fulling pipe layout dwg	N/A (HHI Standard)		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
		5	Electrical heat tracing shall be pre-installed in FACTORY and relevent qualitty inspection recommended by Electrical heat tracing vendor inspection shall be completed before shipping.	Electrical heat tracing shall be pre-installed in FACTORY and relevent qualitty inspection recommended by Electrical heat tracing vendor inspection shall be completed before shipping.	N/A (HHI Standard)		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
		6	Heat tarcing in package equipment shall be provided by package vendor and also all cabling works between junction box and heat tracing shall be provided by vendor and general arrangement & wiring diagram for above shall be provided separately.	Heat tarcing in package equipment shall be provided by package vendor and also all cabling works between junction box and heat tracing shall be provided by vendor and general arrangement & wiring diagram for above shall be provided separately.	N/A (HHI Standard)		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
15	Obstruction Lights	1	Aircraft warning lights shall be installed on all structures 40 meters and above, designed and constructed in accordance with local aviation regulation and project requirement.	The aircraft obstruction warning signalling shall comply with the requirements of NORMAM 05, NORMAM 27 and RIPEAM 72 standards. It shall be composed by warning lighting fixtures in towers, cranes, rigs and helideck. Where it is not possible to install aircraft obstruction lights on obstacles (e.g., flare tower), floodlights shall be used as an alternative solution. Floodlights shall be positioned in order to not dazzle the vision of the pilots during landings and takeoffs. Refer to I-ET-3010.00-5140-700-P4X-002 - SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
		2	Helideck lighting and associated illuminated wind sock shall be as per local aviation authority requirement and ICAO, CAP 437 and CAAP71 standards.	The control panel of this system shall automatically control the signalling and lighting of helideck (helideck perimeter lighting fixtures, touch area lighting, windsock lighting and runway searchlights, status light, etc.). When the helideck is unguarded, or the parameters provided by the Helideck Monitoring System (HMS) are out of safety limits, or the offshore unit is under gas leakage, the status light shall remain on, according to NORMAM 27. This system shall be supplied with 220VAC from essential lighting transformers. For further information, refer to I-ET-3010.00-5140-700-P4X-002 - SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS.	I-ET-3010.00-5140-700-P4X-001		NO	Noted	NO HEAT TRACING IN OUR PACKAGE	Closed
16	Multi Cable Transits (MCT)	1	Wherever required, the supplier shall provide MCT for power, control, lighting, and earthing cables. The MCT insert modules shall allow reconstituting the tightness of the wall, floor or roof.	Wherever required, the supplier shall provide MCT for power, control, lighting, and earthing cables. The MCT insert modules shall allow reconstituting the tightness of the wall, floor or roof.	N/A (HHI Standard)	YES		Noted		Closed
		2	The MCT frame and insert modules shall be of fire resistant materials. The MCT frame material shall be min 316L stainless steel if not specified on porject spec. and welded directly on the floor, wall or roof.	The MCT frame and insert modules shall be of fire resistant materials. The MCT frame material shall be min 316L stainless steel if not specified on porject spec. and welded directly on the floor, wall or roof.	N/A (HHI Standard)	YES		Noted		Closed
		3	All bolts nuts, and accessories shall be 316L stainless steel if not specified on porject spec. and suitable for marine applications.	All bolts nuts, and accessories shall be 316L stainless steel if not specified on porject spec. and suitable for marine applications.	N/A (HHI Standard)	YES		Noted		Closed
		4	Non-magnetic stay-plates and MCT frame body are required within the transits for routing single core AC cables (in trefoil formation).	Non-magnetic stay-plates and MCT frame body are required within the transits for routing single core AC cables (in trefoil formation).	N/A (HHI Standard)	YES		Noted		Closed
		5	The MCT shall have the minimum A60 fire rating. And the rating shall be suitable for fire protection rating as per project requirement.	The MCT shall have the minimum A60 fire rating. And the rating shall be suitable for fire protection rating as per project requirement.	N/A (HHI Standard)	YES		Noted		Closed
		6	MCTs shall be peel-off type (similar to Roxtec ®) and certified for use in the hazardous area by approved international authority.	MCTs shall be peel-off type (similar to Roxtec ®) and certified for use in the hazardous area by approved international authority.	N/A (HHI Standard)		NO		NOT APPLICABLE FOR OUR PACKAGE	Closed
		7	Adequate spare MCT blocks (not bank type) shall be kept for future use.	Each MCT shall have at least 20% of spare blocks.	I-ET-3010.00-5140-700-P4X-002	YES		Noted		Closed
		8	The MCT location shall be arranged on proper location without interference for cable penetration.	The MCT location shall be arranged on proper location without interference for cable penetration.	N/A (HHI Standard)	YES		Noted		Closed
		1	Supplier to proivde suitable termination point for field interface cable and dedicated terminal block (ferrul connection type) and bus bar connection.	Supplier to proivde suitable termination point for field interface cable and dedicated terminal block and bus bar connection. Cables shall be grouped on a properly identified terminal block. Each block shall have at least 20% of reserve terminals. Terminals shall be eye type.	I-ET-3010.00-5140-700-P4X-002	YES		Noted		Closed
		2	Interwirng between componet and terminal block shall be provided to avoid direct connection to componets such as MCBs, PCB card and realys etc..	Interwirng between componet and terminal block shall be provided to avoid direct connection to componets such as MCBs, PCB card and realys etc..	N/A (HHI Standard)	YES		Noted		Closed
		3	In case that Main power cable for incomming outgoing is connected to bus bar, to apply Bus bar hole size shall be as below; - Cable core size 95sqmm and below : Bus bar hole 11mm for 10 Pi terminal lug hole. - Cable core size above 95sqmm : Bus bar hole 14mm for 12 Pi terminal lug hole.	In case that Main power cable for incomming outgoing is connected to bus bar, to apply Bus bar hole size shall be as below; - Cable core size 95sqmm and below : Bus bar hole 11mm for 10 Pi terminal lug hole. - Cable core size above 95sqmm and below : Bus bar hole 14mm for 12 Pi terminal lug hole.	N/A (HHI Standard)	YES		Noted		Closed
		4	If Two lug hole type is mendatory, Two hole space shall be 44.5mm as standard. And Terminal lug, bolt, Nut and accessory suitable for Project requirement shall provied by vendor as equipped on bus bars.	Compression terminal lug, bolt, Nut and accessory suitable for Project requirement shall be provided by vendor as equipped on bus bars. If Two hole lug type is required, two hole space shall be 44.5mm as standard.	N/A (HHI Standard)	YES		Noted		Closed
		5	Special termination bus bar type is not accepted in general. Vendor shall obtain approval from Contractor to apply special type connection. In this case all termination material and accessary shall be provided Vendor as per project cable sizing and specification.	Special termination bus bar type is not accepted in general. Vendor shall obtain approval from Contractor to apply special type connection. In this case all termination material and accessary shall be provided Vendor as per project cable sizing and specification.	N/A (HHI Standard)	YES		Noted		Closed
		6	If terminal block naming procedure is not specifed on project requirement, terminal block name shall follow below requireement for field interface cable. - AC power line : XAC - DC power line : XDC - Control Signal Input: XCI - Control Signal output: XCO	If terminal block naming procedure is not specifed on project requirement, terminal block name shall follow below requirement for field interface cable. - AC power line : XAC - DC power line : XDC - Control Signal Input: XCI (for electric) - Control Signal output: XCO (for electric) For Instrument control • The terminal blocks shall be segregated by each type of electrical signal, i.e., discrete input, discrete output, analog input, analog output, power supply, in order to assure proper segregation of signals. • Numbering procedure shall be approved during detailed engineering	N/A (HHI Standard)	YES		Noted		Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
17	Termination	7	In case of fiber optic cable termination as field cable, Vendor shall comply below requirement - Dedicate cable route for Fiber optic cable in panel or equipment inside shall be provided, ensuring adequate cable bending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data. - Patch box shall be equipped is panel (not ship loose) and to ensure Clear entrance for Fiber optic cable without obstruction. - Patch box to have all termination material (cable gland or fitting for Fiber optic cable, splicing kit, pig tail etc.) - Patch cord cable between Patch box and equipment shall be vendor scope and pre-wiring shall be completed in vendor factory. - Manufacturer test report for internal Patch cord shall be submitted accordinlgy. - Patch cord connect type shall be standard type as per project requirement.	In case of fiber optic cable termination as field cable, Vendor shall comply below requirement. - Dedicate cable route for Fiber optic cable in panel or equipment inside shall be provided, ensuring adequate cable bending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data. - Patch box shall be equipped is panel (not ship loose) and to ensure Clear entrance for Fiber optic cable without obstruction. - Patch box to have all termination material (cable gland or fitting for Fiber optic cable, splicing kit, pig tail etc.) - Patch cord cable between Patch box and equipment shall be vendor scope and pre-wiring shall be completed in vendor factory. - Manufacturer test report for internal Patch cord shall be submitted accordinlgy. - Patch cord connect type shall be standard type as per project requirement. - Test report for patch cord / splicing to be supplied.	N/A (HHI Standard)	YES		Noted	IF APPLICABLE FO OUR PACKAGE	Closed
		8	In case of Patch cord cable termination as field cable, Vendor shall comply below requirement - Dedicate cable route for patch cord in panel or equipment inside shall be provided, ensuring adequate cable vending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data. - Patch cord connect type shall be standard type as per project requirement.	In case of Patch cord cable termination as field cable, Vendor shall comply below requirement. - Dedicate cable route for patch cord in panel or equipment inside shall be provided, ensuring adequate cable bending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data. - Patch cord connect type shall be standard type as per project requirement.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		9	All field cables are allowed to be terminated in only front access of equipment without dismantling any componet or equipment.	All field cables are allowed to be terminated in only front access of equipment without dismantling any componet or equipment.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
		10	For motor termination, The cable entry size and entry direction for all motors, equipments and instruments shall be indicated on vendor drawing and suitable for HHI's cable data accordingly. For motors rated certain level(based on project requirement) above, a space heater shall be used. Separated cable termination box shall be provided for motor power and space heater circuit for all motors for certain level(based on project requirement) above.	For motor termination, The cable entry size and entry direction for all motors, equipments and instruments shall be indicated on vendor drawing and suitable for HHI's cable data accordingly.	N/A (HHI Standard)	YES		Noted		Closed
		11	Size and quantity of terminal block or bus bar shall be suitable for HHI cable size and core Quantity. Especially for bus bar connetion, bus bar hole size and dimension shall be compatible with HHI terminal lug size. All relevant data of cable will be informed later.	Size and quantity of terminal block or bus bar shall be suitable for HHI cable size and core Quantity. Especially for bus bar connetion, bus bar hole size and dimension shall be compatible with HHI terminal lug size. All relevant data of cable will be informed later.	N/A (HHI Standard)	YES		Noted		Closed
		12	The terminaltion space between enclosure and terminal block or bus bar shall be sufficient for external cable connection. In case of cabinet, gap between power cable connetion point and bottom gland plate shall be over 300mm for adequate cable termination.	The terminaltion space between enclosure and terminal block or bus bar shall be sufficient for external cable connection. In case of cabinet, gap between power cable connetion point and bottom gland plate shall be over 300mm for adequate cable termination.	N/A (HHI Standard)	YES		Noted		Closed
		13	1. Terminal block material color for each signal type shall be inline with project requirement. 2) Terminal block and bus bar size shall be suitable for external cable size and q'ty which will be informed later 3. Terminal block for spare wires of multi cables shall be provided by Package vendor. For other methods such as floating of spare wires, it shall be approved via official documentation. 4. Terminal block type for each signal type shall be selected as per project requirement. Ex. knife type, fused type, cage clamp type.....	1. Electrical : Unless otherwise stated in specific equipment technical specification, for control cables, these connectors shall be eye type. Terminal lugs shall be suitable for naval use, shall be anti-vibration type and assembled on support profiles ("C" channels). In order that neither destruction nor deformation of wires forming the cable occurs, terminal lugs shall be of indirect press over the conductor. They shall have a minimum capacity of 20A/600V and shall be made of steatite or melamine insulation, not containing toxic or organic substances. It shall not be accepted more than one cable connected to each terminal. In case of necessity of connection of more than one cable at the same point, it shall be used one terminal lug for each cable and these terminal lugs shall be connected by metallic bridge bars. Jumpers between terminals through external conductors shall not be accepted in terminal blocks. For this purpose metallic bridge bars shall be used. The terminals strips shall be installed in order to guarantee enough space to perform the cable terminations, easy access to terminals and easy reading of the identification rings. The terminals strips shall be numbered with progressive numbers and codified as per electrical diagram indications. 2. Instrument For external Wiring in control panel, Terminal connectors shall be • Made of non-hygroscope material; • Adequate for conductors with 2.5 and 1.5 mm 2 cross section; • Flame retardant; Insulated to 300 V; • Insulated to 300 V; • Of disconnect type (knife type) for external wiring / incoming cable For Juction box Signals to sub-systems, must be segregated in different terminal strips and multicables (where applicable), when installed inside the same junction box. The junction boxes shall be segregated into 3 types: • TYPE 1: for signals using flame-resistant cables • TYPE 2: for signals related to safety interlocks • TYPE 3: for signals related to process control 3. Teminal block Color for instrument • intrinsically safe: Blue • Non intrinsically safe: Hold (To be approved by HHI during detailed engineering) 4. Duct Color for instrument • intrinsically safe: Blue • Non intrinsically safe: Gray 5. Spare terminal block for control panel 20% wired etra terminal block and 10% for future use Note: all cable screen, shielding shall be individually connected to terminal block,	I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-1200-800-P4X-002 I-ET-3010.00-5520-888-P4X-001	YES		Noted		Closed
3. UCP/LCP, DCS/SDS/FGS interface and requirement										
1	**IO segregation**	1	IO segregation shall be provided as per project requirement. (Ex. IS/NIS, Safety, FGS, Analogue/digital, subsystem..)	Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification. Control/ESD(Safety) shall be independent systems only where required by Classification Society.	I-MD-3010.1M-1200-940-P4X-025 I-ET-3010.00-1200-800-P4X-002 I-ET-3010.00-5520-862-P4X-001	YES		Noted		Closed
2	**Voting / Degrading Redundancy**	1	Voting, degrading, redundancy shall be provided as per project requirement. Below Is item list to check but not limited to: 1. IO cards/modules 2) Controllers/PLCs 3. Program and data memory in a PLC 4. Power supplys 5. Communication modules	General: Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification. 1. IO cards (Bidding for redudant, In addition to provide cost saving for single) 2. The PLC CPU racks shall be arranged in redundant half-clusters in hot standby configuration.(In addition to provide cost saving for single) 3. The programming software shall allow visualization, monitoring, modifications, simulation of the application software from the programming tools (PC based) in both on-line and off-line modes. Programming of Package Control System (controllers, HMI, switches, screens and all programmable/configured variables) as well as applicable software shall be supplied fully compatible with Microsoft Windows® based operating system. 4. Power supply shall comply with I-ET-3010.00-5140-700-P4X-003 - ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS. 5. Redundant Gigabit Ethernet (IEEE 802.3an) link shall be provided between Packages UCP and SOS Package Units Data Server. Package. It is part of the panel SUPPLIER's scope to convert and redundantly distribute the different power supplies inside panels, including, where necessary, redundant AC/DC or DC/DC stabilized power supply units for cabinet's internal distribution of 24 VDC. et. 5. VENDOR to comply all requirement onl-MD-3010.1M-1200-800-P4X-001 6. When possible, to standardize equipment, engineering, construction and assembly and commissioning activities the CONTRACTOR shall procure PACKAGES with same PLCs models, same HMIs models. For PACKAGE switches that interfacing with CSS/HCSS and SOS System, the MANUFACTURER shall be the same of SOS System. For all P2, P2S, P2C and P2SC PACKAGE units, CONTRACTOR shall send the properly templates (OPC UA, Modbus TCP/IP and any other protocol developed during detailed design phase) to PACKAGER in order to standardize the memory map presentation and to facilitate the configuration and tests between PACKAGES and CSS/HCSS/SOS Systems. PACKAGE Servers shall be redundant. .	I-MD-3010.1M-1200-940-P4X-025 I-ET-3010.00-5520-862-P4X-001 I-ET-3010.00-5520-888-P4X-001 I-MD-3010.1M-1200-800-P4X-001 I-ET-3010.00-5520-800-P4X-001 I-ET-3010.00-5520-861-P4X-001 I-ET-3010.00-5520-861-P4X-001 I-ET-3010.00-5520-861-P4X-002 I-ET-3010.00-5520-862-P4X-001		NO	Vendor to provide control system with redundant power supply.	NO REDUNDANCY IS INCLUDED IN OUR PACKAGE 14/07/2021 : to be discussed in the meeting HHI20210716] Noted. However, Vendor to follow MR requirement if ther is power supplier in vendor package.	Closed
3	**SIL level Reliability and Availability**	1	For safety system, SIL certified components shall be selected as per project requirement. Ex) SIL relay, SIL certified PLC/ Controller and etc	SIL certification to be provided for Safety PLC and safety relay if applicable.	I-ET-3010.00-5520-888-P4X-001		NO	noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
4	**Wire duct requirement (Including spare and yard cable)**	1	1. Wire duct color shall meet project requirement 2) Wire duct size shall be designed to accommodate whole yard cable. Duct sizing calculation report shall be provided for approval during detailed engineering.	1) Blue plastic ducts for intrinsically safe field wiring, plastic gray ducts for Non intrinsically safe. 2) Maximum 60% of their width are used, considering all spare and future I/O	I-ET-3010.00-5520-888-P4X-001		NO	Vendor to provide	NOT APPLICABLE FOR OUR PACKAGE 14/07/2021: noted HHI20210716] Noted. we will check detail design stage.	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out)		HHI 1st Comment	Remark	Status
						Yes	No			
5	**Time synchronization**	1	Time synchronization shall be provided as per project requirement if applicable.	PACKAGE UNITs´ alarms acknowledgment applied from PACKAGE HMI or from the dedicated workstation shall be synchronized with UNIT SOS. PACKAGE SUPPLIER shall implement acknowledgement feature in PACKAGE UNIT side (controller), so that this acknowledgement can be easily readable from SOS PACKAGE UNIT RTDS. Time synchronism for all A&C systems shall be sent from Time Servers (part of Electrical System), mounted inside Time Server Panel (PN-5140005), through SNTP protocol. This signal shall be sent to SOS LAN, CSS LAN and Package Unit LAN, in order to be available for all A&C systems. For further information, see I-MD-3010.1M-5520-800-P4X-003 - AUTOMATION NETWORK DESCRIPTION. All Automation equipment shall be able to synchronize their clocks using the SNTP packets circulating in the automation rings. Therefore, all Automation network equipment (switches L2 and L3) shall be SNTP compatible and replicate the signal and to make the Synchronization signal reach all workstations and panels connected to the network. Item 5.2.10.2_I-MD-3010.1M-5520-800-P4X-003 is also valid for equipment inside package units, i.e., every package PLC, server and other equipment shall use the SNTP packets for their own synchronization. In order to do so, the switches inside the packages shall also be compatible with SNTP and able to propagate the SNTP packets to the package equipment.	I-ET-3010.00-1200-800-P4X-002 I-MD-3010.1M-5520-800-P4X-001 I-MD-3010.1M-5520-800-P4X-003		NO	Vendor to provide	NOT APPLICABLE FOR OUR PACKAGE 14/07/2021: to be discussed in the meeting HHI[20210716] Noted. However, Vendor to follow MR requirement if needed.	Closed
6	**Watchdog**	1	Watchdog shall be provided as per project requirement if applicable.	Watchdog requirement to be informed in detailed engineering. (If applicable)	I-ET-3010.00-1200-800-P4X-002		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
7	**Control Panel FAT**	1	Factory Acceptance Test for control panel shall be conducted as per project requirement. FAT procedure for control panel shall be submitted for approval.	FAT procedure for UCP shall be provided for review	I-MD-3010.1M-1200-800-P4X-001	YES		Noted	AS PER FAT PROTOCOL FOR THE PACKAGE	Closed
8	**Minimum hard IO interface for DCS**	1	Minimum hard IO interface shall be provided as per project requirement. Ex) Common fault, time sync……and etc	Hardwire interface specified on I-ET-3010.1M-1200-800-P4X-014 shall be provided as minimum. In addition, cabinet common alarm for earth fault, fan failure to be provided.	I-ET-3010.1M-1200-800-P4X-014	YES		Noted		Closed
9	**ESD/SDS IO interface**	1	Any safety related interface between PKG UCP and ICSS(ESD/SDS) shall be hardwired IO.	Hardwire interface specified on I-ET-3010.1M-1200-800-P4X-014 shall be provided as minimum.	I-ET-3010.1M-1200-800-P4X-014	YES		Noted		Closed
10	**FGS IO interface**	1	Any interface between PKG UCP and FGS shall be hardwired IO.	Hardwire interface specified on I-ET-3010.1M-1200-800-P4X-014 shall be provided as minimum.	I-ET-3010.1M-1200-800-P4X-014		NO		NOT APPLICABLE FOR OUR PACKAGE	Closed
11	**Panel color / Size / type**	1	Panel color, size, type shall be inline with project requirement.	1) Internal and external Automation panels color shall be light cream (Munsell notation 2.5 Y 9/4) 2) Maximum size of each section shall be 800 mm x 800 mm x 2,000 mm (length x width x height). 3) Side plates shall not be used between panel sections. Standardized beams and/or struts shall be used for section reinforcement. Doors shall be of single type and constructed with a 14 gauge (or thicker) steel plate, "U" beam reinforcement, grounding bolts, internal hinges, locks and handles. Locks, handles and hinges shall be designed to allow changing of door opening direction without any further piece of equipment. 4) Control Panel requirement specified in clause 4.8 of I-ET-3010.00-5140-700-P4X-002 shall be fulfilled.	I-ET-3010.00-5520-888-P4X-001	YES		Noted but bidder to comply with painting and color.	ACCORIDING TO OUR STANDARDS 14/07/2021: noted	Closed
12	**Line monitoring**	1	Line monitoring to be provided as per project requirement. If diode or resistantance shall be installed on field devices such as detectors for line monitoring, all field device shall be supplied after installing the required items.	For non fail safe, diode and resistor shall be provided for line monitoring. For interface with CSS, detail SPEC of resistor and diode will be proivded by CONTRACTOR and VENDOR shall install them at panel and field devices accordingly	I-ET-3010.00-5520-888-P4X-001		NO		NOT APPLICABLE FOR OUR PACKAGE	Closed
13	**Failure mode alarm**	1	Failure mode alarm shall be provided as per project requirement.	Failure mode alarm specifed on I-ET-3010.00-5520-888-P4X-001 and I-ET-3010.00-1200-800-P4X-002 shall be provided.	I-ET-3010.00-5520-888-P4X-001 I-ET-3010.00-1200-800-P4X-002		NO		NOT APPLICABLE	Closed
14	**Firewall**	1	If applicable, Vendor to install the fire wall as per project requirement.	All firewall required for CSS interface shall be supplied by PKG VENDOR (IEC 62443)	IEC 62443		NO		NOT APPLICABLE	Closed
15	**HMI/Laptop/Software license**	1	HMI/Laptop and software license shall be provided as per project requirement.	Programming of Package Control System (controllers, HMI, switches, screens and all programmable/configured variables) as well as applicable software shall be supplied fully compatible with Microsoft Windows® based operating system. Package supervisory system software shall be supplied to be installed at Engineering Workstation. Package UCP controller software shall be supplied to be installed at Maintenance Workstation. All keys, drivers, manuals and licenses of all software inside Package shall be provided. No software access restrictions will be accepted. The programming software shall allow visualization, monitoring, modifications, simulation of the application software from the programming tools (PC based) in both on-line and off-line modes. Supply for each PACKAGE the special tools, special cables, programmers, converters, licensed software, for allow configuration, maintenance and troubleshooting of PACKAGE equipment as PLCs, HMIs, switches, etc Provide the following PACKAGE software on original specific media: • PLC programming software; • HMI system development software; • The software of monitoring system; • All process recipes of PLC and of HIM; • Any other software necessary to reinstall all the control system inside the PACKAGE. • All devices inside of PACKAGE with Microsoft operational system shall be provide with the software on original specific CD/DVD media and licenses. OEM licenses are not acceptable. Before a Handover (after SIT) of PACKAGE, provide a bootable image of each computer in the PACKAGE. The same software installed in PACKAGE Maintenance Workstations and PN-5500005A/B shall create computer images. Supply for each PACKAGE the special tools, special cables, programmers,converters, licensed software, for allow configuration, maintenance and troubleshooting of PACKAGE equipment as PLCs, HMIs, switches, etc.	I-ET-3010.00-1200-800-P4X-002 I-MD-3010.1M-1200-800-P4X-001 I-MD-3010.1M-1200-800-P4X-001 I-ET-3010.00-5520-800-P4X-001 I-ET-3010.00-5520-861-P4X-001 I-ET-3010.00-5520-861-P4X-002 I-MD-3010.1M-5520-800-P4X-001 I-ET-3010.00-5520-862-P4X-001		NO	noted	THE SOTWARE OF OUR PACKAGE IS A COMMERCIAL SOFTWARE, WE DO DEVELOP AN SPECIFIC ONE FOR OUR PACKAGE. THEREFORE, NO LICENCES ARE SUPPLIED	Closed
15	**HMI/Laptop/Software license**	1	HMI/Laptop and software license shall be provided as per project requirement.	PAS HMI shall comply all requirement including Symbols and color codes for graphic on I-ET-3010.00-5520-800-P4X-001 The PACKAGE HMI screens shall cover all PACKAGE UNIT functionalities, including not only PACKAGE UNIT´s own process, but also its diagnostic information. Data and variables shall be displayed following PACKAGE UNIT Process and Instrumentation Diagrams (P&ID). The PACKAGE HMI screens shall be developed in a way that it is easy for the operator to understand its contents. Real-time values, equipment status changes, alarms and failures shall be highlighted and promptly identifiable. There shall be at least one screen with a general overview of the PACKAGE UNIT main variables. HMI screens shall be in Brazilian Portuguese language and in additional languages requested in the PACKAGE UNIT technical specification P2x PACKAGE UNITs HMIs and dedicated workstation(s) appearance and functionalities shall follow the defined in I-ET-3010.00-5520-800-P4X-001 – SUPERVISION AND OPERATION SYSTEM (SOS) SCREENS, whenever possible, to keep PACKAGE UNITS´ operation screens similar to SOS screens. At least, the following items shall be complied: engineering units, controllers mode (direct/reverse), control valves´ indication, level indication, line colors and open/close/start/stop indications. For P1 type Packages, the operator interface shall have at least: • Push-buttons and leds signaling; • Small electronic graphic units and keyboard; • Video screen and keyboard. For P2 type Packages, the operator interface shall be mounted at the external face of the panel front door. All supervision and control and start/stop functions, local data archiving, maintenance and configuration functions, alarm processing and status displays shall be provided at UCP HMI. Maintenance Workstations (PN-5500005A/B) supplied by CSS shall contain all software needed to configure, program and test CSS controllers and package units´ controllers. All required support and SW for Maintenance Workstations to be supplied by PAS supplier.	I-ET-3010.00-1200-800-P4X-002 I-MD-3010.1M-1200-800-P4X-001 I-MD-3010.1M-1200-800-P4X-001 I-ET-3010.00-5520-800-P4X-001 I-ET-3010.00-5520-861-P4X-001 I-ET-3010.00-5520-861-P4X-002 I-MD-3010.1M-5520-800-P4X-001 I-ET-3010.00-5520-862-P4X-001		NO	noted	THE SOTWARE OF OUR PACKAGE IS A COMMERCIAL SOFTWARE, WE DO DEVELOP AN SPECIFIC ONE FOR OUR PACKAGE. THEREFORE, NO LICENCES ARE SUPPLIED	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out) Yes	Comply (Vendor to fill out) No	HHI 1st Comment	Remark	Status
16	**Cyber security**	1	Cyber security shall comply to project requirement. Below Is item list to be checked but not limited to: -Back-up and restore procedure shall be written and tested. (If required in project specification) -External mass storage device(USB..) shall not be accessible in normal operation. -Access level shall be configured as per project requirement.	18.1.1 The Automation Network is described in I-MD-3010.1M-5520-800-P4X-003 – AUTOMATION NETWORK DESCRIPTION. Automation Network equipment specification is defined in I-ET-3010.1M-5520-800-P4X-004 – AUTOMATION NETWORK REQUIREMENTS. 18.1.2 All A&C systems shall be designed taking into account IEC 62443-2-1 - Industrial communication networks – Network and system security – Part 2-1: Establishing an industrial automation and control system security program and IEC 62443-3-3 - Industrial communication networks – Network and system security – Part 3-3: System security requirements and security levels. 18.1.3 Writing in any process variables externally from Automation Network shall not be allowed. Any external access to Automation Network shall be previously authorized and registered. 18.1.4 The connections from Automation Network to Automation firewall are from Package Unit LAN, CSS LAN and SOS LAN. Any system that shall communicate with Corporate Network, such as FMS, PI, SPCS, AMS and MMS shall one of these connections. Specific policies shall be foreseen in Automation Firewall for these communications. Vendor to comply IEC 62443 18.1.5 Anti-virus, anti-spyware, anti-malware, access register/control, version control, USB ports blocking and other protection means shall be foreseen in all A&C systems. 18.1.6 All anti-virus/anti-malware/anti-spyware to be installed in SOS computers shall be certified by the Supervisory System manufacturer. 18.1.7 All Automation systems shall be supplied cleaned against anti-virus/anti-malware. 18.1.8 Software suppliers shall issue a report containing information on which Security Levels are achieved by their product. * All firewall required for CSS interface shall be supplied by PKG VENDOR (IEC 62443) * Temporary Antivirus A temporary antivirus approved by the Automation System Vendor and compatible with Automation Supervisory and Controller systems shall be installed on all Automation system servers and workstations. Licenses shall start at the beginning of the construction and assembly time period, and shall be renewed for as many times as necessary until the definitive antivirus is installed or until 3 years after the first oil event date (i.e., the actual event date, not its predictions). The license renewals during this time period, the eventual time schedule changes or first oil postponing shall not incur in additional costs to PETROBRAS. VENDOR shall support all requirement for backup management during detailed engineering PACKAGE MANUFACTURER shall issue a document presenting all applications configured, procedures and recommendations for application programs back-up and restore.	I-MD-3010.1M-5520-800-P4X-001 IEC 62443 HHI Standard I-MD-3010.1M-5520-800-P4X-003 I-DE-3010.1M-5520-800-P4X-002		NO	vendor to provide	IN OUR OPINION, IT IS NOT APPLICABLE FOR OUR PACKAGE 14/07/2021: The bilge alarm in protected by a passwword, in order to avoid any changes in the configuration. PLC is also protected by a password. HHI20210716] Noted.	Closed
17	**Surge arrestor**	1	Surge arrestor shall be provided for required eletrical loop as per project requirement.	N/A	N/A				N/A	Closed
18	**Accessaries**	1	Following panel accessaries shall be supplied by vendor according to project requirement. 1. Warming label (power circuit) 2) Protection cover (power circuit) 3. Anti vibration pad 4. Door lock / stopper (hinge) 5. Tag plate 6. Lighting/door switch 7. Power socket 8. Fan / Temperature sensor 9. Earthing monitoring system	Following panel accessaries shall be supplied by vendor according to project requirement. 1. Warming label (power circuit) 2) Protection cover (power circuit) 3. Anti vibration pad 4. Door lock / stopper (hinge) 5. Tag plate 6. Lighting/door switch 7. Power socket 8. Fan / Temperature sensor 9. Earthing monitoring system	I-ET-3010.00-5520-888-P4X-001	YES		Noted	IF REQUIRED AND APPLICABLE	Closed
19	**Space requirement**	1	1. Terminal block, relay, IO card/module, IO module slot, CPU, space spare, and others for future shall be provided as per project requirement. 2) Calculated spare shall be specifed on UCP panel DWG.	External Wiring All terminal blocks(as well as relay) shall be provided with 20% of wired extra terminal connectors and 10% of extra terminal connectors, for future use. Internal Wiring All cable trays shall be sized in such a way that at maximum 60% of their width are used, considering all spare and future I/O.	I-ET-3010.00-5520-888-P4X-001	YES		Noted		Closed
20	**Gland plate**	1	Removable gland plate shall be provided by vendor. 1. Material 2) Hole arrangement 3. spare hole plugging 4. Earth jumper wire to equipment body.	Removable gland plate shall be provided by vendor. 1. Material: ASTM 316L stainless steel, with minimum thickness 2.5 mm. 2. Hole arrangement 3. Spare hole plugging 4. Earth jumper wire to equipment body.	I-ET-3010.00-5520-888-P4X-001	YES		Noted		Closed
		2	Removable Gland Plate shall not be painted and material shall be stainless steel, unless otherwise specified. Gland plate size shall be enough to accommodate all connected cable gland based on HHI's cable information including spare holes.	If Removable Gland Plate is provided, it shall not be painted and material shall be stainless steel, unless otherwise specified. Gland plate size shall be enough to accommodate all connected cable gland based on HHI's cable information including spare holes. For outdoor panel, Removable gland plates or Multi-cable Transits (MCT) shall be fitted for bottom cable entry. Gland plates shall be strong enough to support the necessary glands and cables. The panels shall have mounting legs of 300 mm height to accommodate installation and cable glands or MCTs. Panels shall be supplied with anti-vibration mats to be mounted between panel and mounting legs. If more than two rows (see Figure 1) of cables are needed, cable glands shall not be used, only MCTs.	I-ET-3010.00-5520-888-P4X-001	YES		Noted		Closed
21	**Wire**	1	Wire color, type(flame retardant..), size shall be designed as per project requirement.	Wire color, type(flame retardant..), size shall be designed as per project requirement. External Wiring for instrument • Made of non-hygroscope material; • Adequate for conductors with 2.5 and 1.5 mm 2 cross section; • Flame retardant; • Insulated to 300 V; • Of disconnect type (knife type). Terminal block for elec. shall be suitable for yard interface cable. Incoming and outgoing cables as well as internal wiring shall be supported and routed through plastic gray ducts with covers. In case ofintrinsically safe field wiring, the cables shall run through blue plastic ducts with protective cover. Power supply cables shall run in cable trays segregated from the marshalling termination wiring, low voltage wiring and communication wiring. These terminals shall be covered with protective plastic covers and shall have warning signs.	I-ET-3010.00-5520-888-P4X-001	YES		Noted	ACCORDING TO OUR STANDARDS	Closed
22	**Bay requirement**	1	If internal wiring between panels to be dismantled due to delivery or installation at yard, it shall be specified on DWG and loosed wiring with wire markers/lugs shall be kept inside one of panels.	If internal wiring between panels to be dismantled due to delivery or installation at yard, it shall be specified on DWG and loosed wiring with wire markers/lugs shall be kept inside one of panels.	N/A (HHI Standard)		NO		NOT APPLICABLE FOR OUR PACKAGE	Closed
23	**Network**	1	All required components for Customer network interface shall be supplied by VENDOR. 1) FO Convertor 2) Patch code 3) Patch panel 4) Switch	1 Gb/s shall be the default speed for all automation network, except for the SOS LAN ring, the CSS LAN ring, Package Units LAN ring, where the default speed shall be 10 Gb/s. Switches shall possess 30% spare ports. (30% port shall applied to FO convertor as well) All network devices shall be managed. (managed type switch) All switches shall have port mirroring capabilities in all of its ports (this is required in order to allow the IDS solution to work).	I-MD-3010.1M-5520-800-P4X-003		NO		NOT APPLICABLE FOR OUR PACKAGE	Closed
24	**In Skid Control panel & Remote Control panel**	1	The location of control panel shall be determined as per project requirement.	"VENDOR to bid based on all control panel shall be installed at Equipment skid edge excepting EDG remote control panel, inert gas gen remote control panel and marine growing protection system control panel. In addition, all cabling between JB & elec devices in skid and control panel shall be completed under vendor scope. If required during detailed engineering, control panel shall be supplied as ship loose and all cabling up to skid edge junction box(Vendor scope) shall be completed according to HHI request without cost variation"	I-MD-3010.1M-5520-800-P4X-003	YES		Noted		Closed
25	**System requirement (Incl Training)**	1	Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification. PACKAGER shall submit for PETROBRAS approval the detailed training program. In detailed program shall be presented the course plan, curriculum vitae of instructors, detailed agenda.	Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification. PACKAGER shall submit for PETROBRAS approval the detailed training program. In detailed program shall be presented the course plan, curriculum vitae of instructors, detailed agenda.	I-MD-3010.1M-1200-940-P4X-025 I-MD-3010.1M-1200-800-P4X-001	YES		Noted		Closed
4. Test requirement										
1	**Internal test report**	1	Internal test report stamped by vendor QM/QC shall be provided with FAT notification. All test report shall be inline to FAT procedure.	Internal test report stamped by vendor QM/QC shall be provided with FAT notification. All test report shall be inline to FAT procedure.	N/A (HHI Standard)	YES		Noted		Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out)		HHI 1st Comment	Remark	Status
						Yes	No			
2	**IFAT**	1	IFAT shall be conducted with third party as per project requirement.	For IFAT between CSS and PACK. CSS Supplier will visit PKG vendor shop with test tools. PKG VENDOR shall notice detailed test plan 4 weeks ago with readiness check sheet. For PACKAGE FAT, CONTRACTOR shall provide technical assistance and a portable unit (notebook) fully configured with all the necessary software of SOS, properly configured, in order to test memory maps and HMIs functionalities between PACKAGE and SOS. * IFAT failure; If major or excessive deviations from the requirements are detected during IFAT, HHI may choose to immediately halt the test and depart Supplier's facilities. If HHI halts the test and departs Supplier's facilities, Supplier shall be responsible for the following: • Immediate completion of all internal testing, correction of deficiencies, and rescheduling of the test. Supplier shall be responsible for all costs associated with these corrective activities. • Payment to HHI of purchase order and contract penalties should there be delays in shipments and delivery dates. • 100% of CONTRACTOR/COMPANY's and CONTRACTOR/COMPANY representative's travel and living costs for attending the abandoned tests. • 100% of all costs for staging and conducting the rescheduled IFAT. • If partial delevery is required due to IFAT delay, it is carried out under supplier's responsibility as per CONTRACTOR's request.	I-MD-3010.1M-1200-800-P4X-001	YES*		Noted	FAT ACCORDING TO AGREED PROTOCOL. ONLY CLASS SOCIATY EXPENSES ARE INCLUDED IN THE SCOPE. CONTRACTOR`S REPRESENTATIVES ARE INVITED TO FAT, ALTHOUGH ALL TRAVELLIN G AND LIVING EXPENSES MUST BE COVERED BY THE CONTRACTOR	Closed
3	**SIL**	1	SIL verification / validation to be conducted as per project requirement.	SIL certification for safety devices shall be supplied	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
4	**Heat soak test**	1	Heat soak test for electrical panel shall be carried out as per project requirement if applicable.	HOLD	HOLD		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
5	**FAT**	1	FAT shall be conducted with third party as per project requirement	Inspection and test requirement in I-ET-3010.00-5140-700-P4X-003 clause 7 to be fulfiled. PKG VENDOR shall notice detailed test plan 4 weeks ago with readiness check sheet. * FAT failure; If major or excessive deviations from the requirements are detected during FAT, HHI may choose to immediately halt the test and depart Supplier's facilities. If HHI halts the test and departs Supplier's facilities, Supplier shall be responsible for the following: • Immediate completion of all internal testing, correction of deficiencies, and rescheduling of the test. Supplier shall be responsible for all costs associated with these corrective activities. • Payment to HHI of purchase order and contract penalties should there be delays in shipments and delivery dates. • 100% of CONTRACTOR/COMPANY's and CONTRACTOR/COMPANY representative's travel and living costs for attending the abandoned tests. • 100% of all costs for staging and conducting the rescheduled FAT. • If partial delevery is required due to FAT delay, it is carried out under supplier's responsibility as per CONTRACTOR's request. All test specified on I-MD-3010.1M-1200-800-P4X-001 shall be conducted Control, safety and monitoring equipment and devices (power supplies, workstations, microcomputers, data servers, communication cards, controllers, I/O racks and network devices, among others) shall be designed to operate satisfactorily under the environmental conditions stipulated by the Classification Society and project documents.	I-ET-3010.00-5140-700-P4X-003 I-MD-3010.1M-1200-800-P4X-001 I-ET-3010.1M-1200-800-P4X-001	YES*		Noted	FAT ACCORDING TO AGREED PROTOCOL. ONLY CLASS SOCIATY EXPENSES ARE INCLUDED IN THE SCOPE. CONTRACTOR`S REPRESENTATIVES ARE INVITED TO FAT, ALTHOUGH ALL TRAVELLIN G AND LIVING EXPENSES MUST BE COVERED BY THE CONTRACTOR	Closed
6	**Cyber security**	1	Cyber security verification / validation shall be conducted with third party as per project requirement.	CYBERSECURITY TESTING Two Cybersecurity tests shall be foreseen in order to attest that the measures of this document (especially Sections 6 and 7_I-MD-3010.1M-5520-800-P4X-003) have been correctly implemented: • Cybersecurity Factory Acceptance Test (CFAT) o To be performed immediately after the Factory Acceptance Test, at the Automation System VENDOR facilities. o Performed after functional testing of the system is complete • Cybersecurity Site Acceptance Test (CSAT) o Performed at the location of the system o Performed after operational testing is complete o Prior to releasing the system to operation During these tests, a cybersecurity robustness testing shall also be performed. This test includes vulnerability analysis, penetration testing, and submitting the system to unusual network loads to ensure that it maintains its correct behavior. *DETAILED ENGINEERING DESIGN DOCUMENTS specifed on I-MD-3010.1M-5520-800-P4X-003 _section 9 shall be supplied for review/approval. NETWORK ONE-LINE DRAWING DETAILED NETWORK INTERCONNECTION DIAGRAM IP LIST ASSET LIST Other documents	I-MD-3010.1M-5520-800-P4X-003		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
5. Numbering philosophy										
1	**Cable numbering**	1	Cable tag number shall be allocated as per project requirement.	Cable tag number shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
2	**Instrument numbering**	1	Instrument tag number shall be allocated as per project requirement.	Instrument tag number shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
3	**Hard IO numbering**	1	Hard IO number shall be allocated as per project requirement.	Hard IO number shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
4	**Soft IO numbering**	1	Soft IO number shall be allocated as per project requirement.	Soft IO number shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
5	**Loop numbering**	1	Instrument loop number shall be allocated as per project requirement.	Instrument loop number shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
6	**Tubing numbering**	1	Tubing tag number shall be allocated as per project requirement.	Tubing tag number shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
7	**Terminal block numbering**	1	Terminal block numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation.	Terminal block numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation.	HOLD	YES		Noted		Closed
8	**Wire marker numbering**	1	Wire marker numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation.	Wire marker numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation. All panel wiring shall be identified in both ends, by plastic rings with the terminal block number.	HOLD	YES		Noted		Closed
9	**Junction box numbering**	1	Junction box numbering shall be allocated as per project requirement.	Junction box numbering shall be allocated as per project requirement.	I-ET-3000.00-1200-940-P4X-001		NO		NOT APPLICABLE FOR OUR PACKAGE , NO JUNCTION BOXES	Closed
10	**Unit control panel (UCP) numbering / Local control panel(LCP) numbering**	1	UCP and LCP numbering shall be allocated as per project requirement.	UCP/LCP Numbering to be allocated as per project requirement	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
12	**Elec. Equipment numbering**	1	All electrical equipment tag number shall be in line with project requirement.	All electrical equipment tag number shall be in line with project requirement.	I-ET-3000.00-1200-940-P4X-001	YES		Noted		Closed
13		1								
6. Others										
1	**Design Review meeting**	1	Each PKG E&I design review meeting shall be conducted as pe project requirement.	Each PKG E&I design review meeting for FAT shall be conducted 3 times as minimum. Detail schedule to be confirmed during detail engineering.	N/A (HHI Standard)	YES		Noted		Closed
2	**SIL workshop**	1	PKG vendor shall attend SIL workshop as per project requirement if applicable.	Circular Letter CL-28, Q16 The only SIL/LOPA study that needs to be considered in EPC phase is regarding closed flare system. However, during Detail Design phase, if SELLER deems necessary to carry out a LOPA Analysis to any other system, BUYER shall be consulted. Bidder to quote priced proposal for attendence of SIL workshop as an option.	N/A		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
3	**Cyber security workshop**	1	PKG vendor shall attend cyber security workshop as per project requirement if applicable.	Bidder to quote priced proposal for attendence of cyber security workshop as an option.	N/A		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
4	**Special cable scope**	1	1. Special cable shall be supplied by PKG vendor. Maximum length from skid to control/electrical room is 200m for fixed platform and 350m for FPSO. 2) All cable type which is not specified in project cable specification shall be considered as special cable.	1) Special cable shall be supplied by PKG vendor. Maximum length from skid to control/electrical room is 200m for fixed platform and 350m for FPSO. Length to be confirmed during detail engineering. 2) All cable type which is not specified in project cable specification shall be considered as special cable. 3) The cables of submerged motors must be suitable for installation in the seawater, and / or liquid hydrocarbons environment; cables must be designed to withstand this specific environment with degradation. These cables must be supplied and installed on the motor (at factory) by the motor manufacturer, with the cable length specified in the pump / motor data sheet.	N/A (HHI Standard)		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed

Technical Compliance Sheet

PROJECT :		P-78	
RFQ No.:		HQ-54	
EQUIPMENT NAME :		OILY WATER SEPARATOR	
INSPECTED ON :			
INSPECTED BY :			

Section No	Subjects		Technical requirement	Detail Project Requirement (Fill out by project team)	Relevant document number, If applicable (Fill out by project team)	Comply (Vendor to fill out)		HHI 1st Comment	Remark	Status
						Yes	No			
5	**Standardization**	1	If required, the stardarization shall be applied to PKG accoding to project requirement.	I-ET-3010.00-1200-956-P4X-002 Clause 4.1.1 For field instruments and instrumentation accessories where painting is required I-ET-3010.00-1200-956-P4X-002 – GENERAL PAINTING shall be followed. Instruments of the same type and function shall be of the same manufacturer. (HOLD)HHI considers that the Standardization Requirements on Field Instruments need not be applied to the inside of package when considering the cost impact. KSL to review and confirm. I-ET-3010.00-5140-700-P4X-002 Clause 3.6.1 "Manufacturers shall keep uniformity of equipment and components for the same supply. The same model for lighting fixtures, sockets, junction box and all bulk material shall be used in all Unit modules." (HOLD)Standardization is principally required for all bulk materials but the applicable standardization shall be clarified such as cables, etc. However, HHI considers that the Standardization Requirements on Field Electrical devices (lighting fixtures, sockets, junction box and all bulk material which are located outside package) don't need to be applied to the inside of package when considering the cost impact caused by late selection of Vendor for aformentioned electrical components. KSL to review and confirm. I-ET-3010.00-1200-800-P4X-002 Clause 6.4.4 P2x Supervisory software shall be based on Microsoft Windows® operational system and shall present information through graphic screens. Preferably, P2x Supervisory Software shall be of the same manufacturer and version as the UNIT SOS. PACKAGE SUPPLIER shall liaise with CONTRACTOR in order to identify UNIT SOS Supervisory Software manufacturer and version. (HOLD) UNIT SOS Supervisory Software manufacturer and version shall be confirmed by KSL on an urgent basis in order to inform the related PACKAGERS of the same information before TBE / PO phase.	I-ET-3010.1M-1200-800-P4X-005 I-ET-3010.00-5140-700-P4X-002 I-ET-3010.00-1200-800-P4X-002		NO	Noted	NOT APPLICABLE FOR OUR PACKAGE	Closed
6	**Approved vendor for subvendor**	1	All subvendor shall be selected among project approved vendor list. If approved vendor is not available for certain items, it shall be submitted for approval.	All subvendor shall be selected among project approved vendor list. If approved vendor is not available for certain items, it shall be submitted for approval.	N/A (HHI Standard)		NO		WE HAVE SUBMITTED OUR SUB-SUPPLIER LISTS FOR YOUR REVISION. MOST OF THEM ARE INCLUDED PROJECT VENDOR LIST	Closed
7	**L/L review meeting**	1	1. Lesson Learned meeting shall be conducted as per project requirement. 2) Compliance sheet for L/L shall be submiitted.	1) Lesson Learned meeting shall be conducted during detail engineering. 2) Compliance sheet for L/L shall be submiitted during detail engineering	N/A (HHI Standard)	YES		Noted	IF APPLICABLE	Closed
8	**Machine monitoring system interface**	1	Machine monitoring system shall be provided as per project requirement if applicable.	Comply to I-ET-3010.00-5500-854-P4X-001 and each PKG specification.	I-ET-3010.00-5500-854-P4X-001		NO		NOT APPLICABLE FOR OUR PACKAGE	Closed
9	**Asset management interface**	1	Asset management (monitoring) system interface shall be provided as per project requirement. Ex) Hart mux… All software configulation at field devices shall be conducted by PKG vendor before the delivery. Ex) Instrument tag number configulation at transmitter.	The following field instruments of critical equipment, including also Packages, shall be integrated in the AMS to the full extent: -All transmitters and valve positioners connected to CSS-PCS or CSS-HCS, including the ones that are part of P0 type Packager; -All transmitters connected to CSS-PSD or CSS-HSD, including the ones that are part of P0 type Packager; -All Fire and Gas sensors connected to CSS-FGS or CSS-HFGS, including ones that are part of P0 type packager; -All transmitters and valve positioners connected to Packages type P2 and P2C; -All Fire and Gas detectors connected to Packages, exception to those connected to an addressable system. In the latter case, those may be connected if capable. NOTE 1: All instruments connected to AMS system shall comply with HART FOUNDATION. NOTE 2: For Package type P2 and P2C, PACKAGER shall use its UCP controller with analog modules with HART capability allowing AMS data access through Single Ethernet link. In case UCP's controller I/O module does not possess HART capability, it shall be submitted to PETROBRAS approval the installation of HART multiplexers with Ethernet converter. (Including FO convertor, patch panel, patch cord) A self-paced user training package shall be included with the AMS, plus details of the Manufacturer's scheduled training courses updated through website.	I-ET-3010.00-1200-850-P4X-002		NO		NOT APPLICABLE FOR OUR PACAKGE	Closed
10	**Sunshade**	1	Sunshade shall be supplied as per project requirement.	All electrical/electronical components subjected to direct sun incidence shall have a cover, in order to minimize damage caused to excessive heating of the component's enclosure.	I-ET-3010.1M-1200-800-P4X-005		NO		NOT APPLICABLE, PACKAGE LOCATED INDOORS	Closed
11	**Spare part**	1	Spare part shall be provided as per project requirement.	Packager shall provide a list of spare parts for all electrical equipment, for at least 2 (two) years of continuous operation, including prices and part number codes. Packager shall provide all unusual tools necessary for maintenance, assembly or disassembly of all electrical equipment. Packager shall provide the necessary spare parts for the commissioning and pre-operation periods.	I-ET-3010.00-5140-700-P4X-003	YES		Noted		Closed
12	**FAT**	1	Equipment dimension and internal arrangement shall be inspected based on approved vendor GAD and signed off by Vendor QA prior to FAT. Discrepancy between equipment and vendor GAD shall be marked up in signed drawing.	Equipment dimension and internal arrangement shall be inspected based on approved vendor GAD and signed off by Vendor QA prior to FAT. Discrepancy between equipment and vendor GAD shall be marked up in signed drawing.	N/A (HHI Standard)	YES		Noted		Closed

Attachment **1** E&I tie in document guidance

APPENDIX #11

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APPENDIX #12

TECHNICAL BID EVALUATION TABLE

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TECHNICAL BID EVALUATION FOR OILY WATER SEPARATOR

COMPANY :	BID REFERENCE :	Applicable Specification(s) and corresponding document(s) :	GRADING SYSTEM (2)	with requirements	100
BUYER : HHI	PREPARED BY :			= risky or unclear	50
TAG N° : SAO-5330501	REVISION : 0			X = disqualifying	0
REQUISITION N° : I-RM-3A36.06-5330-667-KES-301				Void	----

Comments :

CODE STATUS (1)

A : ACCEPTABLE ALTERNATIVE
O : OPTIONAL (EXTRA COST)
Y : STATEMENT ACCEPTED
X : NOT ACCEPTED
C : CLARIFCATION REQUIRED
E : BIDDER TO SUBMIT

NO.	DESCRIPTION OF PROJECT SPECIFICATION	SPECIFICATION REQUIREMENT	BIDDER'S COMMENT / RESPONSE AS PER SUBMITTED TQS			
			Detegasa			
			TQS reference :	Status (see code 1)	Score (2)	Formula result
1.0	BID SUBMISSION					----
1.1	Essential Submission Required Document Checking Sheet	Refer to Appendix 11 in the Requisition				----
1.1.1	- Complete description of the scope of supply	Bidder to submit with technical quotation	YES	Y	F	
1.1.2	- Filled-in TBE table	Bidder to submit with technical quotation	YES	Y	F	100
1.1.3	- Deviation / Clarification List	Bidder to submit with technical quotation	YES	Y	F	100
1.1.4	- Prelminary GAD with weight & dimension	Bidder to submit with technical quotation	YES	Y	F	100
1.1.5	- Preliminary Utility List	Bidder to submit with technical quotation	YES	Y	F	100
1.1.6	- Preliminary datasheet	Bidder to submit with technical quotation	YES	Y	F	100
1.1.7	- Preliminary Sizing Calculation	Bidder to submit with technical quotation	YES	Y	F	100
1.1.8	- Preliminary P&ID	Bidder to submit with technical quotation	YES	Y	F	100
1.1.9	- E&I technical compliance check sheet	Bidder to submit with technical quotation	YES	Y	F	100
1.1.10	- Spare part and Special Tool list	Bidder to submit with technical quotation	YES	Y	F	100
1.1.11	- Sub-Vendor List	Bidder to submit with technical quotation	YES	Y	F	100
1.1.12	- Fabrication Schedule Bar Chart with proposed delivery	Bidder to submit with technical quotation	YES	Y	F	100
1.1.13	- Vendor Document Schedule comment	Bidder to submit with technical quotation	YES	Y	F	100
1.1.14	- Reference List	Bidder to submit with technical quotation	YES	Y	F	100
2.0	GENERAL SCOPE OF WORK					----
2.1	System and Equipment detail design	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
2.2	Manufacture, shop assembly, corrosion protection	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
2.3	Identification, inspection, testing	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
2.4	Other engineering service including layout drawing, stress analysis, technical data book, drawings, technical meetings, control system interface, vibration and noise data	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
2.5	Approved vendors list	As per Project Approved Vendor List, Bidder to confirm	PARTIALLY COMPLIANCE	Y	A	50
2.5.1	- Sub vendor list	Bidder to submit their sub-contractor list for approval as part of Vendor documentation, Bidder to confirm	YES	Y	F	100
2.6	Required documentation as per Vendor Data Requirements	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
2.6.1	- Vendor Data to be submitted	As per MR and I-ET-3A36.06-9000-911-KSL-001-DOCUMENT CONTROL PROCEDURE. Bidder to confirm.	YES	Y	F	100
2.6.2	- Final Hadnover Document	As per Material Requisition. Final dossiers shall be available for final review within 1 month of dispatch of equipment. Bidder to confirm.	YES	Y	F	100
2.6.3	- Dispatch of Documents	As per Material Requisition. Bidder to confirm.	YES	Y	F	100
2.6.4	- Inspection Release Certificate	As per Material Requisition. Bidder to confirm.	YES	Y	F	100
2.6.5	- Material Traceability, Marking and Manufacturing Work Traceability	As per Material Requisition. Bidder to confirm.	YES	Y	F	100
2.6.6	- Instruction manuals for operation and maintenance of the equipment/PACKAGE shall be provided in Portuguese language.	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES	Y	F	100
2.7	Complete liaison work with sub suppliers & sub vendors	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES	Y	F	100
2.8	Performance Guarantee & Mechanical Warranty	Bidder to confirm	YES	Y	F	100

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TECHNICAL BID EVALUATION FOR OILY WATER SEPARATOR

COMPANY :	BID REFERENCE :	Applicable Specification(s) and corresponding document(s) :	GRADING SYSTEM (2)	with requirements	100	Comments :	
BUYER : HHI	PREPARED BY :			= risky or unclear	50		
TAG N° : SAO-5330501	REVISION : 0			X = disqualifying	0		
REQUISITION N° : I-RM-3A36.06-5330-667-KES-301				Void	----		

CODE STATUS (1)

A : ACCEPTABLE ALTERNATIVE	
O : OPTIONAL (EXTRA COST)	
Y : STATEMENT ACCEPTED	
X : NOT ACCEPTED	
C : CLARIFCATION REQUIRED	
E : BIDDER TO SUBMIT	

NO.	DESCRIPTION OF PROJECT SPECIFICATION	SPECIFICATION REQUIREMENT	BIDDER'S COMMENT / RESPONSE AS PER SUBMITTED TQS															
											Detegasa							
											TQS reference :	Status (see code 1)	Score (2)	Formula result				
2.9	3D modeling (PDMS convertible, such as step file), native files (STEP/STP)	1) for 30% review, 2) for 60% review, 3) for 90% review. Final as built model. Bidder to confirm									YES	Y	F	100				
2.10	Submission of Native files (AutoCAD) of detailed drawings	Bidder to confirm									YES	Y	F	100				
2.11	Preservation / Packing suitable for seaworthy and marine climate	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.									YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50				
2.12	Kick Off Meeting at BUYER's Office (1 time, 2 working days)	Bidder to confirm									YES, BY VIDEOCALL DUE TO PANDEMIC RESTRICTIONS	Y	A	50				
2.13	Pre-Inspection Meeting at Bidder's Premise (1 working day)	Bidder to confirm									YES, AT DETEGASA PREMISES	Y	F	100				
2.14	HAZOP Meeting at BUYER's Office (1 time, 2 working days)	Bidder to confirm									Accoring to the latest discussion, this is not applicable for our package	Y	F	100				
2.15	Technical Clarification Meeting at BUYER's Office (2 times, 2 working days ea	Bidder to confirm									YES, BY VIDEOCALL DUE TO PANDEMIC RESTRICTIONS	Y	A	50				
2.16	Technical Clarification Meeting at BIDDER's Premises or Conference Call (anytime when required)	Bidder to confirm									YES, BY VIDEOCALL DUE TO PANDEMIC RESTRICTIONS	Y	A	50				
2.17	Material Identification & Traceability, Welding, Inspection	Bidder to confirm									YES	Y	F	100				
2.18	SHIPPED LOOSE ITEM control	As per Material Requisition. Bidder to confirm.									YES	Y	F	100				
2.19	Third Party Certified Spreader Beam (along with shackles, slings) suitable for single point lift for each skid.	Required. Bidder to confirm.									NOT INCLUDED IN OUR SCOPE OF SUPPLY	Y	F	100				
2.20	Supervision service for installation, pre/commissioning & start-up	Required dates' amount and lump sum pricing with daily rates shall be proposed by Bidder.									YES	Y	F	100				
2.21	Punch Close	Bidder to confirm									YES	Y	F	100				
2.22	Nameplates and tag plates as per project requirements.	Bidder to confirm									YES	Y	F	100				
2.22.1	- Equipment shall have nameplates in Brazilian Portuguese language, made of stainless steel AISI 316L, with 3 mm minimum thickness and fixed by stainless steel (AISI 316L) bolts or fasteners on visible and accessible location	Bidder to confirm									YES	Y	F	100				
2.23	First fill of lubricant and other consumables	as required for FAT Package equipment shall be specified and procured with recommended lubricants									YES	Y	F	100				
2.24	Special tools and spare parts	Required as per MR.									YES	Y	F	100				
2.24.1	- Special Tools and equipment required for installation, operation and maintenance, if any	Required									YES	Y	F	100				
2.24.2	- Spare parts for Start-up and commissioning	Bidder to provide									YES	Y	F	100				
2.24.3	- List of 2-year operational spare parts (Priced list)	Bidder to submit									YES	Y	F	100				

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TECHNICAL BID EVALUATION FOR OILY WATER SEPARATOR

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NO.	DESCRIPTION OF PROJECT SPECIFICATION	SPECIFICATION REQUIREMENT	BIDDER'S COMMENT / RESPONSE AS PER SUBMITTED TQS								Detegasa			
											TQS reference :	Status (see code 1)	Score (2)	Formula result
2.24.4	- List of Capital spares (Priced list)	Bidder to submit									IN A LATER STAGE, IN CASE OF ORDER	Y	F	100
2.25	Warning signs in Brazilian Portuguese language shall be provided where risk of personnel injury exist.	Bidder to provide									YES	Y	F	100
2.26	Training	Bidder to provide as per Section 10 in I-ET-3010.1M-5330-667-P4X-001.									YES	Y	F	100
3.0	**DETAILED SCOPE OF SUPPLY**													
3.1	Oily Water Separator Package	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.									YES	Y	F	100
3.1.1	One(1) Completed Package is intended to perform the bilge water suction from the Engine Room Bilge Water Settling Tank, with a flow rate of 5 m³/h with the maximum oil content of 500 ppm, and discharge this bilge water flow rate to sea overboard with a maximum oil content of 15 ppm, attending IMO MEPC 107 (49) requirements and Marpol Annex I – Reg. 21.	Required									YES	Y	F	100
3.1.2	Bilge water oil monitoring system - One(1) sampling collecting point in the Oil Water Separator overboard discharge; - One(1) oil cell with range between 0 and 30 ppm of oil;	Required									YES	Y	F	100
3.1.3	Control Panel (PN-SAO-5336501)	Required									YES	Y	F	100
3.1.4	Automation as per Package Classification "P1" as defined in I-ET-3010.00-1200-800-P4X-002(AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS) and I-ET-3010.1M-1200-800-P4X-014(AUTOMATION INTERFACE OF PACKAGE UNITS PACKAGE INTERFACE)	Required									YES, WITH COMMENTS/DEVIATIONS	Y	A	50
3.1.5	All required sub-equipment and/or componets in the Package based on PACKAGER's proposed technology.	Bidder to advise on detail of proposed technology and list of required sub-equipment and/or componets									YES	Y	F	100
3.1.6	All required valves, PSV and piping	Required									YES	Y	F	100
3.1.7	Junction boxes and control panels, as required;	Required									YES	Y	F	
3.1.8	Drain collector connected to the drain system of the unit;	Required									YES	Y	F	
3.1.9	Complete one(1) common skid assembly with resilient mounting for Oily Water Separator Package. - HHI standard type 4, specified in 5.2.3 Equipment support type selection - Self-supporting structural skid with drip pan, grating for access ways etc. along with earthing (grounding) / bonding.	Required									YES, ALTHOUGH GRATING IS NOT NECESSARY, DUE TO THE DIMENSIONS AND CONFIGURATION OF THE PACKAGE	Y	A	
3.1.10	On-skid access platforms and ladders, including ladder and stairway to upper levels for any skids with multiple levels as applicable – Requirement in Brazil Regulation NR 12 shall be complied with.	Required									YES	Y	F	100
3.1.11	Access steps/platforms or ladders to be mounted outside of the skid edge shall be provided with fixing brackets and bolts/nuts as shipped loose. Mounting pads for the fixing bolts shall be welded at the skid edge.	Required									YES	Y	F	100
3.1.12	All on-skid piping and tubing system including valve and supports terminating in flanged connections at skid edge. (All shipped loose piping spools and valves (if any) shall be completely welded and tested.)	Required									YES	Y	F	100
3.1.13	All on-skid electrical equipment required for the safe, automatic, continuous operation and monitoring of the package including cable trays/supports for BIDDER and BUYER cables and earthing (grounding) within the package.	Required									YES	Y	F	100
3.1.14	All on-skid instrumentation and controls required for the safe, automatic, continuous operation and monitoring of the package, including cabling, cable trays and junction boxes supplied via the platform control system BIDDER in accordance with the approved vendor list. Refer to APPENDIX 14 E&I TIE-IN DOCUMENT GUIDANCE TO BIDDERS and provide required interface documents as per the guidance.	Required									PARTIALLY COMPLIANCE	Y	A	50

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TECHNICAL BID EVALUATION FOR OILY WATER SEPARATOR

COMPANY :	BID REFERENCE :	Applicable Specification(s) and corresponding document(s) :	GRADING SYSTEM (2)	with requirements	100
BUYER : HHI	PREPARED BY :			= risky or unclear	50
TAG N° : SAO-5330501	REVISION : 0			X = disqualifying	0
REQUISITION N° : I-RM-3A36.06-5330-667-KES-301				Void	----

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											TQS reference :	Status (see code 1)	Score (2)	Formula result				
3.1.15	Pre-Inspection Meeting(PIM), Material Identification & Traceability, PMI reports, Welding, Inspection, NDT, Heat Treatment as required and Hydrostatic Testing, Bolt tightening Record, Leak Test, Mechanical Completion(MC), Factory Acceptance Test, Punch Close, Final Inspection, Packing & Preservation and Inspection Release Certificate(IRC)	Required									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	A	50				
3.1.16	Calculation Reports – for system, capacity, selection and etc.	Required									YES	Y	F	100				
3.1.17	Classification Society Certification (Classification & Certification)	Required									YES	Y	F	100				
3.1.18	Bidder shall issue an authorization to Classification Society allowing direct and full access of COMPANY(Petrobras)/HHI to all information regarding certification of project and fabrication of equipment and materials.	Required									YES	Y	F	100				
3.1.19	INMETRO compliance Certification	Required									NO, WE UNDERSTAND IT IS NOT APPLICABLE FOR OUR PACKAGE	Y	F	100				
3.1.20	Surface preparation and painting based on I-ET-3010.00-1200-956-P4X-002 – (PETROBRAS – General Painting Technical Specification).	Required									YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50				
3.1.21	Tagging and Nameplates (dual language: English and Portuguese, respectively) for Package / Equipment / Component	Required									YES	Y	F	100				
3.1.22	Safety sign per specification I-ET-3010.00-5400-947-P4X-002	Required									YES	Y	F	100				
3.1.23	Transport restraints, lifting spreader bars or frames for transportation and installation, complete with slings and shackles, materials, welding and NDE certification and proof load test certification.	Required									NOT INCLUDED IN OUR SCOPE OF SUPPLY	Y	F	100				
3.1.24	Welding and Welding Inspection	As per per I-ET-3010.00-1200-955-P4X-001, I-ET-3010.00-1200-955-P4X-002 and I-ET-3010.00-0000-970-P4X-001.									ACCORDING TO OUR STANDARDS	Y	A	50				
3.1.25	Non Destructive Testing (NDT)	As per I-ET-3010.00-1000-970-P4X-002									ACCORDING TO OUR STANDARDS	Y	A	50				
3.1.26	Bolted joints asembly and management	As per I-ET-3010.00-1200-200-P4X-116									ACCORDING TO OUR STANDARDS	Y	A	50				
4.0	**TECHNICAL REQUIREMENT**																	
4.1	**General**													----				
4.1.1	Area Classification	Non classified compartment									YES	Y	F	100				
4.1.1	Equipment Location	Engine Room as per I-DE-3010.1M-1350-942-P4X-001, Engine Room Layout									YES	Y	F	100				
4.1.2	Design Life	30-year life in a corrosive offshore environment									YES	Y	F	100				
4.1.3	Environment Conditions	As specified in Datasheet(I-FD-3010.1M-5133-661-P4X-001), MOTION ANALYSIS(I-RL-3010.1M-1350-960-P4X-009) and METOCEAN DATA(I-ET-3A36.00-1000-941-PPC-001).									YES	Y	F	100				
4.1.4	Motion & Acceleration	As per I-ET-3A36.00-1000-941-PPC-001 METOCEAN DATA and I-RL-3010.1M-1350-960-P4X-009 MOTION ANALYSIS									YES	Y	F	100				
4.1.5	Utility Consumption incl. Eletric Power Consumption	Bidder to confirm									YES	Y	F	100				
4.2	**Design Conditions (at Bilge Water Settling Tank, TQ-5330501)**																	
4.2.1	DESIGN TEMPERATURE , °C	60									SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50				
4.2.2	PRESSURE , BARG	3.4									SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50				

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TECHNICAL BID EVALUATION FOR OILY WATER SEPARATOR

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			Detegasa			
			TQS reference :	Status (see code 1)	Score (2)	Formula result
4.3	**Operating Conditions (at Bilge Water Settling Tank, TQ-5330501)**					----
4.3.1	OPERATING TEMPERATURE , °C	30	SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50
4.3.2	PRESSURE , BARG	ATM	SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50
4.4	**Capacity**	5 m³/h	SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	
4.5	**Maximum remaining oil content**	15 ppm	SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50
4.6	**Utility Requirement**	as per I-RL-3010.1M-1200-940-P4X-001 GENERAL SPECIFICATION FOR AVAILABLE UTILITIES	SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50
4.7	**Electrical Power Supply**	as per I-ET-3010.00-5140-700-P4X-003 ELEC REQUIREMENTS FOR PKG FOR OFFSHORE UNITS	SHIPYARD´S SCOPE OF SUPPLY Noted and confirmed	Y	A	50
4.8	**Separator**					----
4.8.1	Type / Model	Bidder to confirm	DELTA OWSAN-5	Y	F	100
4.8.2	Q'ty / Configuration	Bidder to confirm	1	Y	F	100
4.8.3	Design Code	Bidder to confirm	ASME (Tanks)	Y	F	100
4.8.4	Material	Bidder to confirm	ASTM A516 Grade 60 (Tanks)	Y	F	100
4.9	**Pump - 1**					----
4.9.1	Service / Purpose	Bidder to confirm	Process pump	Y	F	100
4.9.2	Design Code	Bidder to confirm	Generally API 676	Y	F	100
4.9.3	Type	Bidder to confirm	Positive displacement pump (Screw type)	Y	F	
4.9.4	Rated power	Bidder to confirm	3,6 kW	Y	F	100
4.9.5	Flow rate	Bidder to confirm	Max. 6 m3/h	Y	F	100
4.9.6	IP grade	Bidder to confirm	IP 54	Y	F	100
4.9.7	PSV	Bidder to confirm	YES	Y	F	100
4.9.8	Material	Bidder to confirm	Stainless Steel	Y	F	100
4.9.9	Test	Bidder to confirm	NOT APPLICABLE	A	C	----
4.9.10	Electric motor	as per I-ET-3010.00-5140-712-P4X-001	YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50
4.10	**Pump - 2 (Air operated sludge pump)**					----
4.10.1	Service / Purpose	Bidder to confirm	Not Applicable	Y	F	
4.10.2	Design Code	Bidder to confirm	Not Applicable	Y	F	100

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4.10.3	Type	Bidder to confirm	Not Applicable	Y	F	
4.10.4	Rated power	Bidder to confirm	Not Applicable	Y	F	100
4.10.5	Flow rate	Bidder to confirm	Not Applicable	Y	F	100
4.10.6	IP grade	Bidder to confirm	Not Applicable	Y	F	100
4.10.7	PSV	Bidder to confirm	Not Applicable	Y	F	100
4.10.8	Material	Bidder to confirm	Not Applicable	Y	F	100
4.10.9	Test	Bidder to confirm	Not Applicable	Y	F	100
4.10.10	Electric motor	as per I-ET-3010.00-5140-712-P4X-001	Not Applicable	Y	F	100
4.12	**Monitoring System**					----
4.12.1	Configuration	- One(1) sampling collecting point in the Oil Water Separator overboard discharge; - One(1) oil cell with range between 0 and 30 ppm of oil;	YES	Y	F	100
4.12.2	Detector Model	Bidder to advise	DOM-107	Y	A	50
4.13	**Filter (if required in proposed solution)**					----
4.13.1	Service / Purpose	Bidder to confirm	ABSORPTION TANK	Y	F	100
4.13.2	Filter Type	Bidder to confirm	ABSORPTION FILTER	Y	F	100
4.13.3	Design Code	Bidder to confirm	ASME (TANK)	Y	F	100
4.13.4	Material	Bidder to confirm	MODIFIED CELLULOSE (FILTER) ASTM A516 Grade 60 (TANK)	Y	F	100
4.13.5	Coating(Internal/External)	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100
4.13.6	Operating / design temperature	Bidder to confirm	Max. 50ºC	Y	F	100
4.13.7	Operating / design pressure	Bidder to confirm	2,5 BAR	Y	F	100
4.13.8	Filtration Efficiency	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100
4.13.9	Dimensions	Bidder to confirm	SEE PRELIMINARY DRAWINGS	Y	F	100
4.13.10	Volume	Bidder to confirm	SEE PRELIMINARY DRAWINGS	Y	F	100
4.13.11	Internals	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100
4.13.12	Instruments attached on Filter	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100
4.14	**Tank**					----
4.14.1	Service / Purpose	Bidder to confirm	GRAVITY SEPARATION	Y	F	100
4.14.2	Design Code	Bidder to confirm	ASME (TANK)	Y	F	100
4.14.3	Material	Bidder to confirm	ASTM A516 Grade 60	Y	F	100
4.14.4	Coating(Internal/External)	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100
4.14.5	Operating / design temperature	Bidder to confirm	MAX. 60 ºc	Y	F	100
4.14.6	Operating / design pressure	Bidder to confirm	1,5 bar	Y	F	100
4.14.7	Dimensions	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100
4.14.8	Volume	Bidder to confirm	SEE TECHNICAL SPECIFICATION	Y	F	100

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											TQS reference :	Status (see code 1)	Score (2)	Formula result					
4.14.9	Internals	Bidder to confirm									SEE TECHNICAL SPECIFICATION	Y	F	100					
4.14.10	Instruments attached on Tank	Bidder to confirm									SEE TECHNICAL SPECIFICATION	Y	F	100					
4.15	**Heater (Electrical Heater)**													----					
4.15.1	Service / Purpose	Bidder to confirm									HEATING OF OIL IN THE GRAVITY SEPARATION TANK	Y	F	100					
4.15.2	Design Code	Bidder to confirm									N/A	Y	F	100					
4.15.3	Design Duty , kW	Bidder to confirm									1,5 kW	Y	F	100					
4.15.4	Heater Maker / Model No.	Bidder to confirm									ELECTRIFOR or similar	Y	F	100					
4.15.5	Material	Bidder to confirm									SS316L	Y	F	100					
4.15.6	Coating(Internal/External)	Bidder to confirm									N/A	Y	F	100					
4.15.7	Operating / design temperature	Bidder to confirm									45 ºC	Y	F	100					
4.15.8	Operating / design pressure	Bidder to confirm									3 bar	Y	F	100					
4.15.9	Dimensions	Bidder to confirm									SEE TECHNICAL SPECIFICATION	Y	F	100					
4.15.10	Volume	Bidder to confirm									SEE TECHNICAL SPECIFICATION	Y	F	100					
4.15.11	Internals	Bidder to confirm									SEE TECHNICAL SPECIFICATION	Y	F	100					
4.15.12	Instruments attached on Heater	Bidder to confirm									SEE TECHNICAL SPECIFICATION	Y	F	100					
4.16	**PIPING**													----					
4.16.1	Reference Specification	I-ET-3010.1M-1200-200-P4X-002 (PIPING SPECIFICATION FOR HULL)									Accoding to our standards	Y	A	50					
4.16.2	Material for piping	B14H									SS316L	Y	A	50					
4.16.3	Valves	As per I-ET-3010.1M-1200-200-P4X-002 (PIPING SPECIFICATION FOR HULL)									Accoding to our standards	Y	A	50					
4.16.4	Bolting in piping	As per I-ET-3010.00-1200-251-P4X-001(REQUIREMENTS FOR BOLT MATERIALS) and I-ET-3010.1M-1200-200-P4X-002 (PIPING SPECIFICATION FOR HULL)									Accoding to our standards	Y	A	50					
4.17	**Electrical and Instrument**													----					
4.17.1	E&I technical compliance check sheet	Bidder to complete check sheet in Appendix 14, No.5									YES	Y	F	100					
4.17.2	I-ET-3010.00-5140-700-P4X-003 (ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS)	Bidder to confirm									YES, WITH COMMENTS/DEVIATIONS	Y	A	50					
4.17.3	I-ET-3010.00-5140-700-P4X-001 (SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS)	Bidder to confirm									YES, WITH COMMENTS/DEVIATIONS	Y	A	50					
4.17.4	I-ET-3010.00-1200-800-P4X-002 (AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS)	Bidder to confirm									YES, WITH COMMENTS/DEVIATIONS	Y	A	50					

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4.17.5	I-ET-3010.00-1200-800-P4X-013 (GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS)	Bidder to confirm									YES, WITH COMMENTS/DEVIATIONS	Y	A	50
4.17.6	Package Control Type	P1 as per I-ET-3010.00-1200-800-P4X-002_AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS									YES	Y	F	100
4.17.7	Oil Water Separator Control Panel(PN-SAO-5336501)	Required as per I-ET-3010.00-5520-888-P4X-001(AUTOMATION PANELS)									YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50
4.17.8	HMI	Bidder to confirm as per I-ET-3010.00-1200-800-P4X-002(AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS)									YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50
4.17.9	Signal Interface	As per I-ET-3010.00-1200-800-P4X-002 (AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS and I-DE-3010.1M-5330-944-P4X-001(P&ID for BILGE, SLUDGE, BALLAST (AFT) AND GENERAL SERVICE SEAWATER SYSTEM)									YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50
4.17.10	Type of PLC	Bidder to confirm									SIEMENS S7-1200	Y	F	100
4.18	**Structure steel and bolt&nut**													----
4.18.1	Reference Specification/Code	Bidder to confirm									ACCORDING TO OUR STANDARDS	Y	A	50
4.18.2	Structure steel grade	Bidder to confirm									Carbon steel S275 JR	Y	A	50
4.18.3	Bolt and Nut	As per I-ET-3010.00-1200-251-P4X-001(REQUIREMENTS FOR BOLT MATERIALS)									YES	Y	F	100
4.18.4	Grating	As per I-ET-3010.00-1352-130-P4X-001 (Floor Gratings, Tray systems and Guardrails made of Composite Materials)									NOT APPLICABLE, CONSIDERING THE DIMENSIONS AND CONFIGURATION OF OUR PACKAGE	Y	A	50
4.19	**Noise level , dB(A)**	As per I-ET-3010.1M-1200-300-P4X-002									YES	Y	F	100
4.20	**Vibration**	As per I-ET-3010.1M-1200-300-P4X-002									YES, ACCORDING TO OUR STANDARDS	Y	F	100
4.21	**Size of Overall Skid (L x W x H mm)**	Bidder to confirm									2820 x 2120 x 2197 mm	Y	F	100

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											TQS reference :	Status (see code 1)	Score (2)	Formula result				
4.22	Weight (Net(kg)/ Operating(kg))	Bidder to confirm									Approx. Net Weight: 2500 kg Approx. Operating weight: 3700 kg Subject to be revised durting project execution	Y	A	50				
5.0	INSPECTION AND TESTING													----				
5.1	Inspection and Testing of entire package shall be as per approved ITP	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.2	Third party inspection & Certifying agency	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.3	Material inspection and certification as per project specification	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.4	Mechanial running test as per project specification	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.5	Full load test as per project specification	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.6	Hydrotest for pump	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.7	Auxiliary equipment test	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.8	Complete unit string test as per project specification	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.9	Noist test	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.10	Vibration test	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				
5.11	Instrumentation inspection and testing as per project specification	Bidder to confirm									YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100				

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5.12	Factory and Onboard Acceptance Tests	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES, ACCORDING TO Section 6 along with Table 1 and Table 2 in EXHIBIT V, as both parties agreed in the Deviation list	Y	F	100
6.0	**SURFACE PREPARATION AND COATINGS**					----
6.1	Painting code	As per I-ET-3010.00-1200-956-P4X-002. Bidder to confirm.	YES, ACCORDING TO THE DISCUSSION IN THE TC AND DEVIATION LIST SHEETS	Y	A	50
6.2	Final Color Coading	As per DR-ENGP-I-1.15. Bidder to confirm.	YES	Y	F	100
7.0	**SPECIFICATIONS, CODES AND STANDARDS**					----
7.1	Industrial Codes and Standards	Equipment governed by this Specification shall be designed, manufactured and tested in accordance with the latest edition of the following primary Codes and Standards, including any supplementary revisions or addenda, unless modified specifically by Project Specifications listed in the Requisition and this TBE table.	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
7.2	Package Specifications	As per Appendix 2 in MR	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
7.3	Project Specifications	As per Appendix 3 in MR	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
7.4	Project Drawings	As per Appendix 4 in MR	YES, WITH COMMENTS/DEVIATIONS	Y	A	50
7.5	International Code and Standards	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES	Y	A	50
7.6	Brazilian Regulation	As per I-ET-3010.1M-5330-667-P4X-001 and MR. Bidder to confirm.	YES	Y	A	50
8.0	**GENERAL**					----
8.1	Country of origin	Bidder to confirm	SPAIN	Y	F	100
8.2	Fabrication shop location	Bidder to confirm	SPAIN	Y	F	100
8.3	Delivery Duration	Bidder to confirm	SEVEN (7) MONTHS EXWORKS	Y	F	100
8.4	Transportation method	Bidder to confirm	SEA FREIGHT	Y	F	100
8.5	Reference list of similar installations	Bidder to confirm	YES	Y	F	100
8.6	Packing list management	Packing list shall indicate component tag(project numbering) or part number with drawing number(project numbering)	YES	Y	F	100

PASS (P) OR FAIL (F) if any 0 scoring	**P**
TENDERER TECHNICAL TOTAL SCORING	**16800**
RECOMMENDATION	**Detegasa is technically acceptable**

CONFIDENTIAL

APPENDIX #13

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APPENDIX #14

E&I TIE-IN DOCUMENT GUIDANCE TO VENDOR

APPENDIX #15

QUALITY REQUIREMENTS